

The Photony Electromagnetic Compendium: Volume VIII

Beyond Field-Line Descriptions of Electromagnetic Energy Transport, Coupling,
and Release

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The Photony Electromagnetic Compendium
Volume VIII

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Preface

"I have therefore preferred to seek an explanation of the fact in another direction, by supposing them to be produced by actions which go on in the surrounding medium as well as in the excited bodies." – James Clerk Maxwell, *A Dynamical Theory of the Electromagnetic Field*, 1865.

The material collected in this volume represents the outcome of more than a decade of sustained investigation into the physical meaning of electromagnetic structure, energy transport, coupling, storage, and release. The work began with several persistent questions that arise naturally from the study of classical electromagnetics but are rarely addressed at a mechanical or structural level: What is electric charge as a physical entity? What are electric and magnetic field lines beyond their mathematical representation? How is electromagnetic energy physically organized around conductors? How does that organization move, couple, redistribute, and ultimately become released?

Mainstream electromagnetic theory provides extraordinarily successful predictive tools. Maxwell's equations, transmission-line theory, circuit models, antenna theory, Poynting-vector analysis, and established methods in signal integrity and electromagnetic compatibility allow engineers and physicists to calculate fields, currents, voltages, impedances, coupling, radiation, and energy flow with remarkable accuracy. These methods remain indispensable and are retained throughout this book as the quantitative foundation.

The purpose of this compendium is different. It asks whether the familiar field-based descriptions can be supplemented by a deeper structural interpretation. In particular, the work examines whether electric and magnetic behavior can be understood in terms of underlying physical organization rather than only through abstract field quantities. From this inquiry emerged Photony Theory, which proposes that elemental charge photons and electron-mediated transduction processes provide the structural basis for electric and magnetic phenomena.

Within the Photony framework, dynamic elemental charge photons are associated with current transport, while static elemental charge photons can accumulate at conductor surfaces and are associated with voltage and electric-chain organization. Internal electrons are proposed to act predominantly as transducers of magnetic-chain structures, while surface-associated electrons are proposed to transduce electric-chain structures. The resulting electric-chain and magnetic-chain organizations are treated not as independent abstractions but as complementary manifestations of one electromagnetic energy-transport system.

The chapters that follow develop this interpretation progressively. The discussion begins with the foundational concepts required for a Beyond Field-Line description of electromagnetism. It then examines electric energy-release regimes, magnetic energy-release regimes, and a unified framework for distinguishing and relating them. The treatment proceeds to alternating electromagnetic energy transport in AC systems, where repeated transduction, reduction, reversal, and redistribution become central. The later chapters extend the same framework to signal integrity, electromagnetic compatibility, coupling, common-mode behavior, ingress, egress, and unintended antenna formation.

A recurring objective throughout the book is to preserve the distinction between established electromagnetic relationships and the proposed Photony interpretation. Conventional equations

remain the measurable constraint. Photony is used to ask a different question: what physical organization may underlie the quantities that are already measured successfully as voltage, current, electric field, magnetic field, impedance, stored energy, coupling, and radiation?

This distinction is especially important in the treatment of electromagnetic energy transfer. Conductive, capacitive, inductive, and radiative coupling are retained as the principal engineering mechanisms by which disturbances move from source to victim. The Photony interpretation seeks to describe how elemental charge-photon populations and electron-transduced electric and magnetic structures may be reorganized during those transfers. In this way, familiar engineering concepts such as return paths, reflections, resonance, crosstalk, common-mode conversion, shielding, and antenna behavior can be reconsidered from a common structural perspective.

The phrase “Beyond Field-Line Descriptions” is therefore not intended as a rejection of field theory. It reflects an effort to move from description toward proposed physical interpretation. Field lines remain useful mathematical and visual tools, but this compendium asks what underlying physical processes might correspond to the organization they represent.

The work is also intended to connect areas that are often treated separately. Signal integrity and electromagnetic compatibility are shown to be closely related consequences of electromagnetic energy transport and redistribution. Intended and unintended antennas are treated as different engineering realizations of the same underlying electromagnetic processes. Electric and magnetic energy-release regimes are examined within a common framework. Across these topics, geometry, boundary conditions, conductivity, current, voltage, frequency, and electron response repeatedly determine how electromagnetic organization develops and changes.

This book should therefore be read neither as a replacement for conventional electromagnetics nor as a conventional textbook. It is a compendium of a developing physical interpretation. Its value ultimately depends on whether the proposed structures provide useful explanatory power, remain consistent with established measurements, and lead to distinctions or predictions that can be examined experimentally.

The engineering instruction to “Be the Signal” (Dr. Eric Bogatin) encourages the reader to follow a disturbance through the actual physical interconnect, including its return path, discontinuities, coupling opportunities, and eventual destination. The Photony perspective extends that exercise by asking the reader to “Be the Photon,” following the proposed elemental charge-photon organization through transport, transduction, redistribution, coupling, reduction, reception, and release.

This book offers a structured foundation for that investigation and invites readers to examine electromagnetic phenomena not only in terms of the equations that describe them, but also in terms of the physical organization those equations may represent.

Introduction

Electromagnetism is among the most successful and widely applied areas of physics. Maxwell's equations provide a unified mathematical description of electric and magnetic phenomena, while transmission-line theory, antenna theory, circuit analysis, electromagnetic compatibility, and signal-integrity methods extend that description into practical engineering systems. These theories predict measurable behavior with extraordinary success and remain the quantitative foundation for the work developed in this book [Maxwell \(1865\)](#); [Jackson \(1998a\)](#); [Balanis \(2016\)](#); [Ott \(2009\)](#); [Paul \(2006\)](#).

Yet, predictive success does not eliminate questions of physical interpretation. Electric and magnetic fields are represented mathematically as continuous quantities defined throughout space, and field lines provide useful visual descriptions of their direction and relative magnitude. These representations are indispensable, but they do not by themselves specify a mechanical structure corresponding to a field line, explain what physically constitutes electric charge, or identify a discrete carrier from which macroscopic electromagnetic organization emerges.

Questions of electromagnetic energy transport are especially important in this regard. Poynting's treatment established that electromagnetic energy flow cannot be understood solely by following current within a conductor. Energy transport must also be considered in the electromagnetic region surrounding conductors and sources [Poynting \(1884\)](#). Modern electromagnetic theory captures this through the Poynting vector,

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad (1)$$

which describes the directional flux of electromagnetic energy. This formulation is quantitatively powerful. The question pursued throughout this compendium is complementary: what physical organization might correspond to that energy transport?

The same question arises in circuits and transmission structures. A propagating signal is not adequately understood as a voltage value that instantaneously appears everywhere along a conductor. Once propagation time becomes significant, the signal must be treated as a distributed electromagnetic process involving both signal and return geometry, distributed capacitance and inductance, characteristic impedance, reflections, and energy transport [Bogatin \(2018\)](#); [Paul \(2006\)](#). At still larger electrical dimensions, the same physical system can support resonance, common-mode current, coupling, and radiation. The boundaries between circuit behavior, signal integrity, electromagnetic compatibility, and antenna behavior are therefore determined less by separate physical laws than by geometry, frequency, boundary conditions, and the manner in which electromagnetic energy is organized.

The engineering instruction to "Be the Signal," associated with Eric Bogatin's approach to signal-integrity analysis, captures an important way of thinking about such systems [Bogatin \(2018\)](#). Rather than considering only schematic nodes, the engineer follows the signal through the actual interconnect and asks what impedance it encounters, where its return current flows, where it reflects, what structures it couples to, and where its energy ultimately goes. This perspective becomes increasingly important as edge rates increase and structures that appear electrically insignificant at low frequency become distributed electromagnetic systems.

The present work extends that mode of reasoning into the proposed Photony framework. Once one learns to “Be the Signal,” the next question is whether it is possible to “Be the Photon,” following the proposed underlying electromagnetic organization through transport, transduction, accumulation, redistribution, coupling, reduction, and release.

Photony Theory proposes that electromagnetic phenomena arise from discrete underlying structures involving elemental charge photons and electron-mediated transduction [Martin \(2022\)](#). Within this interpretation, dynamic elemental charge photons are primarily associated with current transport, while static elemental charge photons can accumulate at conductor surfaces and contribute to voltage and electric-chain organization. Internal electrons are proposed to transduce predominantly magnetic-chain structures, while surface-associated electrons are proposed to transduce predominantly electric-chain structures.

This distinction provides a recurring structural theme throughout the book:

dynamic elemental charge-photon transport → internal-electron transduction
→ magnetic chain organization,

while

static surface accumulation → surface-electron transduction
→ electric chain organization.

These proposed processes are not introduced as substitutes for Maxwell’s equations or established engineering relationships. Conventional electromagnetic quantities remain the measurable constraints. The Photony interpretation instead asks whether those quantities correspond to an underlying structural organization that can provide a more tangible physical description of electric and magnetic behavior.

This distinction is important throughout the compendium. When conventional theory describes the electric field, magnetic field, current density, potential, impedance, stored energy, coupling, resonance, or radiation, the corresponding equations remain applicable. Photony proposes what physical processes may underlie those quantities. Agreement with established measurements is therefore required rather than optional in the proposed framework.

One of the principal themes developed in the chapters that follow is the distinction between electric and magnetic energy organization. In conventional electromagnetics, electric and magnetic energy densities are written as

$$u_E = \frac{1}{2} \epsilon E^2, \quad (2)$$

and

$$u_B = \frac{B^2}{2\mu}, \quad (3)$$

for linear media. These expressions describe where electromagnetic energy is stored and how its magnitude depends on the local field quantities. The Photony interpretation develops corresponding structural descriptions in which electric-chain and magnetic-chain organization arise through different electron-transduction processes but remain parts of one electromagnetic system.

This distinction becomes particularly significant during rapid energy release. Electric breakdown is associated primarily with sufficiently strong electric conditions and the organization of charge between surfaces or regions of differing potential. Magnetic energy release, by contrast, is associated with magnetic structure accumulated in conducting systems and plasmas. The chapters

devoted to electric and magnetic energy-release regimes examine these processes separately before developing a unified framework for distinguishing their characteristic conditions.

A second major theme is electromagnetic transport under alternating excitation. Alternating-current systems require continual reversal of local electrical conditions. Conventional theory describes this behavior through sinusoidal and transient voltages and currents, phase relationships, impedance, reactance, propagation, and energy exchange. The Photony interpretation adds repeated electron reorientation, transduction, chain reduction, and renewed transduction as proposed microscopic processes accompanying alternating excitation.

This AC behavior provides an important bridge between energy storage and electromagnetic radiation. An antenna is not a fundamentally separate electromagnetic object from a transmission structure. It is a conductor geometry in which distributed current and voltage conditions allow electromagnetic organization to extend outward and radiate efficiently [Balanis \(2016\)](#). The distinction between intended and unintended antennas is therefore primarily an engineering distinction involving design intent, geometry, excitation, and confinement.

The same reasoning connects antenna behavior to electromagnetic compatibility and signal integrity. Conductors intended to carry signals or power can become unintended electromagnetic structures when their physical dimensions, return paths, asymmetry, or resonances allow electromagnetic energy to extend beyond the intended geometry. Cables, shields, chassis structures, traces, reference planes, connectors, and enclosure features can therefore participate in unwanted coupling and radiation [Ott \(2009\)](#); [Paul \(2006\)](#).

For this reason, the compendium treats signal integrity and electromagnetic compatibility as closely related consequences of electromagnetic transport. Signal integrity concerns whether the intended waveform remains sufficiently faithful as it propagates through its interconnect. Electromagnetic compatibility concerns whether unwanted energy leaves the intended system, enters it, or couples between structures in a manner that interferes with proper operation. Both depend on the physical pathway followed by electromagnetic energy.

A useful system-level representation is

$$\text{source} \rightarrow \text{coupling path} \rightarrow \text{victim}. \quad (4)$$

The coupling path can involve four principal mechanisms:

conductive	capacitive	inductive	radiative
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Conductive coupling involves a shared physical conductor or impedance. Capacitive coupling is associated primarily with voltage, surface charge, mutual capacitance, and dv/dt . Inductive coupling is associated primarily with current, mutual inductance, loop geometry, and di/dt . Radiative coupling occurs when an electromagnetically significant structure launches propagating energy that can be intercepted by another structure [Ott \(2009\)](#); [Paul \(2006\)](#).

These mechanisms also provide a natural way to describe electromagnetic egress and ingress. Egress refers to electromagnetic energy leaving its intended system or boundary. Ingress refers to electromagnetic energy entering a system and producing an unwanted response. A disturbance may pass through more than one coupling mechanism during its path from original source to final victim.

The Photony interpretation seeks to give each of these mechanisms a structural counterpart. Conductive coupling is associated with connected dynamic transport and redistribution. Capacitive coupling is associated primarily with surface accumulation and electric-chain interaction. Inductive coupling is associated primarily with current-driven magnetic chain organization. Radiative coupling is associated with electromagnetic organization that becomes capable of outward propagation and subsequent interaction with a receiving structure.

The chapters of this book develop these ideas progressively rather than treating them as isolated topics. The foundational chapter introduces the Photony concepts required for electromagnetic applications and relates them to established electromagnetic theory. The following chapters examine electric energy-release regimes and magnetic energy-release regimes separately, then bring them together into a unified framework. The treatment then turns to alternating electromagnetic energy transport in AC systems and finally to unintended antennas, signal integrity, electromagnetic compatibility, ingress, egress, and the four coupling mechanisms.

The final synthesis considers these subjects as different manifestations of one underlying question: how electromagnetic energy is organized, transported, stored, redirected, coupled, released, and received.

The phrase “Beyond Field-Line Descriptions” summarizes the purpose of the compendium. Field lines remain useful and conventional electromagnetic theory remains indispensable. The objective is not to discard those descriptions, but to ask whether a deeper structural interpretation can be developed beneath them. Photony Theory proposes one such interpretation and subjects it to the requirement that it remain consistent with the established electromagnetic behavior that has already been measured and successfully modeled.

The value of that interpretation will ultimately depend on more than conceptual appeal. It must provide explanatory coherence across phenomena that are presently described using different engineering frameworks, remain compatible with established quantitative results, and lead toward distinctions or consequences that can be examined experimentally. The chapters that follow develop that case across electromagnetic energy release, AC transport, coupling, radiation, signal integrity, and electromagnetic compatibility.

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Chapter 1

Foundations of Beyond Field-Line Electromagnetics

1.1 Purpose and Scope

Electromagnetism is conventionally described through electric and magnetic fields, charge and current densities, potentials, material properties, boundary conditions, and the equations that relate them. Maxwell's equations provide the central mathematical framework,

$$\nabla \cdot \mathbf{D} = \rho, \quad (1.1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (1.2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (1.3)$$

and

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (1.4)$$

Together with constitutive relationships such as

$$\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}, \quad (1.5)$$

these equations describe electromagnetic behavior with extraordinary quantitative success [Maxwell \(1865\)](#); [Jackson \(1998a\)](#).

The purpose of the Beyond Field-Line approach is not to replace these relationships. It asks a different question. If an electric or magnetic field has measurable energy, exerts forces, propagates, interacts with matter, and transfers

energy between systems, what underlying physical organization might correspond to the field quantities appearing in the equations?

Photony Theory proposes one possible structural answer. It introduces a discrete underlying space, elemental charge photons, energy filaments, electron-mediated transduction, and chain structures that are proposed to underlie electric and magnetic phenomena [Martin \(2022\)](#). Within this framework, conventional fields remain the measurable macroscopic description, while Photony attempts to provide a structural description beneath them.

This distinction will be maintained throughout this compendium:

Conventional electromagnetics provides the quantitative description of observable behavior, while Photony provides a proposed structural interpretation of that behavior.

Agreement with established electromagnetic measurements is therefore a constraint on Photony rather than an optional objective.

1.2 From Field Description to Physical Structure

Field lines are extraordinarily useful. They provide a visual representation of the direction and relative strength of electric and magnetic fields and allow complicated spatial relationships to be understood intuitively. They do not, however, constitute a claim that literal mathematical lines exist independently as physical objects.

A field-line representation can therefore answer questions such as where the electric field is stronger, how a magnetic field circulates around a current, or how electromagnetic energy may be distributed spatially. It does not necessarily answer what the field itself consists of.

The Beyond Field-Line approach begins at that point. It seeks a physical structure that can account for several established properties simultaneously:

- electric and magnetic energy can occupy regions outside conductors;
- electromagnetic disturbances propagate at finite speed;
- geometry and boundary conditions strongly determine electromagnetic behavior;
- current and voltage produce different but related electromagnetic organizations;
- electric and magnetic energy can be stored and subsequently returned or released;
- electromagnetic energy can couple between structures without direct conductive contact;
- electromagnetic radiation can propagate through free space independently of its original source.

Any proposed structural model must ultimately reproduce these established behaviors.

Photony proposes that the field quantities observed macroscopically arise from organized populations of elemental charge photons, whose configurations are established, regulated, and modified through interaction with electrons.

1.3 Planck Space as the Proposed Electromagnetic Substrate

At the foundation of Photony Theory is Planck Space, proposed as a discrete, quasi-static, compressible spatial structure composed of interconnected Planck-scale cells [Martin \(2022\)](#). The

model treats space not as featureless emptiness but as a structured medium through which energy configurations move and within which they can impose local geometric and energetic constraints.

Each Planck cell is conceived as having six nearest-neighbor connections aligned with three mutually orthogonal spatial axes. Propagation at the fundamental scale therefore follows rectilinear nearest-neighbor pathways rather than arbitrary continuous diagonals.

A simplified representation is

$$(\pm x, \pm y, \pm z),$$

corresponding to the six orthogonal ports of a Planck cell.

This structure produces what Photony describes using a Manhattan metric. In a continuous Euclidean representation, the circumference of a circle of radius R is

$$C_{\text{Euclid}} = 2\pi R. \quad (1.6)$$

For an idealized rectilinear Manhattan loop representing the same radial extent,

$$C_{\text{Manhattan}} = 4R. \quad (1.7)$$

Photony introduces geometric compression as part of the reconciliation between preferred geodesic filament motion and the orthogonal connectivity of Planck Space. This interplay leads to the recurring 4π compression relationship used within the theory [Martin \(2022\)](#).

The importance of Planck Space to electromagnetism in this volume is not primarily gravitational. Its electromagnetic role is to provide the proposed substrate through which elemental charge photons, chain structures, and classical photons can exist and move.

The local electromagnetic propagation speed remains constrained conventionally by

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (1.8)$$

in free space, and more generally by

$$v = \frac{1}{\sqrt{\mu \epsilon}} \quad (1.9)$$

for an idealized linear medium.

Within Photony, these equations remain measurable relationships. The structural interpretation associates changes in effective permittivity and spatial compression with changes in the local propagation environment rather than treating c as detached from the properties of the space through which energy propagates.

1.4 Energy Filaments and Elemental Charge Photons

The fundamental energetic object in Photony is the energy filament. The term *filament* is reserved in this compendium for this fundamental energy structure and is not used for macroscopic bundles of electromagnetic chains.

An elemental charge photon consists of a self-organized energy filament configuration. Depending on its motion and organization, an elemental charge photon can participate in static or dynamic electromagnetic behavior [Martin \(2022\)](#).

The theory distinguishes two especially important populations:

dynamic elemental charge photons

 (1.10)

and

static elemental charge photons.

 (1.11)

Dynamic elemental charge photons are proposed to participate in propagating energy transport and electrical current. Static elemental charge photons are proposed to accumulate in localized configurations, especially at conductor surfaces, and contribute to electric potential and electric chain organization.

This distinction becomes central to the electromagnetic framework developed throughout the book:

dynamic population → current behavior
static population → surface voltage behavior

These populations are not proposed as two unrelated kinds of energy. Rather, they represent different organizational states of the same fundamental photonic energy structure.

1.5 Classical Photons and Electromagnetic Radiation

Photony distinguishes an elemental charge photon from the classical photon conventionally associated with electromagnetic radiation.

A classical photon is proposed to consist of a chainlet formed from two interlinked elemental charge photons. Unlike a static elemental charge photon, the classical photon is a propagating energy quantum and cannot exist as a stationary object within the proposed framework [Martin \(2022\)](#).

Its conventional energy remains

$$E = hf, \quad (1.12)$$

and its wavelength is related to frequency by

$$\lambda = \frac{c}{f} \quad (1.13)$$

in free space.

Photony does not alter these measured relationships. It proposes a structural distinction between longer electromagnetic chain organizations and classical photon chainlets at higher frequencies.

A characteristic transition frequency used in the Photony framework is approximately

$$f_t \approx 3.76 \times 10^{13} \text{ Hz}, \quad (1.14)$$

corresponding to a free-space wavelength of approximately

$$\lambda_t \approx 7.98 \text{ } \mu\text{m}. \quad (1.15)$$

Within the proposed framework, frequencies below this transition are predominantly associated with extended electromagnetic chain behavior, while higher-frequency behavior increasingly involves discrete classical photon emission and absorption.

This proposed transition becomes useful later when distinguishing low-frequency conductor behavior from optical, X-ray, and gamma-ray photon interactions.

1.6 Electrons as Transducers and Regulators

One of the most important departures of the Photony interpretation from common intuitive circuit pictures concerns the electron.

Within Photony, electrons are not treated as the primary source of the electromagnetic energy transported through a circuit. Instead, they act predominantly as *transducers and regulators* of elemental charge-photon organization [Martin \(2022\)](#).

The energy population is associated with elemental charge photons. Electrons determine how that population becomes organized into electric or magnetic structures.

This distinction may be represented as energy population $\neq$ electron population, while electron $\rightarrow$ transduction and regulation.

This interpretation becomes particularly important because conventional electron drift speed in a conductor is much slower than the propagation speed of an electrical disturbance. Circuit energy can therefore reach a load long before individual conduction electrons could traverse the complete conductor length..

Conventional electromagnetic theory already explains this through electromagnetic fields and Poynting transport. Photony seeks to provide a structural interpretation of the same result by treating the dynamic elemental charge-photon population as the rapidly propagating energy population and the electrons distributed throughout the conductor as local transducers.

The result is a distributed rather than serial picture of circuit behavior. An electron near the source does not need to travel physically to the load before an electron near the load can respond. The electromagnetic organization propagates through the system and changes the local transduction condition throughout the conductor.

In addition to acting as a transducer or regulator, an electron may also interact with and travel along a previously established photon-chain structure. This proposed chain-riding behavior is distinct from transduction: transduction forms or reorganizes the chain, whereas riding describes subsequent electron motion along an

already existing chain; the distinction is developed further in Section 5.4.9.

7 Electron Orientation and the Motion Rule

1.8 Electron Orientation and the Motion Rule

Photony assigns the electron an intrinsic internal motion referred to as *c-spin*. Under externally imposed electromagnetic excitation, the electron can also acquire an orientation response referred to as extrinsic *x-spin* [Martin \(2022\)](#).

Important distinction: intrinsic *c-spin* $\neq$ external *x-spin*.

The *x-spin* describes a driven end-over-end reorientation associated with alternating electromagnetic excitation. It does not replace the electron's intrinsic structure.

Photony also imposes a motion rule on energy filaments. A filament progresses head first along admissible pathways and does not undergo unrestricted lateral translation or a simple jump-rope rotation. These constraints are intended to preserve continuity within the discrete Planck Space structure.

For electromagnetic applications, the significance of the motion rule is that chain formation, propagation, transduction, and reduction must occur through locally admissible configurations rather than arbitrary geometric jumps.

1.9 Magnetic Chains and Internal-Electron Transduction

Photony proposes that magnetic behavior in a conductor arises predominantly through interaction between dynamic elemental charge photons and internal electrons.

When a current-associated dynamic population passes through a conductor, internal electrons orient in response to the local electromagnetic condition. These electrons are proposed to transduce elemental charge photons into inter-

linked magnetic chains that form closed structures around the current path [Martin \(2022\)](#).

The conceptual sequence is:

dynamic transport → electron alignment
 → magnetic transduction
 → magnetic organization
 (1.16)

The conventional magnetic field surrounding a current remains described by Maxwell's equations and, for suitable simple geometries, by Ampere's law. For a long straight conductor,

$$B(r) = \frac{\mu I}{2\pi r}. \quad (1.17)$$

The magnetic energy density is

$$u_B = \frac{B^2}{2\mu}. \quad (1.18)$$

Photony requires the proposed magnetic chain population to reproduce this measured dependence. Increasing current must therefore correspond to an increase in net magnetic organization consistent with the conventional increase in B and u_B .

Thus,

$$|I| \uparrow \implies |B| \uparrow \implies u_B \uparrow$$

is retained quantitatively, while magnetic chain transduction provides the proposed structural interpretation.

1.10 Electric Chains and Surface-Electron Transduction

Electric behavior is proposed to arise through a complementary transduction mechanism associated primarily with conductor surfaces.

As elemental charge photons accumulate statically at conductor surfaces, the local electric potential increases. Surface-associated electrons are proposed to organize this population into electric chains that extend outward from the conductor [Martin \(2022\)](#).

The sequence is

surface accumulation → electron transduction
 → electric chain growth.
 (1.19)

Conventionally, electric energy density in a linear medium is:

$$u_E = \frac{1}{2}\epsilon E^2 \quad (1.20)$$

The proposed chain population must therefore reproduce the measured dependence of electric energy on field strength.

Increased potential difference generally produces increased surface-charge redistribution and stronger electric fields according to the geometry and boundary conditions of the structure. Photony interprets this macroscopically measured change as increased organization of static elemental charge photons into electric chains.

1.11 Electric and Magnetic Structures as One Electromagnetic System

The distinction between electric and magnetic chains should not be interpreted as implying two independent energy systems.

Both arise from one underlying elemental charge-photon population and are organized by electron transduction according to the local electromagnetic condition.

The central correspondence is

high current → magnetic chain organization,
 surface accumulation → electric chain organization.

In a general electromagnetic system, both forms can be present simultaneously.

This becomes particularly apparent in transmission lines, AC circuits, resonant structures, and antennas, where voltage and current vary with both position and time. One region may be relatively current dominant while another is relatively voltage dominant.

The resulting system is therefore more appropriately represented as

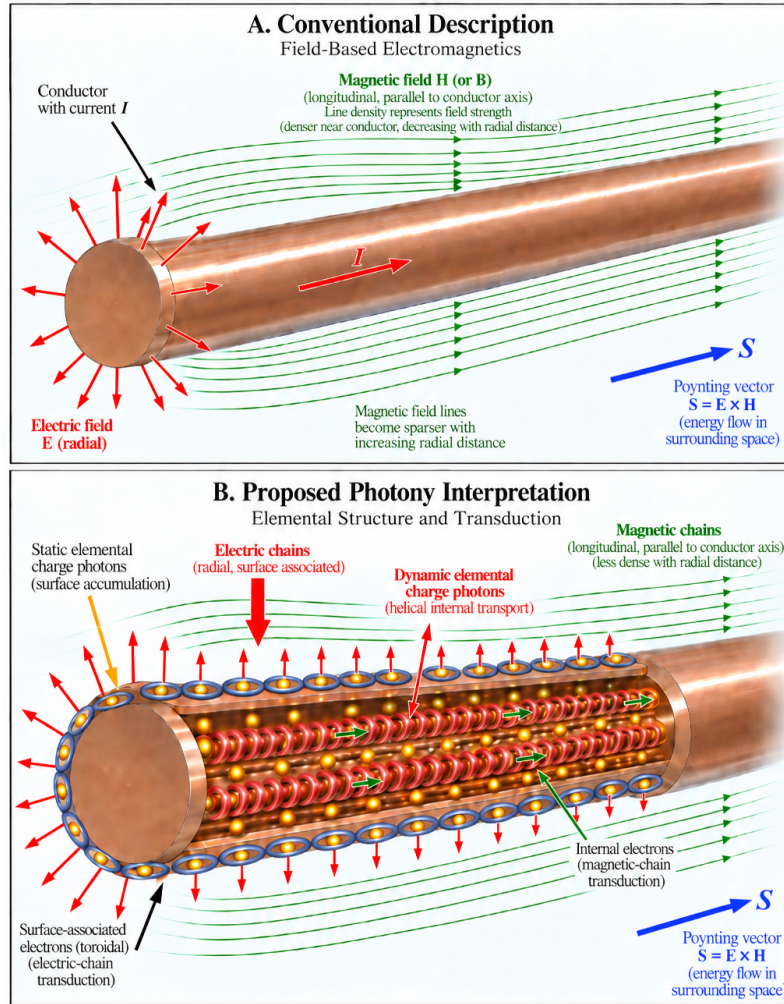


Figure 1.1. Conventional and Photony descriptions of electromagnetic organization around a conductor.
Panel A shows the conventional field-based description for a current-carrying conductor, including the radial electric field E , the longitudinal magnetic field H (or B) with line density indicating field strength (denser near the conductor and decreasing with radial distance), and the Poynting vector $S = E \times H$, which indicates energy flow in the surrounding space.
Panel B shows the proposed Photony interpretation of the same configuration. Dynamic elemental charge photons are depicted as small helices with their axes aligned with the conductor and the Poynting vector. These helices travel within the conductor and interact with internal electrons to produce magnetic-chain organization. Static elemental charge photons reside inside toroidal surface electrons and accumulate at the surface to produce electric chains. The electric and magnetic structures remain coupled components of one electromagnetic system, with substantial energy transport occurring in the surrounding region.

Figure 1.1: Comparison of conventional macroscopic electromagnetic quantities and the proposed Photony description of electromagnetic organization around a current-carrying conductor. Panel A identifies the conventional electric, magnetic, and electromagnetic-energy quantities associated with the conductor. Panel B illustrates the proposed Photony structural interpretation. Internal dynamic elemental charge photons with helical motion are aligned with the conductor axis and associated with internal-electron magnetic transduction. Surface-associated toroidal electrons transduce primarily static elemental charge photons and support radial electric chain organization. The green longitudinal magnetic chain structures (shown as green lines) decrease in density with increasing radial distance, while the red electric chain structures extend outward from the surface.

$$\mathcal{E}_{\text{EM}} = \mathcal{E}_E + \mathcal{E}_B, \quad (1.21)$$

where the conventional total electromagnetic energy contains both electric and magnetic contributions.

Photony interprets these contributions as distinct but coupled chain organizations derived from the same underlying energy population.

Figure 1.1 summarizes the central distinction developed in this chapter. Panel A retains the conventional electromagnetic description used for quantitative analysis, while Panel B presents the proposed Photony structural interpretation of the same physical system. The two panels are intended to represent complementary descriptions of the same electromagnetic behavior rather than competing sets of measurable equations.

1.12 Current in the Photony Framework

Conventionally, electric current is defined as the rate of charge transfer,

$$I = \frac{dQ}{dt}. \quad (1.22)$$

For conduction current density,

$$\mathbf{J} = \sigma \mathbf{E} \quad (1.23)$$

in an idealized Ohmic medium.

Photony does not discard these measurable definitions. It proposes that the rapidly propagating energetic component of current is associated with dynamic elemental charge photons rather than with rapid bulk translation of conduction electrons.

Electrons distributed along the conductor respond locally to the propagating electromagnetic condition. Their role is therefore closer to distributed transduction and regulation than to carrying all transported electromagnetic energy physically from source to load.

This distinction provides a structural interpretation for the well-established fact that electromagnetic signal propagation can approach a substantial fraction of the speed of light

while electron drift velocity remains comparatively small.

Longitudinal Magnetic chain Organization and the Conventional Circumferential Field Description:

A particularly important distinction must be made between the conventional magnetic-field representation surrounding a current-carrying conductor and the physical magnetic chain structure proposed by Photony. This distinction goes directly to the meaning of a "Beyond Field-Line" description.

For a long straight conductor carrying a steady current I , conventional electromagnetic theory gives the magnetic-field magnitude outside the conductor as

$$B(r) = \frac{\mu I}{2\pi r}, \quad (1.24)$$

for the idealized cylindrically symmetric case. The corresponding magnetic field vector is conventionally represented as azimuthal, with field lines drawn as concentric circles surrounding the conductor. The direction is obtained from the right-hand rule, while the magnitude decreases with radial distance r from the conductor.

Photony does not dispute the experimentally established radial dependence of the magnetic-field strength, nor does it discard the Ampere-Maxwell relationship from which the conventional result is obtained. Instead, it questions whether the conventional circular field lines should be interpreted as literal physical trajectories of the underlying magnetic structure.

This distinction is important. A field line is an integral curve constructed from a vector field. It provides an extremely useful representation of the direction assigned to the measured macroscopic field. It does not, by itself, establish that a physical filament, chain, tube, or other material structure must follow exactly the same trajectory.

The proposed Photony interpretation, therefore, separates two questions:

What magnetic field is measured macroscopically?

and

What underlying physical organization may produce that measured field?

For the first question, the conventional electromagnetic description remains unchanged.

For the second, Photony proposes a substantially different geometry.

The Internal Conductor as a Distributed Solenoidal Structure:

Within Photony, current consists primarily of directed transport of dynamic elemental charge photons through a conductor. Internal free electrons respond to this transport by preferentially orienting relative to the current direction. These electrons are proposed to act as electromagnetic transducers, organizing part of the dynamic elemental charge-photon population into magnetic chains [Martin \(2022\)](#).

The important geometric feature is that the transducing electrons are not treated as isolated magnetic sources. A current-carrying conductor contains a very large population of free electrons distributed along its length. When a sustained current establishes a preferred orientation, successive internal electrons can participate in a common magnetic chain structure.

Each toroidal electron can be viewed, within the proposed model, as a small transducing structure through which magnetic chain organization is established. When many such electrons become approximately aligned along the current direction, their individual transduction regions form a distributed adjoined magnetic chain sequence extending along the conductor.

The resulting arrangement is analogous to a solenoid.

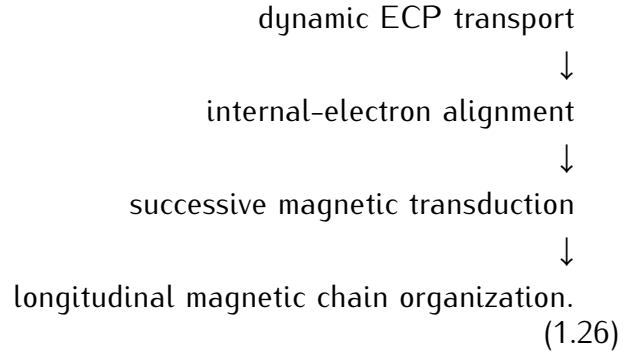
In a conventional elongated solenoid, the magnetic field through much of the interior is predominantly longitudinal. The magnetic flux passes along the interior of the coil, turns through the end regions, and returns outside the coil in the opposite longitudinal direction. The complete magnetic structure is closed, consistent with

$$\nabla \cdot \mathbf{B} = 0. \quad (1.25)$$

Photony proposes that a current-carrying conductor can develop an analogous organiza-

tion at a much smaller structural scale. Instead of macroscopic wire turns establishing the solenoidal organization, aligned internal electrons provide successive transduction sites.

The proposed sequence is therefore



The magnetic chain loops associated with neighboring electrons are proposed to adjoin longitudinally. Rather than remaining as independent microscopic loops around individual electrons, they become parts of a larger continuous magnetic structure extending along the conductor [Martin \(2022\)](#).

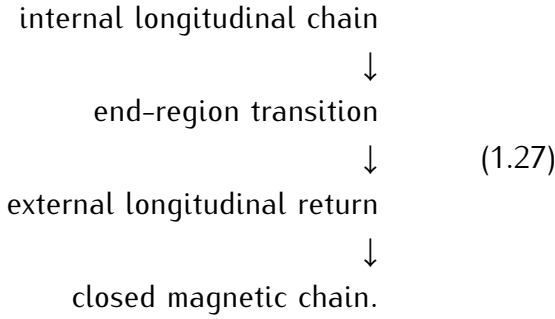
This is the physical interpretation illustrated in Figure 1.1, where the conductor is shown not merely as a source surrounded by abstract magnetic field lines, but as an extended region of electron-mediated magnetic chain transduction.

Internal and External Portions of the Magnetic Chain

A magnetic chain cannot simply begin within the conductor and terminate at its end. Magnetic organization must remain continuous.

Photony therefore treats the extended chain as a closed structure. One portion extends longitudinally through the current-supported interior of the conductor. Near the termination or end region, the structure turns outward. The return portion then extends longitudinally through the region outside the conductor in the opposite direction before ultimately closing the loop.

The proposed geometry is therefore qualitatively analogous to the magnetic structure of an elongated solenoid:



The word “longitudinal” is important. It describes the principal extension of the proposed chain relative to the conductor axis. It does not imply that the magnetic intensity is uniform with radial distance.

Multiple magnetic chains can occupy different radial positions around the conductor. Their population or effective density decreases with increasing distance from the current-supported region. Thus **the proposed physical chains may be predominantly longitudinal while the measured magnetic-field strength remains a radial function.**

A longitudinal magnetic chain structure can possess a radial density distribution. Near the conductor, many chains may occupy a given surrounding region, while at greater radial distance the chain population becomes less dense. The resulting macroscopic magnetic influence therefore decreases with distance even though the dominant longitudinal direction of an individual chain section has not become radial.

For the idealized external region of a long straight conductor, the proposed effective magnetic chain density may therefore be related conceptually to the measured magnetic-field magnitude as

$$\rho_{\text{chain}}(r) \propto B(r) \propto \frac{1}{r}. \quad (1.28)$$

Here, $\rho_{\text{chain}}(r)$ denotes a proposed effective magnetic chain density rather than an independently established measurable quantity. The relationship expresses a radial variation in chain density, not a radial orientation of the chains themselves.

This refines the old Photony Foundations radial-density treatment while preserving its longitudinal-chain concept. The treatment in

Photony Foundations describes magnetic chains as building longitudinally within the conductor as free electrons align, with their loop structures joining longitudinally in a manner analogous to a solenoid and extending outward in surrounding layers [Martin \(2022\)](#).

Why This Does Not Require Rejection of Maxwell's Equations

The distinction between magnetic chain geometry and conventional field-line geometry is essential if the proposed model is to remain compatible with Maxwellian electrodynamics.

Maxwell's equations constrain the observable electromagnetic field. For a steady current density, the relevant relationship is the Ampere-Maxwell law,

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (1.29)$$

For steady DC conditions,

$$\frac{\partial \mathbf{D}}{\partial t} = 0, \quad (1.30)$$

so that

$$\nabla \times \mathbf{H} = \mathbf{J}. \quad (1.31)$$

Equivalently, the integral form is

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I_{\text{enc}}. \quad (1.32)$$

For cylindrical symmetry, this relationship produces the familiar circumferential macroscopic field and its $1/r$ dependence.

Photony does not propose changing this measured relationship.

Instead, it proposes that the quantity represented conventionally by $\mathbf{H}$ or $\mathbf{B}$ may be a macroscopic descriptor of an underlying magnetic chain organization whose physical geometry is more complex than the field-line drawing used to represent the vector field.

That distinction is analogous to many other areas of physics in which a macroscopic field need not reproduce the literal trajectory of every microscopic constituent. The observable field is a collective quantity. A proposed microscopic structure is acceptable only if its collective behavior reproduces the established macroscopic result.

Consequently, the Photony model must satisfy an important requirement:

The proposed longitudinal magnetic chain organization must reproduce the same measurable magnetic forces, energy density, radial field-strength dependence, induction behavior, and Ampere-Maxwell relationships that are described conventionally by the magnetic field.

This requirement prevents the chain interpretation from assigning arbitrary magnetic behavior.

It also identifies an important distinction between a *field vector* and a *physical chain tangent*. Photony does not require that the tangent to an underlying magnetic chain be identical at every point to the direction of the conventional macroscopic **B** vector. If those two quantities were assumed to be identical, a longitudinal chain surrounding a straight current-carrying conductor would indeed conflict with the conventional azimuthal magnetic-field solution.

The proposed interpretation instead requires a mapping between the collective chain organization and the measured macroscopic magnetic field. In this view, the conventional circumferential field direction describes the macroscopic magnetic action surrounding the current, while the proposed longitudinal chain structure describes the physical organization from which that action emerges.

This is a critical Beyond Field-Line distinction.

The Meaning of the Conventional Circumferential Field:

The conventional circular magnetic field surrounding a straight conductor is experimentally validated. Compass orientation, magnetic-force measurements, inductive coupling, Ampere's law, and electromagnetic energy calculations all agree with the established field description.

Photony therefore does not argue that those measurements are incorrect.

The proposed reinterpretation concerns what is inferred from them.

Drawing a circumferential magnetic field line establishes the direction of the conventional magnetic vector field at successive positions around the conductor. It does not necessarily demonstrate that a literal physical magnetic strand circles the conductor independently at every axial position.

Within Photony, that circumferential representation is instead interpreted as the macroscopic manifestation of the collective action of extended closed magnetic chain structures associated with the current-supported electron population.

The distinction is subtle but fundamental. Conventional interpretation:

$$\begin{array}{c} \text{current} \\ \downarrow \\ \text{circumferential magnetic field.} \end{array} \quad (1.33)$$

Proposed Photony interpretation:

$$\begin{array}{c} \text{dynamic elemental transport} \\ \downarrow \\ \text{aligned internal electrons} \\ \downarrow \\ \text{longitudinal closed chains} \\ \downarrow \\ \text{measured macroscopic magnetic field.} \end{array} \quad (1.34)$$

The conventional description begins with the measurable field.

The Photony description asks what physical organization might exist beneath that field.

Why the Solenoid Analogy Is Important:

The solenoid provides a useful bridge because it demonstrates that a closed magnetic structure can contain long regions in which the field is predominantly longitudinal.

Inside a sufficiently long solenoid, the conventional magnetic field is approximately axial. Outside, the field turns and returns so that the overall magnetic structure remains continuous. There is no requirement that the return portion be geometrically identical to the internal portion.

Photony extends this structural idea downward in scale. The aligned current-associated

electron population is proposed to behave collectively as a distributed microscopic transduction structure. Each electron contributes to an adjoining magnetic chain organization, and the succession of aligned electrons allows the chain to extend over distances much larger than the dimensions of an individual electron.

Thus, the conductor itself can be interpreted as a distributed magnetic transducer.

The greater the sustained dynamic elemental charge-photon transport, the greater the number or density of internal electrons participating in magnetic transduction, and the greater the resulting magnetic chain organization. This interpretation must remain quantitatively consistent with the conventional relationships among current, magnetic-field strength, and magnetic energy.

For example,

$$u_B = \frac{B^2}{2\mu} \quad (1.35)$$

remains the conventional magnetic-energy constraint.

Photony therefore cannot assign arbitrary energy to the proposed chain population. Increasing chain organization must correspond to increasing measured magnetic energy in a manner ultimately compatible with Equation 1.35.

Figure 1.1 summarizes this distinction visually. The conventional field quantities describe the measurable macroscopic electromagnetic state, while the longitudinal structures in the Photony representation depict the proposed underlying magnetic chain organization. The two representations therefore operate at different descriptive levels.

A Testable Requirement of the Interpretation

This proposed distinction also creates a clear theoretical obligation.

If longitudinal magnetic chains are the underlying structure while the measured magnetic field around a straight conductor is conventionally azimuthal, then Photony must ultimately specify how the collective state of those chains maps onto the observable vector field $\mathbf{B}$.

Symbolically, one may write

$$\mathbf{B} = \mathcal{M}[\text{magnetic chain organization}], \quad (1.36)$$

where $\mathcal{M}$ represents a presently conceptual coarse-graining or mapping from the proposed microscopic chain organization to the measured macroscopic magnetic field.

Equation 1.36 is not introduced as a completed field equation. It identifies a requirement for further development of the theory.

For the straight-conductor case, that mapping must reproduce

$$B(r) \propto \frac{I}{r}, \quad (1.37)$$

the correct right-hand-rule response, magnetic forces, induction effects, energy density, and all applicable Maxwell boundary conditions.

This is where the proposed interpretation becomes scientifically meaningful. The question is not whether Photony can draw a different microscopic picture. The question is whether that picture can reproduce the complete set of measured electromagnetic consequences while also providing additional physical insight or experimentally distinguishable predictions.

The longitudinal-chain model therefore should not be presented as an exception to Maxwell's equations.

It is proposed as a physical substructure beneath them.

Maxwell's equations remain the macroscopic constraint. Photony proposes what may be physically organized underneath the quantities appearing in those equations.

1.13 Voltage and Surface Accumulation

Voltage conventionally represents electric potential difference,

$$V_{ab} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}. \quad (1.38)$$

Within Photony, voltage is associated structurally with differences in static elemental

charge-photon organization, particularly at conductor surfaces.

When forward dynamic transport is constrained by an open termination, an impedance discontinuity, a capacitive structure, or another boundary condition, an increased fraction of the elemental charge-photon population can become associated with surface accumulation.

The qualitative progression is

restricted transport → surface encounter
 → static accumulation
 → higher potential. (1.39)

This interpretation becomes especially important in capacitors, transmission-line discontinuities, antenna ends, and high-voltage structures.

1.14 Capacitance as Organized Electric Energy Storage

Conventionally, a capacitor stores electric energy according to

$$U_C = \frac{1}{2} CV^2. \quad (1.40)$$

Its current-voltage relation is

$$i_C = C \frac{dv}{dt}. \quad (1.41)$$

Within Photony, the opposing conductor surfaces of a capacitor provide locations for static elemental charge-photon accumulation and surface-electron electric transduction.

Charging therefore corresponds structurally to increasing electric chain organization between and around the conductors. Discharging involves reduction and redistribution of that organization as energy is returned to the surrounding circuit.

Thus,

charging → electric chain growth,
 discharging → electric chain reduction.

The conventional capacitance and energy equations remain the quantitative measures of the process.

1.15 Inductance as Organized Magnetic Energy Storage

The corresponding magnetic storage element is the inductor.

Conventionally,

$$v_L = L \frac{di}{dt}, \quad (1.42)$$

and stored magnetic energy is

$$U_L = \frac{1}{2} LI^2. \quad (1.43)$$

Within Photony, increasing current increases the dynamic elemental charge-photon population participating in internal-electron magnetic transduction. The resulting magnetic chain organization stores electromagnetic energy in the region associated with the current path.

A decreasing current reverses the net transduction balance. The existing magnetic chains are then proposed to undergo reduction, returning energy to the electrical system rather than disappearing without accounting.

The qualitative correspondence is

$$\begin{aligned} \frac{d|I|}{dt} > 0 &\rightarrow \text{net magnetic chain growth,} \\ \frac{d|I|}{dt} < 0 &\rightarrow \text{net magnetic chain reduction.} \end{aligned}$$

This distinction between transduction and reduction becomes essential in later chapters dealing with AC systems and rapid electromagnetic energy release.

1.16 Electromagnetic Energy Transport and the Poynting Vector

One of the most important conventional results for the physical interpretation developed in this book is that electromagnetic energy transport cannot generally be identified solely with energy moving through the material interior of a conductor.

Poynting introduced the electromagnetic energy-flux vector

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad (1.44)$$

whose direction and magnitude describe electromagnetic energy transport per unit area [Poynting \(1884\)](#); [Jackson \(1998a\)](#).

Poynting's theorem can be written as

$$\nabla \cdot \mathbf{S} + \frac{\partial u_{EM}}{\partial t} + \mathbf{J} \cdot \mathbf{E} = 0, \quad (1.45)$$

where

$$u_{EM} = \frac{1}{2} (\epsilon E^2 + \mu H^2) \quad (1.46)$$

for a simple linear medium.

Equation 1.45 expresses local energy conservation. Electromagnetic energy flowing into or out of a region must be accounted for through changes in stored electromagnetic energy or work performed on matter.

Within Photony, the Poynting vector is interpreted as the macroscopic measure of directional transport by the underlying electromagnetic organization. Electric and magnetic chain structures are proposed to provide a physical organization corresponding to the conventional $\mathbf{E}$ and $\mathbf{H}$ quantities.

This interpretation is particularly important near conductors. The conductor guides and constrains the electromagnetic system, while substantial energy transport can occur in the surrounding region. Photony therefore does not picture the wire simply as a pipe through which all power flows internally. The conductor and its electrons organize the electromagnetic pathway.

1.17 Electromagnetic Energy Conservation

Any structural interpretation must preserve energy conservation.

If a volume V is bounded by a surface A , the integral form of Poynting's theorem is

$$\oint_A \mathbf{S} \cdot d\mathbf{A} = -\frac{d}{dt} \int_V u_{EM} dV - \int_V \mathbf{J} \cdot \mathbf{E} dV. \quad (1.47)$$

A nonzero divergence of Poynting flux therefore indicates energy redistribution rather than energy creation or destruction.

Within Photony, electromagnetic energy may be redistributed among several forms, including propagating dynamic elemental charge-photon populations, static surface populations, magnetic chains, electric chains, radiative structures, thermal energy, and mechanical response.

The structural description must therefore obey the same accounting represented conventionally by Equation 1.47.

1.18 Alternating Excitation, Transduction, and Reduction

AC excitation introduces continuous changes in both magnitude and polarity.

For a sinusoidal voltage,

$$v(t) = V_0 \sin(\omega t), \quad (1.48)$$

with

$$\omega = 2\pi f. \quad (1.49)$$

Current may be phase shifted according to the impedance of the system,

$$i(t) = I_0 \sin(\omega t + \phi). \quad (1.50)$$

Photony interprets AC behavior as repeated electromagnetic organization rather than merely the numerical reversal of voltage and current.

The proposed local cycle is

$$\begin{aligned} \text{transduction} &\rightarrow \text{growth} \\ &\rightarrow \text{reduction} \\ &\rightarrow \text{electron reorientation} \\ &\rightarrow \text{renewed transduction.} \end{aligned} \quad (1.51)$$

Electric and magnetic structures need not reach their maxima or reversal conditions simultaneously. Their timing depends on the phase relationship between voltage and current and on the distributed geometry of the system.

This distinction becomes especially important in antennas and resonant transmission structures, where current maxima and voltage maxima can occupy different spatial regions.

1.19 Propagation, Distributed Systems, and Geometry

A conductor ceases to behave adequately as a lumped circuit element when propagation time becomes significant relative to the timescale of the excitation.

The wavelength is

$$\lambda = \frac{v_p}{f}, \quad (1.52)$$

where v_p is the propagation velocity.

As the physical length L becomes a significant fraction of λ ,

$$\frac{L}{\lambda} \not\ll 1, \quad (1.53)$$

voltage and current vary appreciably with position. Transmission-line behavior, reflections, standing waves, resonance, and radiation can then become important [Pozar \(2012a\)](#); [Bogatin \(2018\)](#).

Within Photony, this means that electromagnetic transduction cannot be treated as occurring uniformly along the entire conductor. Different regions contain different combinations of dynamic transport, static accumulation, magnetic transduction, and electric transduction.

Geometry therefore becomes an active part of the electromagnetic mechanism rather than a passive container.

A useful structural progression is

system geometry $\rightarrow$ population distribution
 $\rightarrow$ transduction pattern
 $\rightarrow$ EM organization.

1.20 Return Paths and Electromagnetic Confinement

An electrical signal cannot be fully specified by identifying only its nominal forward conductor. The return path is part of the same electromagnetic structure.

In a well-controlled transmission geometry, signal and return conductors remain close together. The electromagnetic organization is cor-

respondingly concentrated in the region around the pair.

Increasing the separation or interrupting the return structure expands the effective electromagnetic geometry,

return spacing $\uparrow \rightarrow$ loop area $\uparrow$
 $\rightarrow$ EM extent $\uparrow$.

This established engineering principle is important in printed-circuit boards, cables, connectors, shielding structures, and antennas [Paul \(2006\)](#); [Ott \(2009\)](#); [Bogatin \(2018\)](#).

Within Photony, the same result is interpreted as an expansion of the spatial region through which dynamic elemental charge-photon transport and electron-transduced chain organization must be accommodated.

1.21 Frequency-Dependent Current Localization

The current distribution within a conductor also changes with frequency.

For a good conductor, the conventional skin depth is

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}, \quad (1.54)$$

where σ is conductivity.

Increasing frequency decreases δ and progressively concentrates conventional current density toward the conductor surface.

Within Photony, this behavior is interpreted as a change in the spatial distribution of dynamic elemental charge-photon transport and the internal electrons participating in magnetic transduction.

For a round conductor of radius r , a useful frequency scale for the onset of appreciable skin localization is

$$f_{\text{start}} = \frac{1}{\pi\mu\sigma r^2}. \quad (1.55)$$

The exact current distribution depends strongly on geometry. Rectangular traces,

planes, shields, and enclosure surfaces do not reproduce the same localization pattern as cylindrical wires.

This fact will later become important when examining electromagnetic compatibility and unintended coupling.

1.22 The Four Principal Coupling Mechanisms

Electromagnetic energy can move from a source to a victim through four principal engineering coupling mechanisms:

conductive, capacitive, inductive, radiative	(1.56)
---	--------

These mechanisms are conventional EMC classifications [Ott \(2009\)](#); [Paul \(2006\)](#). Photony provides proposed structural interpretations for each.

1.22.1 Conductive Coupling

Conductive coupling occurs when source and victim share a conductor or impedance.

Conventionally,

$$V_{\text{coupled}} = I_{\text{disturbance}} Z_{\text{shared}}. \quad (1.57)$$

Within Photony, this mechanism involves re-distribution of the dynamic elemental charge-photon population through a connected conductive system, accompanied by corresponding changes in surface accumulation and electron transduction.

1.22.2 Capacitive Coupling

Capacitive coupling is driven primarily by changing voltage,

$$i_C = C_m \frac{dv}{dt}. \quad (1.58)$$

Photony associates this mechanism primarily with surface static elemental charge-photon organization and electric chain interaction between source and victim surfaces.

Thus,

large $\frac{dv}{dt} \rightarrow$ more capacitive coupling.

1.22.3 Inductive Coupling

Inductive coupling is associated primarily with changing current,

$$v_M = M \frac{di}{dt}. \quad (1.59)$$

Photony associates this mechanism with current-driven magnetic chain organization and its interaction with electrons in a neighboring current geometry.

Thus,

large $\frac{di}{dt} \rightarrow$ more inductive coupling.

1.22.4 Radiative Coupling

Radiative coupling occurs when an electromagnetically significant structure launches propagating energy that can be intercepted by another structure.

Conventionally, the outward power flow is described through the Poynting vector.

Photony interprets this as outward propagation of an electromagnetic organization that was initially transduced in association with the source but has become capable of continuing independently through free space.

1.23 Electric and Magnetic Energy Release

A major purpose of this compendium is to distinguish electric and magnetic energy-release regimes.

The conventional electric energy density is

$$u_E = \frac{1}{2} \epsilon E^2, \quad (1.60)$$

whereas the conventional magnetic energy density is

$$u_B = \frac{B^2}{2\mu}. \quad (1.61)$$

These forms immediately indicate that electric and magnetic systems can store substantial energy through different macroscopic conditions.

Within Photony, the distinction is structural as well:

electric energy $\leftrightarrow$ electric chains,
magnetic energy $\leftrightarrow$ magnetic chains.

Electric breakdown is therefore treated later as a voltage and electric chain density phenomenon, while magnetic breakdown is associated with sufficiently stressed magnetic organization in high-conductivity systems.

The two regimes can interact, but they should not automatically be treated as the same physical event.

1.24 Intended and Unintended Antennas

An antenna provides one of the clearest examples of the distributed nature of electromagnetic organization.

For an approximately half-wave center-fed dipole, conventional antenna theory predicts relatively high current near the feed region and comparatively high voltage and charge accumulation toward the ends [Balanis \(2016\)](#).

Within Photony, the high-current region is interpreted as a region of strong internal-electron magnetic transduction, while the high-voltage regions support stronger surface-electron electric transduction.

The antenna therefore becomes a distributed transduction structure rather than a single point source.

As the AC excitation evolves, the chain structures are proposed to grow, reduce, reverse, and undergo release. Within the Beyond Field-Line interpretation, radiation is proposed to be a continuation of electromagnetic organization that begins within and on the antenna rather than repeated creation of one field by another.

The same mechanism can occur unintentionally. A cable, shield, chassis structure, trace, or other conductor can behave as an antenna when

its electrical length and excitation support distributed current and voltage and when electromagnetic organization is not adequately confined to the intended transmission geometry.

1.25 Electric and Magnetic Fields Need Not Create One Another Sequentially

Maxwell's curl equations are sometimes described informally as if a changing electric field creates a magnetic field, which then creates another electric field, producing an endless self-generating sequence.

That language can be useful pedagogically but should not be confused with the complete physical solution of Maxwell's equations.

The electromagnetic state is determined by charge, current, boundary conditions, material properties, and the coupled Maxwell system. Electric and magnetic quantities are mutually constrained components of one electromagnetic solution.

Photony adopts the same unified perspective but gives it a proposed structural form. Electric and magnetic chains arise through complementary transduction processes from the same underlying electromagnetic energy population. Outward propagation does not require repeated creation of one chain species by the other.

Instead,

source excitation $\rightarrow$ EM organization
 $\rightarrow$ outward propagation.
(1.62)

The electric and magnetic components remain linked aspects of the same electromagnetic organization.

1.26 Fibrils and Larger Electromagnetic Structures

The terminology of this compendium distinguishes fundamental energy filaments from larger bundles of chains.

The term *filament* is reserved for the fundamental energy filament from which elemental charge photons are proposed to form.

A bundle or organized collection of magnetic chains is instead called a *fibril*. Thus,
 filament → fundamental energy structure
 fibril → organized bundle of magnetic chains

More generally, a fibril is a relatively small fiber- or thread-like structure comprising solar flux tubes. In mainstream solar physics, the term commonly refers to elongated magnetic or plasma features in the chromosphere and related structures, which is why it can sound solar-specific.

In solar applications, the term *fibril* should not be interpreted as having one fixed diameter. Observed fibril widths depend on atmospheric region, magnetic environment, diagnostic wavelength, and instrumental resolution. Chromospheric fibrils and related fine magnetic structures are commonly observed with widths of a few hundred kilometers, while finer coronal strands can occur on scales of tens to roughly one hundred kilometers. Representative observations, therefore, support a broad range extending from approximately 10 to 300 km for fine magnetic substructure, with about 100 km serving as a useful characteristic scale for some coronal fibrils.

A solar fibril should not be pictured as an empty bundle of magnetic structures. Observationally, fibrils are associated with magnetically organized plasma and can undergo collective motion, deformation, and magnetohydrodynamic-wave propagation. Within the Photony interpretation, these behaviors involve the coupled motion and reorganization of the plasma and its containing magnetic chain structure rather than plasma motion relative to stationary magnetic chains. This also applies to magnetohydrodynamic-wave propagation [Saberi et al. \(2025\)](#). In the chromosphere, the plasma is often only partially ionized, so the material contained within and around a fibril can include free electrons, ions, and neutral atoms, particularly hydrogen and helium species [Soler et al. \(2012\)](#). The relative populations depend strongly on temperature, density, height, and local magnetic conditions.

Within the Photony framework, the magnetic chains provide the proposed underlying magnetic

organization, while the fibril is a larger composite magnetic-plasma structure containing both those chains and the plasma matter organized with them. The plasma may include electrons, ions, neutral atoms, and other particulate constituents, with relative populations depending on the local solar environment. The motion of the fibril is therefore treated as the coupled motion of the plasma and the magnetic-chain organization. Plasma does not simply stream through a stationary magnetic-chain framework; rather, the containing chain structure and the plasma it organizes participate together in fibril motion, deformation, oscillation, and wave propagation.

The observed scale should not be treated as a fundamental diameter of a Photony fibril. A fibril is defined structurally as an organized bundle of magnetic chains, and its physical diameter can vary according to the number of participating chains, local magnetic energy density, plasma conditions, and larger-scale magnetic organization. A solar flux tube may therefore contain many fibrils, and each fibril may itself possess unresolved internal substructure. The hierarchy is consequently better understood as

magnetic chains → fibrils
 fibrils → flux tubes or flux ropes

rather than by assigning one universal physical diameter to the fibril itself.

1.27 Conventional Constraints on the Photony Interpretation

The Beyond Field-Line framework is useful only if it remains constrained by measured electromagnetic behavior.

At a minimum, the proposed interpretation must remain consistent with the following established results:

1. Maxwell's equations correctly relate macroscopic electric and magnetic quantities to charge and current.
2. Electromagnetic energy is locally conserved according to Poynting's theorem.
3. Propagation occurs at a finite speed determined by the electromagnetic properties of

the propagation environment.

4. Transmission lines support distributed voltage and current, reflections, standing waves, impedance transformations, and resonance.
5. Electric energy scales with E^2 , while magnetic energy scales with B^2 .
6. Capacitive coupling depends strongly on voltage, geometry, and dv/dt .
7. Inductive coupling depends strongly on current geometry and di/dt .
8. Conductive coupling depends on shared conductive pathways and impedance.
9. Antenna radiation must reproduce established relationships for wavelength, polarization, impedance, field strength, radiation pattern, and Poynting flux.
10. Increasing frequency produces conventional skin effect and geometry-dependent current localization.
11. Differential-mode and common-mode currents can produce substantially different electromagnetic radiation and coupling behavior.
12. Electromagnetic energy cannot appear without a corresponding source or reduction of previously stored energy.

Photony presently provides a proposed structural interpretation of these phenomena, not a complete replacement calculation framework. Quantities such as antenna gain, radiation resistance, transmission-line impedance, capacitance, inductance, propagation constant, and field amplitude continue to be calculated using conventional electromagnetic theory.

This distinction is scientifically important. The structural interpretation should ultimately become more predictive, not less constrained.

1.28 From “Be the Signal” to “Be the Photon”

A useful engineering discipline in high-speed electronics is to “Be the Signal.” The phrase encourages the engineer to follow an electrical disturbance through its actual physical environment rather than assuming that a schematic connection adequately describes its behavior [Bogatin \(2018\)](#).

The signal encounters impedance, return geometry, discontinuities, dielectric interfaces, connectors, coupling regions, terminations, and possibly radiating structures.

The Photony extension is to “Be the Photon.” At each stage, one asks:

- Where is the dynamic elemental charge-photon population moving?
- Where does forward transport encounter a geometric or electrical constraint?
- Where does static surface accumulation increase?
- Which electrons are participating predominantly in magnetic transduction?
- Which electrons are participating predominantly in electric transduction?
- Where are chains growing?
- Where are previously transduced chains being reduced?
- Where does electromagnetic organization remain confined?
- Where does it couple into another structure?
- Where can it separate sufficiently from the source to propagate outward?

This thought process provides the conceptual foundation for the chapters that follow.

The intent is not to replace the engineer’s conventional analysis but to add a structural level beneath it.

Chapter 2

Electric Energy-Release Regimes

Purpose: To examine how electric energy is stored, loaded, and rapidly released through breakdown within the Beyond Field-Line and Photony framework.

2.1 Introduction

Electric energy-release events occur across natural, industrial, and engineered systems, including electrostatic sparks, dielectric failure, surface flashover, lightning, electrical arcs, and deliberately generated ignition discharges. Although these phenomena differ greatly in scale and application, they share a common physical progression. Electric energy can accumulate while charge redistribution is restricted and can then be transferred rapidly when an insulating, resistive, or incompletely conducting region develops a substantially more conductive pathway.

The defining feature of the electric regime is therefore not simply the visible discharge that follows breakdown. The pre-threshold state is equally important. Charge separation, electric-field loading, geometry, permittivity, conductivity, surface condition, leakage pathways, and the rate at which charge or energy is supplied determine whether the system remains stable, redistributes charge gradually, undergoes localized discharge, or progresses toward complete breakdown.

Charge separation can arise through contact and separation of materials, electrostatic induction, movement of liquids or powders, spraying, material-handling processes, or direct application of voltage. Once charge is separated, the resulting electric field can persist only to the extent permitted by the surrounding electrical and material environment. Conductivity determines

how readily charge can redistribute, while geometry can concentrate electric stress into localized regions even when the total potential difference remains moderate.

Conventional electrostatics, dielectric theory, gas-discharge physics, and plasma physics provide the established quantitative descriptions of these processes. Charge, voltage, electric-field strength, capacitance, conductivity, dielectric strength, pressure, gap distance, current, and deposited energy remain the measurable quantities that determine observable behavior. Electric breakdown is therefore not governed by voltage alone. Local field concentration and the surrounding material environment are equally important.

Within the proposed Photony interpretation, surface-associated electrons transduce predominantly static elemental charge photons into linked electric chains that provide a proposed physical organization underlying conventional electric-field descriptions. The term *static* refers to the configuration and motion state of the elemental charge photon rather than to the waveform of the macroscopic source. Electric chain organization may therefore occur under electrostatic, steady, alternating, pulsed, or transient electrical conditions.

Electric-field loading is interpreted as increasing energy and organizational stress within this charge-supported electric chain system. When established material or discharge processes cause a portion of the surrounding region

to become substantially more conductive, localized discharge or complete breakdown may occur. Within the Photony interpretation, this conductivity transition corresponds to rapid reorganization or loss of continuity of the pre-breakdown electric chain structure.

An important distinction must be maintained between the breakdown event and the energy subsequently transferred through the new conducting path. An isolated electrostatic discharge may be limited primarily by stored charge and capacitance. By contrast, a breakdown path connected to an external source can continue to receive energy and may develop into a sustained arc. The severity of an electric energy-release event therefore depends not only on the breakdown threshold but also on the electrical and material energy reservoirs available after that threshold has been crossed.

Electric breakdown is also not inherently a failure. The same fundamental transition can occur unintentionally in electrostatic discharges, insulation faults, flashovers, lightning, and ignition hazards, or it can be produced deliberately in spark plugs, burner igniters, flare-ignition systems, and related devices. Accidental and purposeful breakdown can therefore be examined within the same physical framework.

The central proposition developed in this chapter is that voltage breakdown represents a threshold-driven transition in which a charge-supported electric organization can no longer be maintained within the available insulating, resistive, or weakly conducting environment and a substantially more conductive pathway develops. Conventional electromagnetic, dielectric, discharge, and plasma theories remain the quantitative basis for describing the measurable system, while Photony provides a proposed chain-level interpretation of the pre-threshold loading and post-threshold reorganization.

Terminology: Photony theory, electric energy release, voltage breakdown, electric breakdown, electric chains, static electricity, electrostatic discharge, dielectric breakdown, spark discharge, electrical arcing, lightning, surface-electron transduction, ignition, electrical fire

2.2 Electric-Field Structures Relevant to the Electric Regime

Chapter 1 established the broader Photony framework, including energy filaments, elemental charge photons, electron transduction, and electric and magnetic chain organization. The present section therefore concentrates only on the structures required specifically for electric energy storage, loading, and voltage breakdown.

Photony theory does not replace the established mathematical descriptions of electric fields. Maxwell's equations, electromagnetic energy densities, dielectric models, discharge physics, and plasma diagnostics remain the principal quantitative descriptions of the systems considered here. Photony is applied as a complementary physical interpretation in which conventional electric-field representations correspond to organized chains of elemental charge photons.

The distinction between conventional electric and magnetic field-line representations and the proposed chain-based interpretations is illustrated in Figure 2.1. Conventional field lines indicate field direction and geometry, whereas the Photony interpretation proposes linked physical chain structures underlying the corresponding electric and magnetic fields.

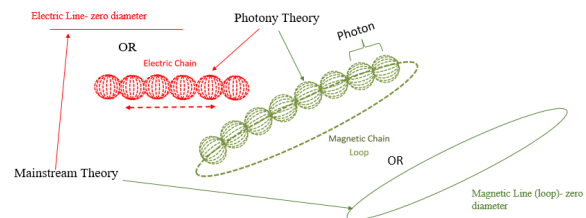


Figure 2.1: Comparison between conventional electric and magnetic field-line representations and the proposed Photony chain-based interpretations. Electric chains are shown as linked structures associated with surface-electron transduction, whereas magnetic chains are shown as closed-loop structures associated with internal-electron transduction. The conventional field lines remain geometrical representations of the corresponding macroscopic fields.

The figure does not imply that conventional field-line descriptions are incorrect. The two representations serve different purposes. Conventional field lines describe the direction and geometry of electric and magnetic fields at the macroscopic level, whereas the Photony interpretation proposes underlying physical chain structures whose organization depends on the associated electron-transduction pathway. In this chapter, the electric chain pathway is the principal subject. The magnetic chain pathway is developed in the next Chapter.

2.2.1 Elemental Charge Photons in the Electric Regime

As established in Chapter 1, an elemental charge photon is a closed filament configuration associated with the fundamental structure identified with electric charge. Two principal forms are distinguished. A static elemental charge photon is a closed ground-state loop associated with stored charge and electric potential, while a dynamic elemental charge photon is a reconfigured helical form possessing forward motion and associated with charge transport and current.

The terms *static* and *dynamic* describe the configuration and motion of the elemental charge photon rather than the waveform of the macroscopic source. Static elemental charge photons are associated primarily with stored charge and electric chain formation, whereas dynamic elemental charge photons participate primarily in current-associated magnetic chain formation.

For the electric regime developed in this chapter, the static population is therefore of primary importance.

2.2.2 Surface-Electron Transduction and Electric chain Formation

The distinction between electric and magnetic organization emerges most directly at the electron-transduction stage. Photony theory uses the term **transduction** for the interaction through which electrons organize participating elemental charge photons into larger field-associated chain structures [Martin \(2026b\)](#).

For the electric regime considered here, the essential pathway is **surface-electron transduction**.

Surface-associated electrons transduce predominantly static elemental charge photons into linked electric chains. These chains extend outward from, or between, charge-supporting regions and provide the proposed physical organization underlying conventional electric-field descriptions. The process may be represented schematically as

$$\hat{T}_E(\Phi_{\text{static}}) \longrightarrow C_E \quad (2.1)$$

where $\hat{T}_E$ denotes surface-electron transduction, Φ_{static} denotes the participating static elemental charge photons, and C_E denotes an electric chain. Equation 2.1 is a structural representation and does not replace conventional electrostatic or electromagnetic field equations.

The term *surface-associated electron* is used broadly enough to include electrons associated with charge-supporting boundaries and material interfaces.

In a conductor approaching electrostatic equilibrium, excess charge is redistributed primarily toward the surface, where the available surface-associated electron population establishes the conditions for the transduction of predominantly static elemental charge photons into electric chains. At dielectric interfaces, limited conductivity and charge trapping can maintain localized surface or interfacial charge for much longer periods, allowing the associated electric chain organization to persist and, under continued loading, become increasingly concentrated.

Real materials may also exhibit polarization, finite leakage, surface contamination, adsorbed species, defects, and charge trapping. These established material processes remain part of the macroscopic description. The Photony interpretation adds the proposal that the available surface-associated electron population participates in organizing the electric chain structure associated with the resulting electric field.

Electric chain formation, therefore, depends on both charge availability and the surrounding electrical environment. Material conduc-

tivity, permittivity, geometry, surface condition, charge distribution, leakage pathways, and the rate at which charge or energy is supplied all affect whether electric chain organization can accumulate or is rapidly redistributed. A highly conductive path generally permits charge to redistribute readily, whereas an insulating or resistive region can allow charge separation and electric-field energy to persist.

2.2.3 Electric Chains and Electric-Field Organization

An **electric chain** is an extended photon-chain structure produced through surface-electron transduction of predominantly static elemental charge photons. Collections of electric chains provide the proposed physical organization underlying an electric field.

Electric and magnetic chains share the same underlying filament and photon hierarchy, but their larger-scale organizations differ. Magnetic chains form closed loops. Electric chains, by contrast, may terminate on like-spinning free electrons. Their endpoints therefore have direct physical significance within the proposed electric chain interpretation.

Electric chains may extend between oppositely charged regions, from an isolated charged body toward surrounding charge-supporting structures, or through dielectric regions whose polarization and boundary conditions shape the macroscopic field. Their orientation and spatial concentration vary with the charge distribution and geometry of the system.

Conventional electric-field lines indicate field direction and are commonly drawn with greater graphical density where the field is stronger. These field lines are mathematical and visual representations rather than physical objects in conventional electrodynamics. Photony theory retains the measurable electric field $\mathbf{E}$ and its established mathematical description but proposes that the field-line representation corresponds to an underlying population of physical electric chains.

A conceptual relation between electric chain density and measurable electric-field strength may be written as

$$E = \beta_E \frac{N_E}{A} \quad (2.2)$$

where N_E is the number of electric chains intersecting a representative area A , and β_E is a proportionality factor associated with the proposed chain organization, local geometry, and material environment. Equation 2.2 is not presented as a replacement for Gauss's law or as an experimentally calibrated constitutive relation. It provides a conceptual bridge between measurable electric-field strength and the proposed density of electric chain organization.

Geometry is particularly important because electric-field loading is rarely uniform in practical systems. A broad, smooth conducting surface may support a relatively distributed charge density, whereas sharp edges, points, small radii of curvature, narrow gaps, electrode irregularities, and dielectric interfaces can strongly concentrate the local electric field. Conventional electrostatics accounts for these effects through charge distributions and boundary conditions. In the Photony interpretation, the same conditions correspond to local concentration and altered spacing of electric chains.

This distinction becomes important when evaluating breakdown. A system may have a moderate total potential difference while a small region experiences a much larger local electric field because of geometry or material inhomogeneity. Voltage alone is therefore not sufficient to characterize electric loading or breakdown susceptibility.

2.2.4 Electric-Field Energy, Loading, and Structural Response

The electric chain interpretation provides a proposed physical basis for distinguishing stored electric-field energy from the processes through which that energy is accumulated, redistributed, and released. Conventional electromagnetic theory quantifies electric-field energy density through

$$u_E = \frac{1}{2} \epsilon E^2 \quad (2.3)$$

where ε is the permittivity of the medium. For an ideal lumped capacitive system,

$$U_E = \frac{1}{2} CV^2. \quad (2.4)$$

Photony does not alter these measurable relationships. Instead, it proposes that stored electric-field energy is physically organized through electric chains whose arrangement reflects charge distribution, geometry, and surrounding material.

A charge-separated electric system may therefore contain substantial stored energy before any rapid discharge begins. Additional loading may develop through greater charge separation, continued surface-charge accumulation, increased potential difference, geometric field concentration, reduced electrode separation, material polarization, or continued energy delivery from an external source.

The amount of energy available during a later discharge is not determined by voltage alone. Capacitance, charge, geometry, source impedance, resistance, inductance, and continuing source power may influence the magnitude and duration of the release.

As electric loading develops, the system does not need to proceed directly to complete failure. Charge may redistribute, leak through finite resistance, remain accommodated by dielectric polarization, or undergo localized corona or partial discharge.

The principal responses can be summarized as

$$\begin{aligned} \text{electric loading} \longrightarrow & \text{stable accommodation} \\ & \text{or localized discharge} \\ & \text{or path breakdown.} \end{aligned} \quad (2.5)$$

Voltage breakdown occurs when an insulating, resistive, or incompletely conducting region can no longer sustain the imposed electric field and a substantially more conductive pathway develops. Depending on the medium, geometry, and available source, the transition may appear macroscopically as a spark, streamer channel, dielectric puncture, surface flashover, lightning channel, or electrical arc.

Within the Photony interpretation, breakdown corresponds to localized failure or rapid reorganization of electric chain continuity when the previously charge-supported organization can no longer be maintained. Stored electric-field energy may then be transferred rapidly into charge-carrier motion, ionization, radiation, heating, plasma formation, acoustic emission, and material change.

The formation of a conducting pathway does not necessarily mark the end of the energy-release process. If the system is an isolated charged body or capacitor, the available stored charge may be depleted rapidly and the discharge may terminate. If an external source remains connected, however, the newly formed pathway may permit substantially greater current to flow and may develop into a longer-duration electrical arc.

2.3 Static Electricity: Formation, Storage, and Redistribution

Static electricity provides one of the clearest examples of electric energy accumulating before a rapid release event. The term *static* in this section refers to a macroscopic condition in which separated charge persists for an appreciable time rather than flowing continuously as current. This usage should be distinguished from the Photony term *static elemental charge photon*, which refers to the configuration and motion state of the elemental charge photon itself.

Static charge can be generated through several mechanisms, including contact and separation of dissimilar materials, electrostatic induction, movement of liquids and particles, spraying, material transfer, and the motion of people or objects across insulating surfaces. The resulting charge may remain localized, redistribute through available conducting paths, leak gradually through the surrounding material, or discharge abruptly when the local electric field becomes sufficiently large to initiate a conducting pathway.

From the Photony perspective, these processes alter the population and distribution of

surface-associated charge and, therefore, the electric chain organization supported by that charge. Static-electricity generation does not require electric chains to arise from a fundamentally different structure than other electric fields. Rather, it provides conditions in which charge separation and surface-electron transduction can persist long enough for substantial electric-field organization to develop [Sobarzo et al. \(2025\)](#).

2.3.1 Contact, Separation, and Charge Transfer

One of the most familiar routes to static electrification is contact between two materials followed by their separation. Although this process is often described as frictional or triboelectric charging, friction is not strictly required. Contact establishes an interface at which charge transfer or redistribution may occur, and subsequent separation leaves the two surfaces with a net charge imbalance.

Repeated rubbing can increase charging because it produces repeated contact, separation, deformation, and renewal of the interacting surface regions. However, the magnitude and polarity of the resulting charge depend on the materials, surface conditions, contamination, humidity, temperature, contact area, separation rate, and previous charging history. The microscopic origin of contact electrification is not described by a single mechanism applicable to all material combinations. Electron transfer, ion transfer, material transfer, surface-state effects, and adsorbed species may contribute under different conditions.

The basic macroscopic result is, nevertheless, well established. After separation, one body may contain an excess of negative charge, while the other contains a corresponding deficiency. If the materials or surrounding environment do not provide a sufficiently conductive return path, the charge imbalance can persist.

For an approximately isolated charged body,

$$Q = CV \quad (2.6)$$

where Q is stored charge, C is effective capacitance relative to the surroundings, and V

is electric potential. This relation demonstrates why a relatively small quantity of charge can produce a large potential difference when effective capacitance is small.

Within the Photony interpretation, charge separation alters the surface-associated electron population available for transduction. Predominantly static elemental charge photons are organized into electric chains whose distribution reflects the resulting charge configuration. Continued charge transfer can therefore increase the number or concentration of electric chains and increase associated electric-field energy.

The process may be summarized compactly as

$$\begin{aligned} \text{contact} &\rightarrow \text{charge transfer} \\ &\rightarrow \text{separation} \\ &\rightarrow \text{charge imbalance} \\ &\rightarrow \text{electric organization.} \end{aligned} \quad (2.7)$$

This sequence does not specify a unique microscopic charge-transfer mechanism. It identifies the macroscopic progression that produces the charge-separated state required for later electric-field buildup and possible discharge.

2.3.2 Electrostatic Induction

Static electrification may also occur without direct charge transfer from the initially charged object. In electrostatic induction, an external electric field redistributes mobile charge within a nearby conductor or polarizable material. Opposite charge is attracted toward the region nearest the external source, while like charge is displaced farther away.

If the induced object remains electrically isolated and the external source is removed, the redistributed charge generally returns toward its previous configuration. If a temporary conducting connection to ground or another reservoir is provided while the inducing field is present, however, charge may enter or leave the object. Removal of the grounding path followed by removal of the inducing source can leave the object with a net charge.

In the Photony interpretation, an externally imposed electric chain organization changes the

distribution of surface-associated electrons on the nearby body. This redistribution changes the local transduction conditions and produces a corresponding reorganization of electric chains.

Nearby grounded structures, conductors, tools, personnel, and equipment can therefore strongly alter electric-field geometry even without direct contact with the initially charged object. Such structures change capacitance, boundary conditions, and local field concentration and may either reduce charge accumulation or create preferred locations for discharge.

2.3.3 Flow, Spraying, and Material-Handling Electrification

Static charge generation is not limited to solid objects rubbed together. Industrial systems provide many examples in which charge develops during movement of liquids, gases containing droplets or particles, powders, granular solids, films, belts, and other materials.

When a liquid moves through pipes, hoses, filters, valves, or nozzles, repeated contact between the liquid and surrounding surfaces can produce charge separation. The transported liquid may carry one polarity downstream while an opposing charge remains on the containing surface. Similar charging can occur during pouring, pumping, filtration, atomization, spraying, and splash filling.

Powders and granular materials can become charged through repeated particle-particle and particle-surface contacts. Pneumatic transport, mixing, grinding, conveying, filling, and emptying operations may therefore produce substantial charge separation. Because many powders and polymeric materials are poor conductors, accumulated charge may persist long enough to produce large potential differences.

The balance between charge generation and charge dissipation may be represented as

$$\frac{dQ}{dt} = \dot{Q}_{\text{gen}} - \dot{Q}_{\text{loss}} \quad (2.8)$$

where $\dot{Q}_{\text{gen}}$ is the rate of charge generation or delivery and $\dot{Q}_{\text{loss}}$ represents charge removal through conduction, leakage, grounding, ionization, or other dissipative processes.

When

$$\dot{Q}_{\text{gen}} > \dot{Q}_{\text{loss}}, \quad (2.9)$$

the net stored charge increases and the electric field may continue to build.

Within the Photony interpretation, this condition corresponds to continued growth or concentration of the charge-supported electric chain organization because surface-electron transduction is being driven faster than the system can redistribute or reduce accumulated electric loading.

The persistence of static electricity strongly depends on conductivity. Charge deposited on a good conductor can rapidly redistribute across the available conducting surface and, if a suitable connection exists, flow to ground or another charge reservoir. Charge deposited on a highly resistive material may remain localized for much longer periods.

A useful conventional measure of the characteristic rate at which free charge relaxes within a material is

$$\tau = \frac{\epsilon}{\sigma} \quad (2.10)$$

where ϵ is permittivity and σ is electrical conductivity. A small relaxation time indicates that charge redistributes or dissipates rapidly, whereas a large relaxation time permits charge separation to persist.

The principal static-electricity generation and retention pathways are summarized in Figure 2.2.

2.3.4 Charge Retention, Leakage, and Electrical Relaxation

Grounding and bonding provide engineered means of reducing persistent potential differences in conducting equipment [Occupational Safety and Health Administration \(2026\)](#). These measures are especially important in systems containing flammable gases, vapors, liquids, or powders because a small spark can provide sufficient ignition energy under suitable conditions.

Within the Photony interpretation, a short dielectric relaxation time favors rapid surface-electron redistribution and continual adjustment

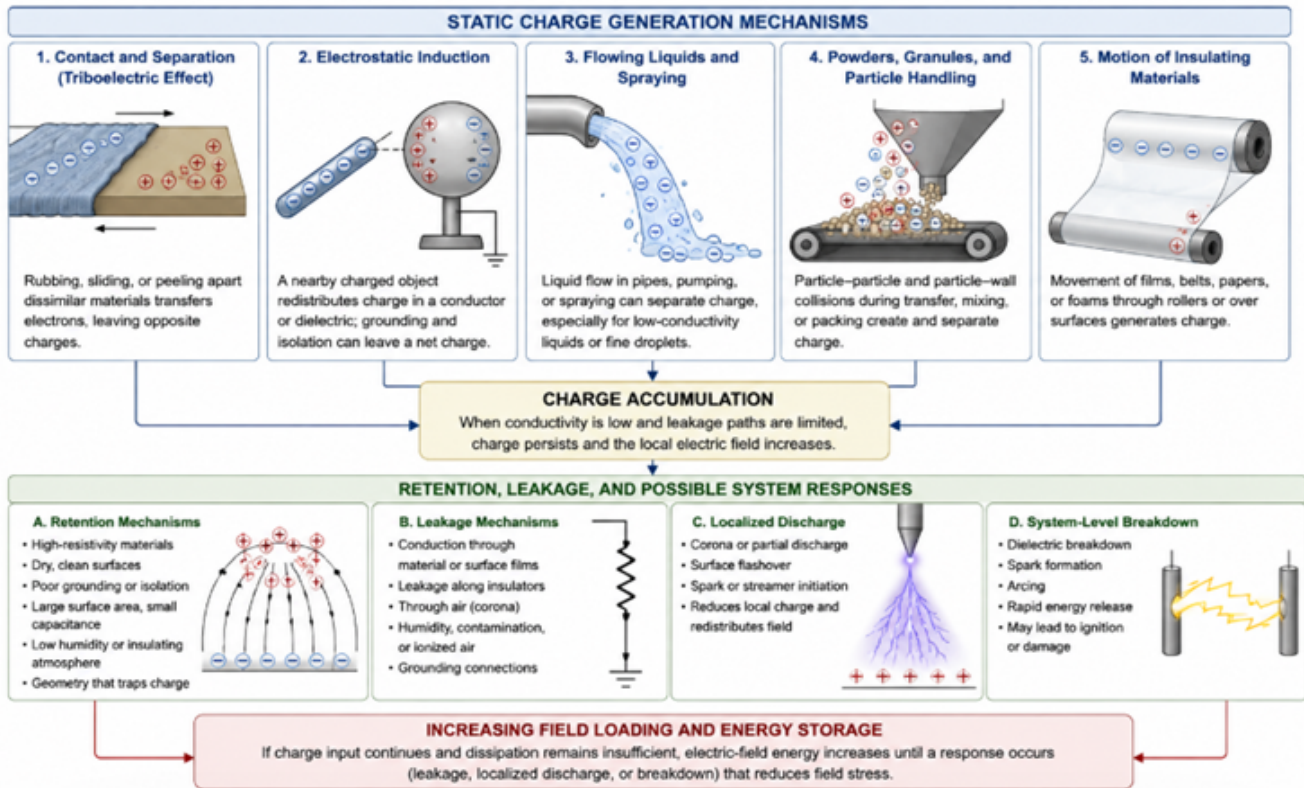


Figure 2.2: Major pathways through which static charge may be generated and retained. Contact and separation, electrostatic induction, liquid flow and spraying, powder or particle transfer, and motion of insulating materials can produce persistent charge separation when dissipation is limited. [Lin et al. \(2020\)](#) Continued charge accumulation can increase electric-field loading until the system responds through leakage, localized discharge, or breakdown.

of the electric chain organization, limiting sustained local loading. A long relaxation time allows charge-supported electric chain organization to persist and become increasingly concentrated as additional charge is supplied.

2.3.5 From Static Charge Accumulation to Discharge

Static electricity becomes an energy-release problem when accumulated charge produces an electric field capable of initiating ionization, dielectric failure, surface flashover, or another substantial increase in conductivity. The transition is controlled by the local field rather than voltage alone.

The progression is

$$\begin{aligned} \text{charge generation} &\rightarrow \text{charge separation} \\ &\rightarrow \text{charge retention} \\ &\rightarrow \text{electric loading} \\ &\rightarrow \text{local threshold} \\ &\rightarrow \text{discharge.} \end{aligned} \quad (2.11)$$

Not every charging event reaches the final stage. Charge may dissipate through leakage, grounding, corona, environmental ionization, or contact with another conductor before complete breakdown occurs.

Within the Photony framework, the same progression is interpreted as the formation and increasing loading of a surface-supported electric chain organization, followed, when accommodation becomes insufficient, by localized chain reorganization or failure and the development of a more conductive pathway.

2.4 Electric-Field Loading and Voltage Breakdown

Electric breakdown is not controlled by voltage alone. The relevant physical state depends on how electric potential is distributed through space and through the surrounding material. Electrode geometry, separation distance, permittivity, conductivity, pressure, temperature,

surface condition, waveform, pulse duration, and availability of free charge carriers can all influence whether a system remains insulating, develops localized discharge, or forms a conducting pathway.

Voltage breakdown is therefore best treated as a progression from electric-field loading toward a threshold condition rather than as a single universal field value.

Within the Photony interpretation, these conventional loading variables alter the density, spacing, continuity, and local organization of electric chains. Breakdown is proposed to occur when the existing charge-supported electric chain organization can no longer be maintained within the available insulating, resistive, or incompletely conducting environment, and a substantially more conductive pathway develops.

2.4.1 Electric Stress, Geometry, and Field Concentration

The electric field is related to spatial variation of electric potential:

$$\mathbf{E} = -\nabla V. \quad (2.12)$$

For an approximately uniform field between parallel electrodes,

$$E \approx \frac{V}{d}, \quad (2.13)$$

where d is electrode separation.

Real breakdown configurations are rarely perfectly uniform. Local curvature can strongly enhance the field near points, edges, scratches, voids, protrusions, electrode defects, or dielectric interfaces.

A useful conceptual expression is

$$E_{\text{local}} = \gamma_E E_{\text{avg}}, \quad (2.14)$$

where γ_E represents a geometry-dependent field-enhancement factor. This emphasizes that the local threshold condition may be reached well before the average field becomes critical.

In the Photony interpretation, local enhancement corresponds to increased electric chain concentration within a restricted region.

2.4.2 Breakdown Is Material and Environment Dependent

No single breakdown field applies to all materials and conditions. Gases, liquids, solids, free-space gaps, contaminated surfaces, and composite insulation systems respond differently because the mechanisms by which conductivity increases are different.

In gases, breakdown depends strongly on pressure, electrode separation, gas composition, polarity, field uniformity, and availability of initial charge carriers. In solid dielectrics, defects, voids, interfaces, thermal history, aging, and local field enhancement may dominate. In liquids, impurities, bubbles, dissolved gases, electrode condition, and streamer formation can be important. Along surfaces, contamination, moisture, roughness, and creepage distance can lower the effective breakdown threshold.

The dielectric strength of a material is therefore a useful engineering quantity but not an immutable physical constant.

From the Photony perspective, the surrounding medium determines how readily surface-associated charge can redistribute and how effectively the existing electric chain organization can be maintained.

2.4.3 Gas Breakdown: Electron Avalanches and Streamer Formation

Gas breakdown provides a particularly useful example of the transition from an initially weakly conducting medium to a substantially more conductive discharge path. In the established description, a free electron may be accelerated by the applied electric field and may ionize a neutral atom or molecule. Additional electrons can then undergo similar acceleration and ionizing collisions, producing an electron avalanche [Meek \(1940\)](#).

The process may be summarized as

$$\begin{aligned} \text{free electron} &\rightarrow \text{field acceleration} \\ &\rightarrow \text{collision ionization} \\ &\rightarrow \text{carrier multiplication.} \end{aligned} \quad (2.15)$$

Avalanche multiplication refers to multiplication of free charge carriers in the gas. It does

not describe multiplication of Photony electric chains.

As the avalanche grows, space-charge effects may permit the discharge to evolve into a self-propagating streamer. A streamer is a rapidly propagating ionization front that extends a conductive or partially conductive region into previously weakly ionized gas.

Within the Photony interpretation, avalanche and streamer processes describe changes occurring in the gas through which the pre-existing electric chain organization extends. Increasing conductivity changes the material environment capable of maintaining that organization.

The conventional avalanche and streamer mechanisms therefore describe how the gas acquires conductivity, while Photony proposes how the associated electric-field organization responds structurally to that change.

2.4.4 Paschen-Type Behavior and Gap Dependence

For relatively uniform gas gaps, conventional gas-discharge theory relates breakdown behavior to both gas pressure and electrode separation. The breakdown voltage varies nonlinearly with the product pd , where p is gas pressure and d is gap distance.

A general representation is

$$V_B = f(pd, \text{gas, geometry, polarity}). \quad (2.16)$$

The physical basis is the balance between electron acceleration, collision frequency, ionization probability, and the distance available for carrier multiplication.

Practical systems can depart substantially from ideal Paschen behavior. Strongly nonuniform electric fields, very small gaps, surface conditions, electrode geometry, and short-duration voltage pulses can modify the observed breakdown voltage. [Hourdakakis, Simonds, and Zimmerman](#) reported departures from conventional Paschen behavior in submicrometer air gaps [Hourdakakis et al. \(2006\)](#).

Within Photony, pressure and gap dependence do not alter the photon-chain composition of the pre-breakdown electric field. These variables determine how readily the gas through

which the electric chain organization extends can undergo the conventional transition from weak conduction to substantially greater conduction.

2.4.5 Corona and Partial Discharge

Electric breakdown does not necessarily produce a conducting path across the entire electrically stressed region. Localized forms of electrical activity can occur while the remainder of the insulation or gas gap continues to sustain the applied potential difference [Van Brunt \(1994\)](#).

Corona commonly develops near a small-radius electrode, sharp point, edge, or other regions of strong field concentration. Partial discharge is a broader insulation-system phenomenon in which breakdown occurs within only part of the insulating structure.

Corona and partial discharge are therefore examples of electric energy release that do not immediately constitute complete conducting-path breakdown. Repeated partial discharge can progressively alter insulation through erosion, heating, chemical change, or other degradation.

Within the Photony interpretation, increased conductivity within the affected region changes the material environment through which a local portion of the electric chain structure is maintained. That portion may reorganize rapidly while the larger charge-supported electric chain system remains substantially intact.

2.4.6 Spark Formation and Breakdown

A spark occurs when electrical breakdown establishes a transient conducting channel through a region that was previously insulating, resistive, or weakly conducting.

For an isolated capacitive system,

$$U_{\text{pre}} = \frac{1}{2} C V_{\text{pre}}^2 \quad (2.17)$$

and the reduction in stored electric energy may be represented as

$$\Delta U_E = \frac{1}{2} C (V_{\text{pre}}^2 - V_{\text{post}}^2). \quad (2.18)$$

During the spark, part of this energy may be transferred into charge-carrier motion, ionization, excitation, radiation, gas heating, electrode heating or erosion, acoustic emission, and chemical processes.

The essential transition is not merely a reduction in voltage but a creation of a substantially more conductive path through the previously insulating or weakly conducting region.

Within the Photony interpretation, spark formation is a rapid structural response of the pre-existing electric chain organization to the abrupt change in conductivity and boundary conditions.

2.4.7 From Spark to Sustained Arc

A spark may terminate after the available stored charge is redistributed and the conducting channel loses conductivity. If an external source continues to supply current, however, the newly formed conducting path may remain sufficiently conductive to develop into a sustained arc.

Within Photony, the initial conducting path results from reorganization of the pre-breakdown electric chain structure. As substantial directed current develops, internal free electrons associated with that current transduce dynamic elemental charge photons into magnetic chains. A sustained arc can therefore evolve from an initiating electric-breakdown regime into a coupled electric-magnetic energy-transfer state.

2.4.8 Solid and Liquid Dielectric Breakdown

The formation of a conducting path is not limited to gases. Solid and liquid dielectrics can also undergo localized or complete electrical breakdown.

In solid insulation, breakdown may initiate at voids, defects, interfaces, regions of high electric stress, or locations altered by thermal, mechanical, chemical, or electrical aging. Repeated partial discharge can erode or chemically modify insulation and progressively alter the local electric-field distribution.

Liquid dielectrics follow different mechanisms. Strong electric stress near electrodes, impurities, suspended particles, bubbles, dissolved

gases, local heating, and fluid motion can contribute to development of localized conductive structures.

At the macroscopic level, these routes share a common progression:

insulating state → local conductivity change
 → conductive-path growth
 → rapid energy transfer. (2.19)

Within Photony, electric chains must be distinguished from conductive, carbonized, ionized, or otherwise modified material pathways that develop during breakdown. The material processes change the environment and boundary conditions under which the electric chain organization is maintained.

2.4.9 A Conceptual Electric Breakdown Threshold

Electric breakdown should not be treated as the consequence of a single universal critical voltage. Established breakdown behavior depends on the combined electrical, material, geometric, and temporal state of the stressed region.

These dependencies may be collected conceptually as

$$\Xi_E = \Xi_E(E_{\text{local}}, \sigma, \epsilon, d, p, T, G, S, \tau_p) \quad (2.20)$$

where E_{local} is local electric-field strength, σ is conductivity, ϵ is permittivity, d is characteristic distance, p is pressure where relevant, T is temperature, G represents geometry, S represents material or surface condition, and τ_p is the characteristic duration of electrical stress.

Breakdown is proposed to occur when the combined electric loading reaches the maximum level that the local material and geometry can sustain. In conceptual form,

$$\Xi_E \geq \Xi_{E,\text{crit}}, \quad (2.21)$$

where Ξ_E represents the combined electric loading, and $\Xi_{E,\text{crit}}$ represents the corresponding critical threshold.

This expression is not an established constitutive law and does not replace Paschen

relations, dielectric-strength measurements, streamer criteria, or material-specific breakdown models.

Within the Photony interpretation, $\Xi_{E,\text{crit}}$ marks conceptually the condition at which the surrounding medium can no longer maintain the pre-breakdown electric chain organization in its existing form and a substantially more conductive pathway begins to develop.

A quantitative relation between a proposed chain-level measure and experimentally established breakdown criteria has not yet been demonstrated.

2.4.10 Summary of the Electric Breakdown Sequence

Electric breakdown is best understood as a family of material-dependent pathways leading from electric-field loading toward a substantial increase in conductivity.

The common macroscopic progression is

loading → local response
 → higher conductivity (2.22)
 → breakdown.

Complete electric breakdown occurs when an insulating, resistive, or incompletely conducting region develops a substantially more conductive pathway capable of significant charge transfer.

A further distinction must be made between the formation of that conducting path and the energy subsequently transferred through it. In an isolated electrostatic or capacitive system, the discharge may terminate after the available stored charge is redistributed. If an external source remains connected, continued current can sustain the path and develop the initial breakdown into a longer-duration arc.

2.5 Natural Electric Energy-Release Regime: Lightning

Lightning provides a large-scale natural example of the electric energy-release regime. Charge separation within a thunderstorm establishes strong electric fields within the cloud, between

different cloud regions, and between the cloud and its surrounding environment.

Unlike a laboratory gap between fixed electrodes, lightning initiation cannot be characterized by a single applied voltage or uniform breakdown field. The atmosphere is spatially nonuniform, charge distribution evolves with time, and conductive or partially conductive regions may develop progressively over large distances.

A leader is a longer-lived, partially conducting discharge channel that advances through the atmosphere in successive extensions and provides a region from which further streamer activity can develop. Streamers and leaders progressively extend conductive or partially conductive regions through the air before a high-current connection becomes established [Hill et al. \(2011\)](#).

The essential distinction is between *conducting-path formation* and the *energy subsequently transferred through that path*. Leader and streamer development establishes or extends portions of the conducting pathway. Once an effective electrical connection develops between regions of substantially different electric potential, much larger charge transfers can occur.

Within the Photony interpretation, storm-scale charge separation establishes an extensive charge-supported electric chain organization. Streamer and leader processes describe changes in the conductivity of the atmosphere through which that organization extends.

The formation of the atmospheric conducting pathway is classified here as an electric-breakdown transition because a region that was initially weakly conducting becomes capable of substantial charge transport.

As large current subsequently develops through the established lightning channel, the physical regime becomes increasingly coupled. Within Photony, internal free electrons associated with directed current transduce dynamic elemental charge photons into magnetic chains. Mature lightning can therefore contain both electric and magnetic organization, even though the threshold event establishing the conducting pathway is classified as electric breakdown.

2.6 Purposeful Electrical Breakdown and Ignition

Electric breakdown is not inherently undesirable. In many engineered systems, a sufficiently strong electric field is deliberately established so that an insulating or weakly conducting gap undergoes breakdown at a selected location and time.

The physical transition is fundamentally similar to accidental breakdown. Electric-field loading increases until the intervening region develops a substantially more conductive pathway. The important differences are that electrode geometry, timing, available energy, source impedance, and discharge location are intentionally designed.

Within Photony, purposeful breakdown begins with the same surface-electron transduction and electric chain organization described for other electric fields. Rather than avoiding the threshold, the engineered system is designed to reach it reproducibly and control the energy transferred after the conducting path forms.

2.6.1 Controlled Spark Ignition

Spark ignition systems deliberately produce a high electric field across a small electrode gap. When the local field becomes sufficient to ionize the intervening gas, breakdown creates a transient plasma channel through which charge can flow.

The ignition system need not supply the energy released during combustion. Its function is to deposit sufficient energy, at the correct location and time, to initiate a self-sustaining chemical reaction.

Automotive spark-ignition systems provide a familiar example. Capacitive-discharge systems store energy primarily in a capacitor before rapidly applying it to the ignition circuit. Inductive systems store energy in a magnetic field associated with current through an ignition coil and release that energy when current is interrupted. Although their energy-storage stages differ, both can ultimately generate electric-field breakdown across the ignition gap.

This distinction is important. A magnetic

energy-storage stage may generate the high voltage, but the final event at the ignition gap is electric breakdown because a previously non-conducting gaseous region becomes conducting.

2.6.2 Source-Supplied Arc and High-Energy Ignition

Some ignition systems require more sustained energy deposition than a brief spark can provide. High-voltage igniters, burner systems, flare igniters, and related devices may therefore establish an arc or repeated discharge so that energy continues to be delivered after the initial breakdown.

Breakdown establishes the conducting channel, but the connected electrical source determines whether that channel extinguishes or develops into a longer-duration arc [Wang et al. \(2021\)](#).

The energy delivered after breakdown is

$$W_{\text{delivered}} = \int_{t_0}^{t_1} V(t)I(t) dt. \quad (2.23)$$

The deposited energy may be substantially greater than the electric-field energy stored locally in the gap immediately before breakdown.

From the Photony perspective, the transition begins as an electric breakdown event. After the conducting channel becomes established, directed charge transport increases, and current-associated magnetic organization becomes increasingly important. Purposeful arcing, therefore, illustrates how an initiating electric regime can evolve into a coupled electric-magnetic discharge state.

2.7 Comparison of Electric and Magnetic Energy-Release Regimes

Electric and magnetic energy-release regimes share a broad structure: energy is stored, loading increases, the existing organization responds through finite accommodation, and a threshold transition permits rapid energy conversion. The

similarity should not obscure the physical differences between the two regimes.

The electric regime considered in this chapter is associated primarily with surface-associated electrons transducing predominantly static elemental charge photons into electric chains. Charge separation and restricted redistribution allow the resulting electric organization to persist across an insulating, resistive, or incompletely conducting region. Breakdown occurs when that region can no longer sustain the imposed electric organization, and a substantially more conductive pathway develops.

The magnetic regime developed in the following chapter begins with directed charge transport in an already conducting environment. Internal free electrons associated with current transduce dynamic elemental charge photons into magnetic chains. Magnetic breakdown concerns the loss of stability of that pre-existing current-supported organization rather than the initial creation of a conducting path.

The principal distinctions are summarized in [Table 2.1](#).

The most important distinction is the state of the conducting pathway before threshold. In electric breakdown, the pathway required for substantial current transfer is initially absent, weak, resistive, or incomplete. The threshold event creates or greatly enlarges that pathway. In magnetic breakdown, current already flows through the system, and the threshold concerns the stability of the existing current-supported structure.

Table 2.1: Principal distinctions between electric and magnetic energy-release regimes within the Photony interpretation.

Characteristic	Electric Regime	Magnetic Regime
Primary elemental charge-photon state	Predominantly static	Dynamic
Electron-transduction pathway	Surface-associated free electrons	Internal free electrons associated with current
Chain organization	Electric chains may terminate on like-spinning free electrons	Magnetic chains form closed loops
Typical conductivity environment	Insulating, resistive, or incompletely conducting region	Conducting, current-supported medium
Principal stored-energy condition	Charge separation and electric-field energy	Current-supported magnetic-field energy
Threshold event	Formation or substantial enlargement of a conducting pathway	Destabilization of an existing energetic magnetic organization
Conducting path before threshold	Absent, weak, resistive, or incomplete	Already present and supporting current

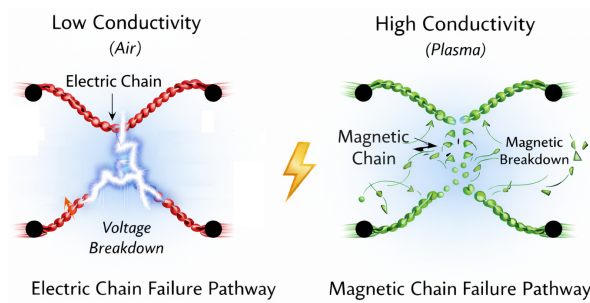


Figure 2.3: Conceptual comparison of electric and magnetic energy-release regimes within the Photony interpretation. Electric breakdown begins in an initially insulating, resistive, or incompletely conducting region and creates or substantially enlarges a conducting pathway, requiring reorganization of the pre-breakdown electric chain structure. Magnetic breakdown occurs in an already conducting, current-supported environment and involves destabilization of the existing magnetic chain organization. Air and plasma are shown as illustrative environments rather than universal requirements for the two regimes.

Electric-field energy can accumulate while charge remains separated and redistribution is inhibited. Magnetic energy, by contrast, is associated with current-supported organization in a system where directed charge transport is al-

ready present.

The two regimes can nevertheless become strongly coupled. An electric breakdown event may create a conducting channel through which substantial current subsequently flows. Once that current develops, magnetic organization becomes important. Conversely, changing magnetic organization can induce electric fields and voltage stresses elsewhere in a system.

A single event may, therefore, contain both electric and magnetic stages, even though the threshold mechanism at a particular location can still be classified according to the local physical conditions.

2.8 Discussion

The electric energy-release regimes examined in this chapter share a common physical progression despite substantial differences in scale, material environment, and application. Charge separation and restricted redistribution permit electric-field energy to accumulate, while geometry, material properties, conductivity, pressure, and surface conditions determine how that loading is distributed.

The distinction between electric loading and conducting-path formation is central. Voltage

alone does not define the breakdown condition, nor does the occurrence of breakdown determine the total energy ultimately released. An isolated electrostatic discharge may be limited primarily by the charge and capacitance present before breakdown, whereas a newly formed conducting path connected to an external source can support a much larger and longer-duration arc.

The same framework applies to both accidental and purposeful electrical breakdown. Electrostatic sparks, insulation failure, flashover, lightning, and unintended arcs represent cases in which a conducting pathway develops under uncontrolled or naturally evolving conditions. Spark plugs, burner igniters, flare igniters, and related systems deliberately create the same general transition at a selected location and time.

Within the Photony interpretation, the pre-breakdown electric field is associated with surface-electron transduction of predominantly static elemental charge photons into electric chains. Increasing electric loading is interpreted as increasing or concentrating this charge-supported chain organization. When established material or discharge processes cause the surrounding region to become substantially more conductive, the pre-existing electric chain organization is proposed to undergo rapid reorganization or loss of continuity.

The resulting framework connects static charge accumulation, localized discharge, dielectric failure, sparks, arcs, lightning, and electrical ignition through a common loading-and-threshold description while preserving the established physics specific to each phenomenon.

2.9 Conclusions

Electric energy-release events ranging from electrostatic discharge and dielectric breakdown to lightning, electrical arcing, and deliberate spark ignition can be described within a common loading-and-threshold framework. Charge separation and restricted redistribution permit electric-field energy to accumulate, while local geometry, material properties, conductivity, and the surrounding environment determine whether that loading is accommodated, produces local-

ized discharge, or progresses to complete breakdown.

A principal conclusion is that voltage breakdown should not be identified with voltage alone. The decisive condition is local electric stress within the actual material and geometric environment. Breakdown occurs when an initially insulating, resistive, or incompletely conducting region develops a substantially more conductive pathway.

The energy released after breakdown is also not determined solely by pre-breakdown voltage. An isolated electrostatic discharge may be limited by the stored charge and capacitance of the system, whereas a conducting path connected to an external source can support substantially greater source-supplied energy and develop into a sustained arc.

Electric breakdown is not inherently a failure. The same general physical transition can occur accidentally in insulation faults, electrostatic discharges, and ignition hazards or be generated deliberately in spark plugs, burner igniters, and other high-voltage ignition systems.

Within the Photony interpretation, the pre-breakdown electric field is associated with surface-electron transduction of predominantly static elemental charge photons into electric chains. When established material or discharge processes cause the surrounding region to become substantially more conductive, the associated charge-supported electric chain organization is proposed to undergo rapid reorganization or loss of continuity. This chain-level interpretation remains complementary to the established quantitative treatment provided by electrostatics, dielectric physics, gas-discharge physics, and plasma physics.

Electric and magnetic energy-release regimes therefore share the broad features of stored energy, increasing loading, finite accommodation, threshold behavior, and rapid energy conversion, but they differ in their pre-threshold physical states. Electric breakdown creates or substantially enlarges a conducting pathway through a previously less conductive region, whereas magnetic breakdown concerns instability within an already conducting, current-supported magnetic organization.

Chapter 3

Magnetic Energy-Release Regimes

Purpose: To examine the storage, loading, instability, and rapid release of magnetic energy from pre-existing current-supported magnetic structures

3.1 Introduction

Magnetic energy-release events occur in conducting materials, laboratory plasmas, astrophysical plasmas, and especially in the solar atmosphere. Solar flares provide a particularly important example. They release large quantities of stored magnetic energy over short timescales, producing electromagnetic radiation, particle acceleration, plasma heating, bulk plasma motion, and, in some cases, coronal mass ejections.

Magnetic reconnection is widely used to describe the associated change in magnetic connectivity and the conversion of stored magnetic energy into observable forms. An important physical question, nevertheless, remains: what initiates the earliest rapid energy release before the larger magnetic structure has completed the reconfiguration described as reconnection?

This chapter examines *magnetic breakdown* as a proposed threshold-driven phenomenon that may provide an incipient physical mechanism for magnetic energy release. Magnetic breakdown is defined here as localized structural failure and fragmentation of a pre-existing, current-supported magnetic organization within a highly conducting environment. It is proposed to occur when increasing magnetic chain density, curvature, compression, interaction, twist, stretching, or confinement exceeds the ability of the magnetic structure to remain stable or to continuously reassign to admissible pathways.

The resulting fragmentation is proposed to

release stored magnetic energy into radiation, accelerated particles, plasma motion, and heat while changing the conditions under which subsequent magnetic reassignment and macroscopic reconnection proceed.

The proposed mechanism is developed through the Photony framework. Internal free electrons associated with current transduce dynamic elemental charge photons into magnetic chain loops that provide a proposed physical organization underlying magnetic fields. Individual magnetic chains are organized into magnetic fibrils, while groups of fibrils can form larger structures such as flux tubes, coronal loops, and flux ropes.

The formation, organization, confinement, and continued maintenance of these current-supported structures embody substantial magnetic energy. Additional loading may develop through increasing field strength, current concentration, chain density, curvature, compression, stretching, twist, interaction, and confinement.

Magnetic breakdown is therefore fundamentally different from the voltage breakdown considered in Chapter 2. Electric breakdown begins in an insulating, resistive, or incompletely conducting region and creates or substantially enlarges a conducting pathway. Magnetic breakdown begins with substantial current and magnetic organization already present. The proposed instability concerns the loss of stability of that existing magnetic organization.

This distinction is especially important in the

solar atmosphere. Solar active regions contain pre-existing currents, magnetic fibrils, flux tubes, coronal loops, and flux ropes before a flare begins. Localized electric fields can arise during a flare and can contribute strongly to particle acceleration, but their presence alone does not demonstrate voltage breakdown. Classification must instead consider the temporal sequence of magnetic stress, radiation, particle acceleration, electric field development, plasma motion, current redistribution, and large-scale magnetic reconfiguration.

For clarity, three related terms are distinguished throughout this chapter. *Magnetic reconfiguration* is the general term for a change in the geometry, organization, connectivity, or energy distribution of a magnetic structure. *Magnetic reconnection* refers to the conventional plasma-physics description of changing magnetic connectivity and associated energy conversion. *Magnetic reassignment* is the Photony interpretation in which magnetic chains or fibrils are redirected into new admissible pathways. Reassignment may occur without destructive breakdown, or it may occur after magnetic breakdown as surviving portions of the magnetic organization reorganize.

The central hypothesis developed in this chapter is that magnetic breakdown may initiate the earliest rapid energy release as a stressed system enters magnetic reassignment and reconnection. This hypothesis does not replace established electromagnetic, magnetohydrodynamic, reconnection, or kinetic models. Those theories remain the quantitative foundation for magnetic energy, current sheets, plasma behavior, particle acceleration, heating, and magnetic reconfiguration. Photony provides a proposed structural interpretation whose validity must ultimately be determined by observation and experiment.

Terminology: magnetic energy release, magnetic breakdown, magnetic chains, fibrils, dynamic elemental charge photons, internal free electrons, internal electron transduction, magnetic loading, magnetic free energy, magnetic reassignment, magnetic reconfiguration, magnetic reconnection, structural accommodation, chain continuity, frag-

mentation, current-supported magnetic structure, flux tube, coronal loop, flux rope, current sheet, precursor emission, solar flare, plasma instability

3.2 Magnetic Structures Relevant to Energy Release

Chapter 1 established the broader Photony framework, including Planck Space, energy filaments, elemental charge photons, classical photons, and electron transduction. The present section therefore concentrates on the structures required specifically for magnetic energy storage, loading, reassignment, and breakdown.

Photony does not replace established mathematical descriptions of magnetic fields. Maxwell's equations, electromagnetic energy densities, magnetohydrodynamic models, reconnection theory, kinetic models, and plasma diagnostics remain the principal quantitative descriptions. The proposed chain interpretation adds a structural description through which magnetic chain organization may be discussed in terms of continuity, density, curvature, compression, interaction, reassignment, and fragmentation.

The distinction between a conventional magnetic field-line representation and the proposed chain interpretation is illustrated in Figure 3.1.

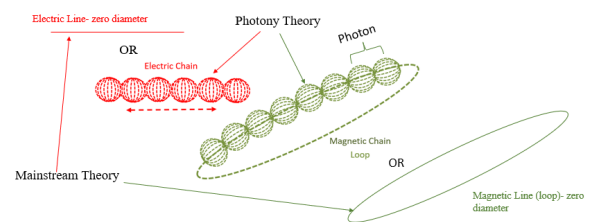


Figure 3.1: Comparison between a conventional magnetic field-line representation and the proposed Photony chain-based interpretation. The field line is a geometrical representation of the magnetic field, whereas the magnetic chain is proposed as a linked physical organization formed through electron transduction.

The figure does not imply that the conventional field-line description is incorrect. The two representations serve different purposes. The

field line describes magnetic direction and geometry at the macroscopic level, whereas the chain representation proposes a physical structure capable of density, curvature, continuity, reassignment, and fragmentation.

3.2.1 Dynamic Elemental Charge Photons

As established in Chapter 1, elemental charge photons occur in static and dynamic forms. Static elemental charge photons are associated primarily with stored charge and electric chain formation. Dynamic elemental charge photons possess directed filament motion and participate in current-associated magnetic chain formation.

The word *dynamic* describes the configuration and motion state of the elemental charge photon. It is not a synonym for alternating current. Dynamic elemental charge photons can therefore participate in magnetic chain organization associated with steady, alternating, pulsed, or transient current.

For the magnetic regime considered in this chapter, the dynamic population is of primary importance.

3.2.2 Internal-Electron Transduction

Electrons serve as transducers and regulators of photon-chain organization. In the magnetic regime, internal free electrons associated with current interact with dynamic elemental charge photons and organize them into extended magnetic chains according to electron distribution, current-supported geometry, surrounding conductivity, and local energy flow.

For the present chapter, the essential pathway is internal-electron transduction:

$$\hat{T}_B(\Phi_{\text{dynamic}}) \longrightarrow C_B, \quad (3.1)$$

where $\hat{T}_B$ denotes internal-electron transduction, Φ_{dynamic} represents participating dynamic elemental charge photons, and C_B represents a magnetic chain.

This is a structural representation and does not replace conventional electromagnetic field equations.

Internal-electron transduction also introduces a finite local organizational capacity. Electrons can process, align, redirect, and maintain only those magnetic chain configurations that remain compatible with local geometry, conductivity, chain density, current, and rate of energy delivery. As the required transduction approaches or exceeds the available organizational capacity, crowding or discontinuity may develop, increasing the likelihood of magnetic reassignment or breakdown.

Figure 3.2 illustrates the complementary processes of transduction and reduction.

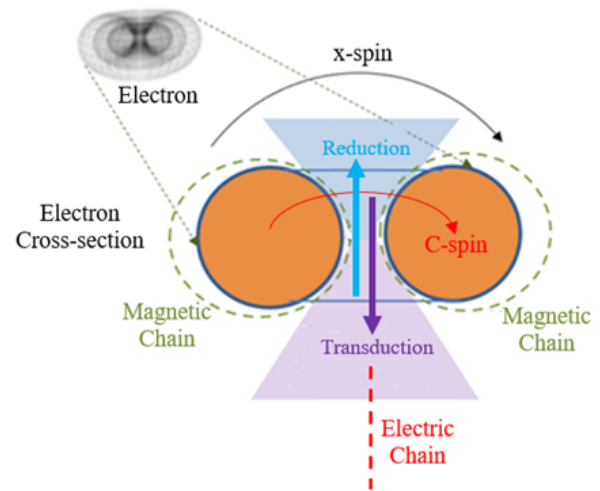


Figure 3.2: Electron transduction and reduction in the Photony framework. During internal-electron transduction, free electrons associated with current receive and organize dynamic elemental charge photons into linked magnetic chains represented macroscopically as magnetic field lines. During reduction, the same electrons decrease, redirect, or disassemble part of the magnetic chain population as the supporting current or elemental charge-photon supply declines. When transduction demand approaches or exceeds the available local organizational capacity, chain crowding or discontinuity may increase the likelihood of magnetic reassignment or breakdown.

3.2.3 Magnetic Chains and Fibrils

A magnetic field is conventionally associated with electric current, magnetic flux density, and forces on moving charges or magnetic materials.

Photony retains these measurable descriptions while proposing that magnetic field lines correspond physically to linked magnetic chain loops.

Magnetic chains are formed through the transduction of dynamic elemental charge photons by internal free electrons associated with electric current. They form closed loops and do not terminate at isolated magnetic charges. Their continuity must therefore be maintained through loop closure, reassignment, or collective structural change.

A conceptual relation between magnetic chain density and measurable magnetic flux density may be expressed as

$$B = \beta_B \frac{N_B}{A}, \quad (3.2)$$

where N_B is the number of magnetic chains intersecting a representative area A , and β_B is a proportionality factor associated with chain geometry, alignment, spacing, and the local medium.

This relation is not proposed as a replacement for conventional magnetic flux density or as an experimentally calibrated constitutive equation. It provides a conceptual bridge between measurable magnetic-field magnitude and proposed magnetic chain organization.

A *fibril* is an organized bundle of magnetic chains. In solar applications, plasma-containing fibrils may assemble into larger magnetic structures such as flux tubes, coronal loops, and twisted flux ropes. These fibrils are not empty magnetic structures. They contain magnetically guided plasma, and in cooler portions of the solar atmosphere, they may also contain partially ionized material.

The hierarchy relevant to this chapter is therefore,

magnetic chains → fibrils
→ larger magnetic structures.

High conductivity favors this organization because abundant mobile electrons are available to support current and internal-electron transduction. This is particularly important in solar plasma, where organized current-supported magnetic structures exist before flare onset.

3.2.4 Magnetic Energy and Structural Stability

Conventional electromagnetic theory quantifies magnetic energy density as

$$u_B = \frac{B^2}{2\mu}, \quad (3.3)$$

where μ is magnetic permeability.

Photony does not alter this measurable relation. It proposes that magnetic energy is physically organized through magnetic chains, fibrils, and the larger current-supported structures assembled from them.

A magnetic organization can remain stable only while its geometry, density, curvature, continuity, electron support, and surrounding environment remain mutually compatible. Increasing field strength does not by itself require breakdown. Additional energy can be accommodated through changes in chain number, spacing, curvature, volume, current distribution, or magnetic reassignment.

Three broad responses are, therefore, possible:

magnetic loading → accommodation
or reassignment (3.4)
or breakdown.

During stable accommodation, continuity is maintained through expansion, deformation, altered chain spacing, or current redistribution.

During magnetic reassignment, chains or fibrils are redirected into alternative admissible pathways while sufficient continuity is retained.

Magnetic breakdown is proposed to occur only when neither stable accommodation nor continuous reassignment can preserve the local magnetic organization. The structure then crosses a threshold, loses magnetic chain continuity, and undergoes localized fragmentation.

3.3 Magnetic Energy and Release Pathways

Magnetic fields store measurable physical energy within current-supported systems. A central

distinction must be made between controlled reduction of a magnetic field and threshold-driven magnetic breakdown.

A magnetic field may decrease because its supporting current is intentionally reduced, because energy is transferred through an established pathway, or because the current-supported magnetic organization becomes unstable. Only the last condition constitutes the proposed magnetic-breakdown regime.

Classification cannot therefore be based only on final outcomes such as radiation, heating, particle acceleration, current change, plasma motion, or magnetic reconfiguration. The pre-event magnetic organization and temporal sequence are essential.

3.3.1 Conventional Magnetic Energy

The magnetic energy density is

$$u_B = \frac{B^2}{2\mu}. \quad (3.5)$$

For a magnetic structure occupying volume V ,

$$E_B = \int_V \frac{B^2}{2\mu} dV. \quad (3.6)$$

Only part of this total energy is generally available for rapid release. The relevant reservoir is magnetic free energy associated with the departure of the current-supported structure from a lower-energy admissible configuration.

These equations quantify stored magnetic energy. They do not, by themselves, specify what physical organization supports the field or what microscopic sequence initiates instability. Photony adds a proposed structural interpretation while retaining the conventional equations as the quantitative energy description.

3.3.2 Controlled Field Reduction

Stored magnetic energy may be released gradually or rapidly without magnetic breakdown. Controlled reduction occurs when a source, current system, plasma flow, or mechanical configuration decreases the process that established and sustained the magnetic field.

In the Photony interpretation, internal free electrons that transduce dynamic elemental charge photons into magnetic chains also regulate their controlled reduction. As supporting current or energy input decreases, the electrons reduce, redirect, or disassemble part of the magnetic chain population.

The complementary processes can be summarized as

transduction $\rightarrow$ chain formation or maintenance
reduction $\rightarrow$ controlled chain decrease

Current change, heating, mechanical forces, plasma motion, or radiation may accompany this process. These effects do not establish magnetic breakdown as long as energy follows an established transfer pathway and the magnetic organization does not undergo threshold-driven fragmentation.

3.3.3 Externally Imposed Current Reduction

Rapid current reduction does not automatically constitute magnetic breakdown. Current may decrease because of a source change, boundary change, plasma expansion, mechanical displacement, or externally initiated interruption.

The sequence is

external change $\rightarrow$ current reduction
 $\rightarrow$ field reduction (3.7)
 $\rightarrow$ energy transfer.

Magnetic breakdown requires a different initiating condition. Evidence must indicate that a pre-existing current-supported magnetic structure became internally unstable and began releasing stored magnetic energy before, or as the incipient cause of rapid current and field change.

The proposed breakdown sequence is

magnetic instability $\rightarrow$ chain fragmentation
 $\rightarrow$ energy release
 $\rightarrow$ reorganization. (3.8)

3.3.4 Threshold-Driven Magnetic Release

Threshold-driven release occurs when a current-supported magnetic configuration can no longer remain stable under increasing loading.

In the Photony interpretation, structural loading may increase through magnetic chain density, fibril curvature, compression, stretching, twist, interaction, current concentration, or confinement. The magnetic structure may initially respond through expansion, deformation, current redistribution, or non-destructive chain reassignment.

Breakdown becomes possible when loading exceeds both the capacity for stable accommodation and the capacity for continuous reassignment into alternative admissible pathways.

The sequence may be represented compactly as

increasing loading $\rightarrow$ accommodation
 $\rightarrow$ reassignment limit (3.9)
 $\rightarrow$ fragmentation.

Localized fragmentation releases part of the stored magnetic energy. Observable products may include electromagnetic radiation, accelerated particles, plasma heating, plasma motion, current redistribution, and magnetic reconfiguration.

3.3.5 Conductivity and Existing Magnetic Structure

High conductivity is a defining environmental condition for the proposed magnetic-breakdown regime because it permits substantial current and organized magnetic structures to exist before rapid release begins.

Conductivity alone does not determine whether breakdown occurs. Magnetic geometry, current density, plasma density, temperature, composition, confinement, interaction, event history, and rate of energy input must also be considered.

The required regime can be summarized as

high conductivity + pre-existing current
+ magnetic loading
→ possible breakdown. (3.10)

The word *possible* is important. A highly conducting magnetic structure may remain stable, expand, or reassign without breakdown if sufficient accommodation capacity remains.

3.3.6 Temporal Classification

Magnetic breakdown, reassignment, and reconnection can produce overlapping observable effects. Each may involve current change, electromagnetic emission, plasma heating, particle acceleration, magnetic motion, and reconfiguration.

Classification by final outcome alone is therefore unreliable.

The principal question is which physical organization first became unstable and initiated rapid release. A magnetic-breakdown classification requires evidence that a pre-existing current-supported magnetic structure crossed a local instability threshold and thereby initiated the principal rapid energy conversion.

Relevant measurements may include magnetic-field strength, current-density estimates, magnetic shear, fibril or loop curvature, compression, interaction, precursor radiation, particle acceleration, plasma heating, magnetic-field change, plasma motion, and timing of large-scale reassignment or reconnection.

3.3.7 Relationship to Magnetic Reconnection

Figure 3.3 illustrates the proposed relationship between magnetic breakdown and magnetic reconnection.

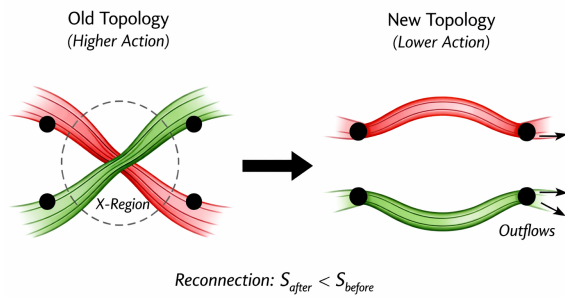


Figure 3.3: Schematic before-and-after representation of reconnection as a result of magnetic breakdown. Before onset, stressed current-supported magnetic chains or fibrils approach a highly loaded interaction region. Localized magnetic breakdown is proposed to fragment part of the pre-existing magnetic organization and release stored magnetic energy. Surviving chains and fibrils are then reassigned into new pathways, producing the larger connectivity change identified macroscopically as magnetic reconnection.

The figure is intended as a structural guide. It does not replace conventional current-sheet, diffusion-region, inflow, or outflow descriptions of reconnection.

The functional sequence is

$$\begin{aligned}
 &\text{stressed structure} \rightarrow \text{local breakdown} \\
 &\quad \rightarrow \text{initial release} \\
 &\quad \rightarrow \text{reassignment} \\
 &\quad \rightarrow \text{reconnection.}
 \end{aligned} \tag{3.11}$$

These stages need not be widely separated in time. Reassignment and developing reconnection may begin immediately after the local threshold crossing and may overlap with continuing magnetic breakdown.

3.4 Solar-Flare Energy Release

Solar flares are rapid releases of energy stored primarily in stressed magnetic structures throughout the solar atmosphere. Observable

consequences include radiation across a broad spectrum, plasma heating, nonthermal particle acceleration, plasma motion, magnetic reconfiguration, and, in some events, coronal mass ejection.

The established solar-physics framework attributes flare energy primarily to magnetic free energy accumulated before onset. Photospheric and subphotospheric motions can progressively stress magnetic structures extending through the photosphere, chromosphere, transition region, and corona. Flux emergence, shearing, twisting, braiding, convergence, and interaction between neighboring magnetic structures can increase current density, magnetic shear, curvature, compression, and stored magnetic energy.

Magnetic reconnection provides the principal conventional framework for describing the associated connectivity change and conversion of magnetic energy. The present framework accepts that role while separating two questions:

1. How does the larger magnetic system change connectivity and relax toward a lower-energy configuration?
2. What physical instability initiates the earliest rapid release of stored magnetic energy?

The magnetic-breakdown hypothesis addresses the second question.

3.4.1 Magnetic Free Energy

The energy available to a flare is associated principally with the stressed magnetic organization. Magnetic energy density remains

$$u_B = \frac{B^2}{2\mu}, \tag{3.12}$$

and the energy within a volume V is

$$E_B = \int_V \frac{B^2}{2\mu} dV. \tag{3.13}$$

Only part of the total magnetic energy is normally available for explosive release. The relevant reservoir is magnetic free energy associated with departure from a lower-energy reference configuration.

Magnetic shear, twist, curvature, current concentration, interaction between flux systems, and confinement by surrounding magnetic structures can increase this available energy.

Within the Photony interpretation, the same measurable magnetic energy is organized through current-supported magnetic chains assembled into fibrils, flux tubes, coronal loops, and flux ropes.

The plasma contained in these structures is guided and confined by the magnetic organization. Solar fibrils contain plasma composed principally of hydrogen and helium, along with electrons and trace heavier ions. In cooler chromospheric structures, a portion of the material may remain neutral or partially ionized.

The energy released during a flare remains magnetic-field energy. Radiation, accelerated particles, plasma heating, outflow, and eruptive motion are treated as products and carriers of released magnetic energy rather than as its original source.

3.4.2 Current Sheets and Reconnection

In the conventional reconnection picture, magnetic structures of different orientations are driven together and form a localized region of concentrated current. Outside the nonideal region, magnetic field and conducting plasma remain approximately coupled. Within the reconnection region, magnetic connectivity changes, and plasma is accelerated away from the interaction region.

Figure 3.4 shows the conventional Sweet-Parker geometry.

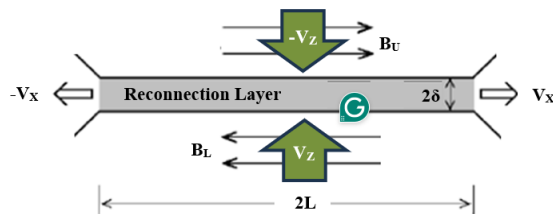


Figure 3.4: The Sweet-Parker Model

The Sweet-Parker model provides a foundational resistive MHD description. Oppositely di-

rected magnetic fields converge toward a long, thin current sheet. Plasma enters through the vertical inflow boundaries with velocity V_z , magnetic connectivity changes in the central region, and plasma leaves through the horizontal outflow regions with velocity V_x .

In highly conducting solar plasma, the Lundquist number can be very large, making steady resistive reconnection too slow by itself to explain many explosive flare timescales. This limitation has motivated Petschek-type, Hall, collisionless, kinetic, plasmoid-mediated, turbulent, and other rapid reconnection models.

Magnetic breakdown is not intended to replace those models. The narrower question is what threshold first causes a gradually stressed magnetic system to enter rapid energy release.

3.4.3 Precursors and Earliest Release

Solar flares frequently exhibit an interval of magnetic loading and localized precursor activity before the strongest impulsive emission and large-scale reconfiguration.

Potential indicators include localized brightening, extreme-ultraviolet and X-ray emission, increasing magnetic shear, flux emergence or cancellation, converging magnetic structures, current concentration, fibril interaction, loop deformation, localized heating, and early particle acceleration.

No single precursor identifies a universal mechanism. The important variables are temporal ordering and spatial coincidence.

The proposed sequence is

- magnetic loading → precursor change
 - initial release
 - current-sheet growth
 - large-scale reconnection.
- (3.14)

Observations have associated flare onset with strongly sheared magnetic regions and have shown reconnection signatures beginning locally before expanding over a larger system [Hagyard and Smith \(1988\)](#); [Moore et al. \(2011\)](#).

Early EUV and soft X-ray activity during filament eruption provides an additional observable precursor [Sterling et al. \(2007\)](#).

The sequence does not require current sheets to form only after the first energy release. Current concentration and current-sheet thinning may already be developing. The question is whether the earliest rapid energy conversion can be explained entirely by the already identified reconnection process or whether an additional local structural threshold is required.

3.4.4 The Proposed Incipient Mechanism

Conventional reconnection theory describes magnetic topology change, flux transport, current-sheet evolution, nonideal plasma behavior, and energy conversion with substantial explanatory power.

Magnetic field lines, however, are conventionally mathematical integral curves rather than physical material objects. The Photony interpretation, therefore, asks what proposed physical magnetic organization accumulates structural loading and what loses continuity at the first threshold crossing.

Magnetic breakdown is proposed as a possible answer.

The functional distinctions are
breakdown → threshold-driven energy release
reassignment → structural reorganization
reconnection → macroscopic connectivity change

These processes may overlap. Smooth reassignment may occur without destructive breakdown. Localized breakdown may occur without immediate large-scale reconnection. During an explosive flare, continuing breakdown may coexist with reassignment and reconnection.

3.5 Magnetic Breakdown

Magnetic breakdown is defined here as the localized, threshold-driven loss of structural continuity within a pre-existing, current-supported magnetic organization.

The affected magnetic structure already exists in a conducting environment and stores mag-

netic energy before onset. Breakdown therefore does not create the first conducting path. It destabilizes magnetic chains, fibrils, or larger structures that have become unable to accommodate additional loading or to reassign continuously into stable pathways.

This definition distinguishes magnetic breakdown from controlled field reduction, non-destructive magnetic reassignment, and conventional magnetic reconnection.

The defining event is the first local loss of stable magnetic chain continuity.

3.5.1 Required Preconditions

The principal proposed preconditions are:

1. **Conducting environment.** The medium must contain sufficient mobile charge carriers to support current before rapid release begins.
2. **Pre-existing current-supported magnetic structure.** Magnetic chains, fibrils, flux tubes, coronal loops, flux ropes, current layers, or interacting magnetic systems must already be present.
3. **Stored magnetic free energy.** The magnetic configuration must depart from a lower-energy admissible state.
4. **Continued structural loading.** Energy or mechanical stress must continue to accumulate through processes such as flux emergence, shearing, twisting, convergence, interaction, compression, or stretching.
5. **Limited accommodation or reassignment capacity.** The structure must be unable to relieve sufficient loading through expansion, redistribution, deformation, or continuous reassignment.

No single condition is sufficient by itself. A strong field does not automatically imply breakdown, nor does the existence of a current sheet.

The sequence is

magnetic structure $\rightarrow$ loading
 $\rightarrow$ capacity limit (3.15)
 $\rightarrow$ breakdown.

loading that can be relieved through continuous reassignment. Then,

$$L_B < C_A \rightarrow \text{stable accommodation,}$$

$$C_A \leq L_B < C_R \rightarrow \text{reassignment,}$$

and

$$L_B \geq C_R \rightarrow \text{breakdown.}$$

These quantities are effective local capacities rather than universal constants.

A dimensionless structural utilization ratio may be defined as

$$\Xi_B = \frac{L_B}{C_R}. \quad (3.17)$$

The proposed interpretation is then

$$\Xi_B < 1 \rightarrow \text{continuity remains available,}$$

$$\Xi_B \geq 1 \rightarrow \text{breakdown becomes possible.}$$

The boundary $\Xi_B = 1$ is conceptual rather than an independently calibrated physical constant. Chain-level constitutive relations required to compute it from first principles have not yet been experimentally established.

3.5.4 Incipient and Continuing Breakdown

Magnetic breakdown is proposed as both an incipient and potentially continuing process.

The first localized fragmentation may initiate rapid release, but the event need not end with that first failure. Current redistribution, plasma motion, loss of confinement, and magnetic reorganization may transfer stress to neighboring fibrils or adjacent portions of the same structure.

A cascading sequence is therefore possible:

$$\begin{aligned} \text{local breakdown} &\rightarrow \text{energy redistribution} \\ &\rightarrow \text{adjacent loading} \\ &\rightarrow \text{additional thresholds.} \end{aligned} \quad (3.18)$$

Such a process could allow an initially localized release to develop into a flare-scale event

3.5.2 Structural Loading

Magnetic loading is multidimensional and cannot be reduced to field magnitude alone. Two regions with similar B may differ greatly in current density, shear, curvature, twist, compression, confinement, interaction, and ability to expand or reassign.

Therefore, a conceptual loading function is introduced as follows:

$$L_B = F \left(B, J, \kappa, \tau, C, S, I_f, \Gamma, \dot{E}_{\text{in}} \right), \quad (3.16)$$

where B is magnetic flux density, J is current density, κ represents curvature, τ represents twist, C represents compression or crowding, S represents stretching or shear, I_f represents interaction between neighboring fibrils or magnetic systems, Γ represents external confinement, and $\dot{E}_{\text{in}}$ represents continuing energy input.

Equation 3.16 is not a completed constitutive law. It identifies classes of variables that a future quantitative magnetic-breakdown criterion may need to include.

The loading variables can reinforce one another. A strongly curved fibril may remain stable when isolated but become unstable when compressed by a neighboring magnetic structure. A twisted flux rope may accommodate further energy through expansion until the overlying magnetic structure limits that motion. A current layer may thin while remaining coherent and then fragment when several loading variables act together.

3.5.3 Accommodation, Reassignment, and Breakdown

A stressed magnetic organization has three proposed responses: stable accommodation, non-destructive reassignment, and breakdown.

Let C_A represent the effective local capacity for stable accommodation and C_R the maximum

without requiring the entire active region to cross a single global threshold simultaneously.

Continuing breakdown should not be confused with macroscopic reconnection. Reconnection describes changing magnetic connectivity as the event develops. Continuing breakdown describes the proposed repeated local loss of magnetic chain continuity while that larger reorganization proceeds.

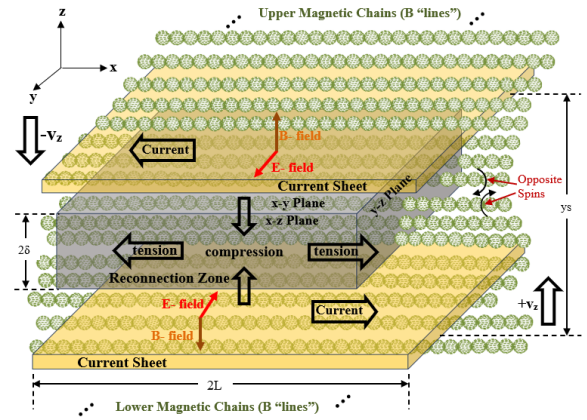


Figure 3.5: Chain-based interpretation of the Sweet-Parker reconnection geometry. Oppositely directed magnetic fibrils converge toward the central current-sheet region, where magnetic chain compression, reassignment, and possible localized fragmentation are proposed to occur. Plasma and magnetic structures enter through the inflow regions and leave through the opposing outflow regions.

3.5.5 Relationship to MHD Instabilities

Established flare and eruption models include kink instability, torus instability, tearing, plasmoid instability, current-sheet disruption, and transitions from slow to fast reconnection. Magnetic breakdown should not be treated simply as another name for any of these mechanisms.

An ideal MHD instability may alter the position, shape, or confinement of a flux rope while preserving the frozen-in condition during its early development. That instability may increase curvature, stretching, interaction, or current-sheet formation and thereby drive the structure toward magnetic breakdown without being identical to the breakdown event.

Figure 3.5 presents the Photony chain interpretation of the conventional Sweet-Parker geometry.

The figure does not alter the conservation laws, inflow and outflow relations, or diffusion-region geometry of the Sweet-Parker model. It adds the proposed Photony interpretation of the physical organization represented conventionally by magnetic field lines.

Tearing and plasmoid instability concern current-sheet instability and fragmentation within established plasma physics. Such behavior may provide a macroscopic or intermediate-scale manifestation of conditions under which magnetic breakdown is proposed to occur. It should not be declared equivalent to chain-level breakdown without observational or experimental evidence.

3.5.6 Localized Fragmentation

Magnetic breakdown is expected to begin locally rather than through the simultaneous failure of an entire magnetic system. Loading, current density, curvature, shear, plasma composition, and confinement vary spatially.

The first threshold crossing should therefore occur where magnetic loading is concentrated or where accommodation and reassignment capacity are locally reduced.

Within the Photony interpretation, local magnetic chain continuity is lost, and the affected structure fragments into elemental charge photons, classical photons, surviving chainlets, and larger chain sections.

Possible observable signatures include:

- localized electromagnetic emission,
- rapid localized heating,
- early nonthermal particle acceleration,
- abrupt plasma motion or a localized jet,
- rapid current-density or magnetic-field change,
- current-sheet thinning or fragmentation,
- expansion into a larger reconnection or eruptive system.

The proposed fragmentation products are summarized in Figure 3.6.

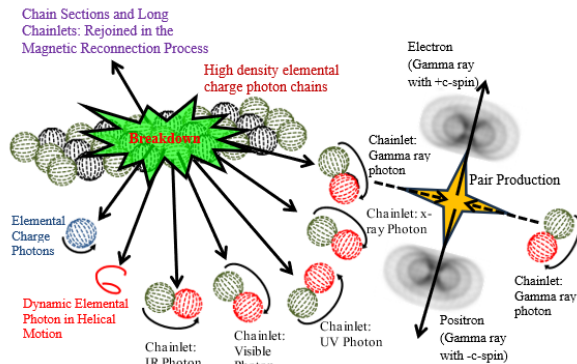


Figure 3.6: Schematic representation of the released products associated with magnetic chain collision or localized fragmentation. In the Photony interpretation, colliding or overloaded magnetic chains may release stored magnetic energy in the forms of elemental charge photons, classical photons, surviving chainlets, and larger chain sections. The emitted classical photons may span a broad spectrum, including visible light as well as infrared, ultraviolet, X-ray, and gamma-ray radiation. At observable scales, the release may also be associated with particle acceleration, localized heating, plasma motion, and subsequent magnetic reassignment or reconnection.

At observable scales, the released energy may therefore appear as radiation, accelerated particles, plasma heating, bulk plasma motion, current redistribution, and magnetic reconfiguration.

3.5.7 Operational Identification Criteria

For magnetic breakdown to function as a distinct scientific classification, it must be identifiable through operational criteria rather than inferred only from the final appearance of an event.

A candidate event should satisfy the following conditions:

1. A conducting, current-carrying magnetic structure exists before rapid energy release.
2. Magnetic free energy and structural loading are concentrated before onset, as indicated by quantities such as magnetic shear, curvature, twist, current density, compression, interaction, or confinement.
3. The earliest reproducible energy-release signature is spatially associated with the stressed magnetic structure.
4. Initial release begins as the system enters large-scale magnetic reassignment and reconnection and is detectable before the macroscopic reconfiguration is fully established.
5. The event does not require formation of the first conducting path through an initially insulating or strongly resistive region.
6. Subsequent evolution is consistent with transfer of pre-existing magnetic energy into radiation, particle acceleration, heating, plasma motion, current redistribution, and lower-energy magnetic configurations.

Failure to satisfy these conditions would weaken or exclude a magnetic-breakdown classification.

3.6 Limitations and Scope

The magnetic-breakdown framework remains a proposed interpretation rather than an experimentally established mechanism.

Its present application is centered primarily on solar flares because solar active regions contain the required combination of pre-existing current, magnetic free energy, magnetically organized plasma, strong structural loading, precursor activity, and rapid magnetic reconfiguration.

The framework may eventually be relevant to stellar flares, planetary magnetospheres, magnetotail current systems, laboratory plasmas, pulsed-power systems, and high-current conductors. Each environment would require its own treatment of geometry, conductivity, composition, confinement, loading, and diagnostics.

Magnetic chains and their internal fragmentation are not presently resolved directly by electromagnetic or plasma instruments. Evaluation must therefore begin with macroscopic measurements such as magnetic-field strength, current-density proxies, magnetic shear, fibril curvature, radiation, particle distributions, temperature, plasma motion, and the timing of current-sheet and reconnection development.

The magnetic-loading function and structural utilization ratio introduced in this chapter are likewise preliminary. They identify potentially important variables but do not yet constitute a calibrated predictive threshold law.

Laboratory testing would require a pre-existing current-supported magnetic structure subjected to controlled increasing loading, with synchronized measurement of current, magnetic geometry, radiation, particle distributions, temperature, and plasma motion. Particular attention should be directed toward determining whether a localized energy-release event can be resolved before the larger reconnection geometry is fully developed.

3.7 Discussion

The central distinction developed in this chapter is between initial magnetic structural failure,

subsequent magnetic reassignment, and macroscopic magnetic reconnection.

Conventional theory continues to quantify magnetic energy, current-sheet formation, non-ideal plasma behavior, magnetic-flux transfer, particle acceleration, heating, and plasma outflow. The Photony interpretation adds a proposed physical organization beneath the conventional field-line representation and identifies threshold-driven loss of that organization as a possible incipient release mechanism.

Magnetic loading may increase through current concentration, shear, curvature, compression, twist, interaction, stretching, and confinement. A magnetic structure may initially respond through deformation, expansion, current redistribution, or non-destructive reassignment. Breakdown becomes possible when these responses can no longer maintain local structural continuity.

The proposed sequence is therefore

magnetic loading $\rightarrow$ capacity limit
 $\rightarrow$ local breakdown
 $\rightarrow$ reconnection growth.

(3.19)

The current sheet need not be absent before initial release. Current concentration, sheet thinning, and limited reconnection may already be developing. The proposed distinction concerns the local threshold that changes the system from gradual evolution into rapid energy release.

Because loading varies throughout a magnetic system, the first threshold crossing is expected to be localized. Initial disruption can then alter neighboring currents, confinement, and magnetic stresses, potentially allowing an affected region to expand.

This framework gives distinct roles to three processes:

magnetic breakdown → initial structural failure
magnetic reassignment → reorganization
magnetic reconnection → connectivity change

The distinction is functional rather than necessarily temporal. The processes can occur in rapid succession or overlap substantially.

3.8 Conclusions

Magnetic breakdown is proposed as a localized, threshold-driven loss of continuity within a pre-existing, current-supported magnetic organization. In the Photony interpretation, magnetic chains assemble into fibrils and larger plasma-containing structures that already embody substantial stored magnetic energy.

Additional loading through current concentration, curvature, compression, twist, interaction, stretching, and confinement may drive such structures toward instability. Magnetic breakdown becomes possible when the local organization can no longer accommodate the loading or reassign continuously without fragmentation.

The resulting local failure is proposed to release part of the stored magnetic energy into electromagnetic radiation, particle acceleration, plasma heating and motion, current redistribution, and subsequent magnetic reconfiguration.

Magnetic breakdown is distinct from mag-

netic reassignment and magnetic reconnection. Breakdown identifies the proposed threshold-driven structural failure. Reassignment describes redirection of surviving chains and fibrils. Reconnection describes the larger macroscopic change in magnetic connectivity. These processes may overlap, but they need not represent the same physical stage.

The proposed mechanism is not yet experimentally confirmed. Its evaluation requires time-resolved observations or controlled experiments capable of showing that localized energy release begins within a highly loaded, pre-existing magnetic structure as the system enters reassignment and reconnection and before the full macroscopic reconfiguration is complete.

Conventional electromagnetic, plasma, MHD, and reconnection theories remain the quantitative foundation. The proposed Photony framework provides an additional structural interpretation and a testable hypothesis concerning the earliest threshold-driven release of stored magnetic energy.

Chapter 4

A Unified Framework for Electric and Magnetic Energy Release

Purpose: To unify electric and magnetic energy-release regimes within a common threshold framework and identify the physical conditions that determine whether stored electromagnetic energy is released through electric breakdown, magnetic reassignment, or magnetic breakdown.

4.1 Introduction

Chapters 2 and 3 examined electric and magnetic energy-release regimes separately. Electric breakdown was associated with the rapid development or substantial enlargement of a conducting pathway through an initially insulating, resistive, or incompletely conducting region. Magnetic breakdown, by contrast, was associated with loss of stability within a pre-existing, current-supported magnetic organization in an already conducting environment. Although the two regimes begin from different physical conditions and produce different structural responses, both can be treated within a broader framework of electromagnetic loading, instability, threshold crossing, and energy release.

The purpose of the present chapter is to develop that unified framework. The central proposition is that electric breakdown, magnetic reassignment, and magnetic breakdown can be represented as distinct post-threshold branches of a common instability process without treating the three phenomena as physically identical. The commonality lies in the progression toward loss of stability. The distinction lies in the physical environment, the organization carrying the stored energy, and the post-threshold pathway available to the system.

Conventional electromagnetics remains the

quantitative foundation of this description. Maxwell's equations, electric and magnetic energy densities, conductivity, current density, electromagnetic energy flow, material properties, plasma parameters, and established breakdown and reconnection models continue to determine measurable behavior. The Photony framework is introduced as a proposed structural interpretation that distinguishes electric chain and magnetic chain organization, electron-mediated transduction, structural loading, reassignment, and loss of continuity.

A useful first distinction is between *threshold crossing* and *regime selection*. Reaching an instability threshold identifies that the existing electromagnetic organization can no longer remain in its previous stable state. It does not, by itself, determine what physical process follows. A weakly conducting gas, dielectric, or resistive region may reach instability while substantial charge separation and electric loading are present. A highly conducting plasma may reach instability while substantial current and an extended magnetic organization already exist. The threshold identifies loss of stability, while the surrounding environment helps determine the accessible post-threshold response.

Within the unified framework, the effective instability condition is represented conceptually by

$$\Theta(\mathbf{r}, n) \geq \Theta_{\text{crit}}, \quad (4.1)$$

where $\Theta(\mathbf{r}, n)$ represents an effective local loading or instability functional and Θ_{crit} represents the corresponding critical stability boundary. This relation is not introduced as a universal constitutive law. It provides a compact representation of the condition that the local electromagnetic organization has reached or exceeded its admissible stability limit.

The threshold condition is common to the regimes considered in this chapter, but their post-threshold evolution differs. Environmental support must therefore be treated separately from the threshold itself. The effective parameter Σ is used to represent the combined environmental conditions that influence whether the unstable system follows an electric or magnetic branch. These conditions include conductivity, mobile-electron support, structural density, confinement, geometry, transduction capacity, and the ability of the local system to sustain an organized current-supported magnetic structure.

For an unstable state, conditions are represented schematically by

$$\Sigma < \Sigma_{\text{crit}}$$

favor the electric-breakdown branch. In this regime, substantial current cannot yet be supported throughout the affected region, and the principal post-threshold transition is the formation or enlargement of a conducting pathway.

Conditions represented by

$$\Sigma \geq \Sigma_{\text{crit}}$$

select a magnetic post-threshold regime. In this case, current and magnetic organization are already established before instability. A second selection must then determine whether the magnetic organization can continue through structural reassignment or whether continuity is lost.

This second magnetic-branch decision is represented by the admissibility indicator A . If an admissible post-threshold continuation remains available, the system may undergo magnetic reassignment. If no such continuation remains

available, magnetic breakdown becomes the proposed failure branch. The three principal outcomes are therefore

electric breakdown,
magnetic reassignment,
magnetic breakdown.

The distinction between reassignment and breakdown is especially important. Both occur after the magnetic regime has been selected, and both may follow the same general instability threshold. Magnetic reassignment preserves sufficient structural continuity for the magnetic organization to move into a new admissible configuration. Magnetic breakdown represents the contrasting condition in which the loaded magnetic structure can no longer maintain continuity through such reorganization.

The framework consequently separates the electromagnetic event into three logical stages:

$$\begin{aligned} \text{loading} &\rightarrow \text{threshold crossing} \\ &\rightarrow \text{branch selection} \\ &\rightarrow \text{energy release.} \end{aligned} \quad (4.2)$$

This separation is important because similar observable products can arise from different initiating mechanisms. Radiation, heating, current change, particle acceleration, plasma motion, and structural reconfiguration do not, by themselves, identify whether an event began as electric breakdown, magnetic reassignment, or magnetic breakdown. Classification should instead begin with the pre-threshold state and follow the temporal sequence through the first instability and subsequent branch selection.

The framework also distinguishes *regime initiation* from later electromagnetic coupling. An event may begin through electric breakdown and then develop substantial current, magnetic organization, heating, and radiation after a conductive path has formed. A source-fed arc is an important example. The later presence of strong magnetic effects does not change the fact that the initiating threshold event occurred through the electric branch. Conversely, a magnetic event can generate localized electric fields during current redistribution or reconnection without therefore becoming a voltage-breakdown event.

All three post-threshold branches share the same basic energetic requirement. The released energy is the difference between the relevant pre-event and post-event electromagnetic energy states,

$$\Delta E = E_{\text{pre}} - E_{\text{post}}. \quad (4.3)$$

The branches, therefore, differ principally in how the energy was organized before threshold crossing, what environmental support was available, what structure became unstable, and what post-threshold continuation remained physically accessible. The unified framework developed in this chapter provides a common language for comparing those regimes while preserving their important physical differences.

Terminology: electromagnetic energy release, electromagnetic loading, instability threshold, threshold crossing, critical threshold, Θ , Θ_{crit} , environmental selection, Σ , Σ_{crit} , regime selection, post-threshold branch, electric breakdown, magnetic reassignment, magnetic breakdown, admissible continuation, admissibility indicator, A , electric chains, magnetic chains, magnetic fibrils, static elemental charge photons, dynamic elemental charge photons, electron transduction, surface-electron transduction, internal-electron transduction, conductivity, current support, structural continuity, structural loading, conducting pathway, magnetic reconfiguration, magnetic reconnection, stored electromagnetic energy, electric-field energy, magnetic-field energy, energy release

4.2 Electric and Magnetic Field Organization

The unified threshold framework requires a physical distinction between electric and magnetic structures before a common instability condition can be applied. Chapters 1 through 3 developed these structures in detail. The present section therefore retains only the aspects needed to establish the branch-selection framework.

Photony theory is used here as a complementary structural interpretation of electric and magnetic field organization. Maxwell's equations, electromagnetic energy density, Poynting

energy flow, dielectric-breakdown theory, magnetohydrodynamics, and plasma diagnostics remain the principal quantitative descriptions of the systems considered [Jackson \(1998a\)](#); [Griffiths \(2017\)](#); [Poynting \(1884\)](#).

Within the proposed Photony framework, conventional field-line representations correspond to organized electric chain and magnetic chain structures. This interpretation allows the corresponding electromagnetic organization to be discussed in terms of continuity, density, curvature, loading, confinement, electron transduction, reassignment, and breakdown.

Electric chain and magnetic chain organization arise through different electron-mediated transduction pathways. Electric chains are associated primarily with surface or interface electrons and static elemental charge photons. Magnetic chains are associated primarily with internal free electrons and dynamic elemental charge photons. Because the two structures are supported differently, they also approach instability under different physical conditions.

Electric chains are most naturally associated with regions in which charge separation can persist. Their continuity depends strongly on surface support, interface conditions, separation geometry, electric loading, and the ability of surface-associated electrons to sustain the organized structure. As loading increases, a local limit can be reached at which that organization can no longer be maintained.

Magnetic chains have a different topology and support structure. They form closed loops and are sustained by distributed internal-electron transduction in conducting media. Their stability therefore depends on collective variables such as current support, chain density, curvature, twist, compression, confinement, and interaction with neighboring magnetic structures.

When a loaded magnetic chain organization approaches instability, it may still preserve continuity by moving into another admissible configuration. If that continuation remains available, magnetic reassignment can occur. If no admissible continuation remains, magnetic breakdown becomes possible.

4.2.1 Electron Transduction and the Two Pathways

Within the proposed Photony framework, electrons act as transducers that organize elemental charge photons into extended structures. The resulting organization depends on the electron population, the local environment, and whether the participating elemental charge photons are predominantly static or dynamic.

The electric pathway can be represented schematically as

$$\Phi_{\text{stat}} \xrightarrow{T_{\text{surf}}} C_E, \quad (4.4)$$

where Φ_{stat} represents the available static elemental charge-photon population, T_{surf} represents surface or interface electron transduction, and C_E represents electric chain organization.

The corresponding magnetic pathway is

$$\Phi_{\text{dyn}} \xrightarrow{T_{\text{int}}} C_B, \quad (4.5)$$

where Φ_{dyn} represents dynamic elemental charge photons, T_{int} represents internal-electron transduction, and C_B represents magnetic chain organization.

These expressions are structural representations rather than conventional constitutive equations.

In the electric pathway, transduction is concentrated primarily at surfaces or interfaces where charge separation is maintained. In the magnetic pathway, a distributed population of internal electrons can support current-associated transduction throughout a conducting volume.

This difference helps explain the characteristic spatial scales of the two regimes. Electric instability may remain strongly localized around a surface, interface, dielectric region, or gap. Magnetic instability can involve an extended current-supported structure because its electron support is distributed throughout a conducting region.

Transduction capacity is also relevant to both regimes. It represents the ability of the available electron population to form, maintain, redirect, or reorganize the required elemental charge-photon structures under the prevailing conditions. Within the proposed framework, this capacity is limited by the local availability of el-

emental charge photons in the proximity of the transducing electrons.

As transduction demand approaches the locally available supply, transduction saturation may be approached. The system can then become less able to maintain or reorganize the required chain structure. This transduction limitation contributes to structural loading but does not by itself determine which post-threshold regime will occur.

4.2.2 Topology and Continuity

The distinction between electric and magnetic organization is also topological. In conventional electromagnetism,

$$\nabla \cdot \mathbf{B} = 0, \quad (4.6)$$

which expresses the absence of isolated magnetic charge and the continuous character of magnetic flux.

Electric fields, by contrast, can originate or terminate on electric charge. Within the Photony interpretation, the corresponding structural distinction is that electric chains may terminate at charge-supporting surfaces or localized charge concentrations, whereas magnetic chains form closed loops.

The two continuity requirements can therefore be summarized as

$$\begin{aligned} C_E &: \text{local continuity,} \\ C_B &: \text{collective continuity.} \end{aligned} \quad (4.7)$$

This is a structural classification rather than a new field equation.

Electric loading is strongly influenced by charge separation, surface conditions, electric chain density, and the geometry of the region between charge-supporting boundaries. Magnetic loading is distributed through an already current-supported structure and may increase through chain density, curvature, twist, compression, confinement, and interaction.

These differences establish why the same general condition of instability does not require the same physical outcome.

4.3 Electromagnetic Energy Storage and Loading

Electric breakdown, magnetic reassignment, and magnetic breakdown all require an available energy reservoir. The magnitude of stored energy, however, does not by itself select the release regime.

Conventional electromagnetic theory provides the common quantitative starting point. For a linear medium, electric energy density is

$$u_E = \frac{1}{2} \epsilon E^2, \quad (4.8)$$

and magnetic energy density is

$$u_B = \frac{B^2}{2\mu}. \quad (4.9)$$

When both contributions are present,

$$u_{EM} = \frac{1}{2} \epsilon E^2 + \frac{B^2}{2\mu}. \quad (4.10)$$

These relations remain the measurable macroscopic description of electromagnetic energy [Jackson \(1998a\)](#); [Griffiths \(2017\)](#).

The distinction between electric and magnetic release regimes cannot be established merely by comparing u_E and u_B . Most electromagnetic systems contain both electric and magnetic energy, and their relative magnitudes can change with time.

Energy density answers how much electromagnetic energy is stored locally. Regime selection instead depends on how that energy is organized, what physical structure supports it, what form of loading is developing, and what continuation remains available when instability occurs.

For a finite volume V , the total stored electromagnetic energy is

$$U_{EM} = \int_V u_{EM} dV. \quad (4.11)$$

This volume dependence is important when comparing phenomena of very different spatial scale. A localized electrical discharge and an extended magnetically organized plasma can both contain strongly loaded regions while differing enormously in total stored energy.

Within the proposed Photony interpretation, the same electromagnetic reservoir can be represented structurally by

$$u_{ch} = \epsilon_E n_E + \epsilon_B n_B, \quad (4.12)$$

where n_E and n_B represent effective local densities of electric and magnetic chains and ϵ_E and ϵ_B represent their corresponding mean energies per chain segment [Martin \(2026b\)](#).

Equation 4.12 does not introduce an additional energy reservoir. It is the proposed chain-level representation of the electromagnetic energy already described by Equation 4.10.

No one-to-one calibrated relationship between field magnitude and exact chain count is presently assumed. Conventional values of E , B , stored energy, current, and material parameters remain the measurable quantities.

4.3.1 Electric Loading

Electric loading is most readily understood where charge separation is maintained. As additional charge accumulates, the electric field can increase and the stored electric energy density rises as E^2 .

Within the proposed structural interpretation, this condition corresponds to increasing electric chain organization and increasing demand on the surface or interface electrons that maintain it.

A large value of u_E does not itself constitute breakdown. The electric system becomes unstable only when the loaded organization reaches a local condition that can no longer be maintained.

The familiar capacitor relation,

$$U_E = \frac{1}{2} CV^2, \quad (4.13)$$

illustrates that substantial energy may be accumulated without conduction through the dielectric or gap. The later formation of a conducting pathway is a separate threshold-dependent event.

4.3.2 Magnetic Loading

Magnetic loading begins from an already conducting state. Sustained current supports magnetic organization and allows magnetic energy

to increase without requiring the formation of a new conductive pathway.

For a simple inductive system,

$$U_B = \frac{1}{2}LI^2. \quad (4.14)$$

The corresponding field energy can be expressed as

$$U_B = \int_V \frac{B^2}{2\mu} dV. \quad (4.15)$$

Within the proposed Photony interpretation, increasing current and magnetic field strength are associated with increasing magnetic chain organization. Structural loading, however, also depends on geometry. Curvature, twist, compression, crowding, confinement, and interaction between neighboring magnetic structures can increase even while the magnetic organization remains coherent.

Magnetic stability therefore cannot be inferred from B alone. The current-supported organization must be considered as a loaded structure.

4.4 Electromagnetic Energy Transport

Stored electromagnetic energy contributes to instability only through the complete energy balance of the system. Energy can be supplied to a loaded region, redistributed within it, removed from it, or converted locally into other forms.

The conventional quantity describing electromagnetic energy transport is the Poynting vector,

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (4.16)$$

For a linear isotropic medium,

$$\mathbf{S} = \frac{1}{\mu} \mathbf{E} \times \mathbf{B}, \quad (4.17)$$

and in free space,

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (4.18)$$

The electromagnetic power crossing a surface A is

$$P_{EM} = \int_A \mathbf{S} \cdot d\mathbf{A}. \quad (4.19)$$

These standard relations connect stored energy with the regions in which loading, transport, conversion, and release occur [Poynting \(1884\)](#); [Jackson \(1998a\)](#); [Griffiths \(2017\)](#).

Local energy conservation is expressed by Poynting's theorem,

$$\frac{\partial u_{EM}}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (4.20)$$

This relation separates three important processes: change of stored electromagnetic energy, net energy transport through the region, and local transfer of electromagnetic energy to charged matter.

If

$$\nabla \cdot \mathbf{S} = 0, \quad (4.21)$$

there is no net Poynting-flux imbalance at that instant. This does not require $\mathbf{S}$ to vanish. Energy may flow continuously through the region while inflow and outflow remain balanced.

When

$$\nabla \cdot \mathbf{S} \neq 0, \quad (4.22)$$

there is a net local electromagnetic energy-flow imbalance. Through Equation 4.20, this must be associated with change in stored electromagnetic energy, transfer to matter, or both.

In an electric system approaching breakdown, electromagnetic energy can be delivered into a localized stressed region while charge separation and electric-field energy continue to increase. After breakdown begins, energy can be redirected rapidly into conduction, ionization, heating, radiation, and mechanical processes.

A magnetic system can be loaded over a larger volume and for a longer interval. Poynting transport can continue to supply an established current-supported magnetic structure while its field strength, current, curvature, compression, confinement, and interactions evolve. Once instability occurs, energy transport can be redirected as magnetic reassignment or breakdown develops.

The essential conclusion is that neither stored energy nor Poynting flux alone establishes instability. The threshold problem depends on whether continued storage and transport drive the organized system toward its support limit.

4.5 Conductivity and the Approach to Instability

Conductivity provides an important macroscopic connection between electromagnetic loading and the form of organization that can persist.

For a simple linear conductor,

$$\mathbf{J} = \sigma \mathbf{E}, \quad (4.23)$$

where σ is electrical conductivity.

For a homogeneous linear medium, charge redistribution can be characterized by the relaxation time

$$\tau_r = \frac{\varepsilon}{\sigma}. \quad (4.24)$$

When conductivity is relatively low, τ_r is comparatively long and separated charge can persist. Electric loading can therefore develop before a large conduction current forms.

When conductivity is high, charge imbalance is redistributed more rapidly and distributed current can be supported. This favors the development and maintenance of current-supported magnetic organization.

The limiting tendencies can be stated simply as

$$\begin{aligned} \text{low } \sigma &\rightarrow \text{electric loading,} \\ \text{high } \sigma &\rightarrow \text{magnetic loading.} \end{aligned}$$

These are regime tendencies, not instability conditions. Low conductivity does not itself cause electric breakdown, and high conductivity does not itself cause magnetic instability.

This distinction is important. Conductivity influences the environment in which the system is loaded. A separate criterion is required to determine whether the loaded configuration has actually reached instability.

4.6 The Common Instability Threshold

The unified framework separates the accumulation of electromagnetic loading from the threshold at which the existing organization can no longer remain stably admissible.

The effective loading functional is written

$$\Theta(\mathbf{r}, n), \quad (4.25)$$

where $\mathbf{r}$ represents spatial position and n identifies the evolutionary state of the system.

The stable regime is represented by

$$\Theta < \Theta_{\text{crit}}, \quad (4.26)$$

the critical boundary by

$$\Theta = \Theta_{\text{crit}}, \quad (4.27)$$

and the post-threshold condition by

$$\Theta > \Theta_{\text{crit}}. \quad (4.28)$$

The quantity Θ is an effective normalized loading functional rather than a new independently measurable physical variable. Its purpose is to combine the principal contributions that move the local system toward loss of stability [Martin \(2026b\)](#).

A schematic decomposition is

$$\Theta = \Theta_{\tau} + \Theta_{\kappa} + \Theta_c + \Theta_{\text{tr}} + \Theta_{\sigma}, \quad (4.29)$$

where Θ_{τ} represents tension-related loading, Θ_{κ} curvature loading, Θ_c compression, Θ_{tr} transduction demand, and Θ_{σ} environmental support associated with conductivity.

These terms are normalized contributions. Equation 4.29 does not imply that tension, curvature, compression, transduction demand, and conductivity are dimensionally equivalent raw physical quantities.

The value of this formulation is that instability need not result from one variable reaching an isolated limiting value. Several moderate contributions may act together to drive the total loading state toward the critical boundary.

For example, a magnetic structure may remain below its limiting chain density while increasing curvature, compression, confinement, and transduction demand collectively cause instability. Likewise, an electric structure can approach breakdown through a combination of field strength, electric chain density, surface-transduction demand, and unfavorable gap geometry.

The threshold should therefore be regarded as a boundary in a multidimensional loading

space rather than as a universal critical value of E , B , J , or σ .

The unified threshold condition is

$$\Theta(\mathbf{r}, n) \geq \Theta_{\text{crit}}. \quad (4.30)$$

This condition states that the previous configuration has reached or exceeded its admissible stability limit. It deliberately does not specify the physical post-threshold outcome.

That outcome requires a second stage of analysis.

4.7 Environmental Regime Selection

Once the common instability threshold (Θ_{crit}) has been crossed, the next question is which post-threshold environment the system can support.

A weakly conducting gas or dielectric may reach instability after substantial charge separation and localized electric loading. A highly conducting plasma may reach instability with substantial distributed current and extensive magnetic organization already present.

The effective environmental-selection parameter is represented by

$$\Sigma(\mathbf{r}, n). \quad (4.31)$$

The parameter Σ is not proposed as a new fundamental quantity comparable to conductivity, electron density, electric field, or magnetic field. It is an effective selection parameter used to collect the principal environmental conditions influencing whether an unstable state is supported predominantly through an electric or magnetic post-threshold regime.

In schematic form,

$$\Sigma = \Sigma(\sigma, n_e, \rho_c, \kappa, \Gamma, C, \dots), \quad (4.32)$$

where σ is conductivity, n_e represents mobile-electron support, ρ_c represents effective structural density, κ represents curvature, Γ represents relevant transduction or source-loss behavior, and C represents confinement or structural support.

Equation 4.32 is intentionally schematic. The variables need not contribute linearly or independently, and a calibrated functional form remains to be developed for particular physical systems.

The distinction between Θ and Σ is fundamental:

$$\begin{aligned} \Theta &\rightarrow \text{stability state,} \\ \Sigma &\rightarrow \text{regime support.} \end{aligned}$$

Thus, if

$$\Theta < \Theta_{\text{crit}}, \quad (4.33)$$

there is no need to select a post-threshold branch.

Only when

$$\Theta \geq \Theta_{\text{crit}} \quad (4.34)$$

does Σ become relevant to the subsequent evolution.

This separation prevents conductivity from being used as a substitute for the instability threshold. A highly conducting plasma need not be unstable, and a weakly conducting dielectric need not be near breakdown.

4.7.1 Electric Branch Selection

The electric branch begins from a pre-threshold state in which charge separation remains important and distributed current support is limited.

Such conditions occur in gases, dielectric materials, interfaces, surface regions, and incompletely conducting pathways. Electric loading is typically concentrated around a region across which substantial electric stress is maintained.

Once instability occurs, the characteristic transition is the formation or rapid enlargement of a conducting pathway. This includes dielectric breakdown, sparks, arcs, electrostatic discharge, and lightning, as developed in Chapter 2.

Let Σ_{crit} represent the effective boundary separating predominantly electric from predominantly magnetic environmental support. The electric branch can then be represented compactly as

$$\begin{aligned} \Theta &\geq \Theta_{\text{crit}}, \\ \Sigma &< \Sigma_{\text{crit}}, \end{aligned} \quad (4.35)$$

which selects the electric-breakdown regime.

The parameter Σ_{crit} should not be interpreted as a universal conductivity value. It represents a combined environmental boundary involving conductivity, electron support, geometry, confinement, and related conditions.

4.7.2 Magnetic Branch Selection

The magnetic branch begins from a fundamentally different state. The medium is already sufficiently conducting, substantial current is already present, and extended magnetic organization is already established.

The corresponding selection condition is

$$\begin{aligned}\Theta &\geq \Theta_{\text{crit}}, \\ \Sigma &\geq \Sigma_{\text{crit}}.\end{aligned}\tag{4.36}$$

This selects the magnetic threshold regime, but it does not yet distinguish between reassignment and breakdown.

High conductivity permits the magnetic organization to exist, but it does not determine its post-threshold fate. An additional structural question must be answered: can the unstable magnetic configuration move into another admissible state while preserving sufficient continuity?

4.7.3 Magnetic Admissibility

Let A represent the effective availability of an admissible post-threshold magnetic continuation.

If

$$A = 1,\tag{4.37}$$

an admissible continuation remains available.

The magnetic organization can therefore undergo reassignment while preserving sufficient structural continuity.

If

$$A = 0,\tag{4.38}$$

no such continuation remains available and magnetic breakdown becomes the failure branch.

Thus, for a magnetic threshold state,

$$A = 1 \implies \text{reassignment},\tag{4.39}$$

whereas

$$A = 0 \implies \text{magnetic breakdown}.\tag{4.40}$$

The indicator A is branch logic rather than a new fundamental physical variable. In a real system, admissibility depends on geometry, confinement, structural density, curvature, electron support, transduction capacity, and the availability of the elemental charge photons required for the proposed reorganization.

Magnetic reassignment and magnetic breakdown therefore do not require different initial instability thresholds. Both begin after the common threshold has been crossed. Their distinction lies in whether structural continuity can still be preserved through reorganization.

4.8 Unified Post-Threshold Branch Structure

The preceding sections divide electromagnetic evolution into three stages: loading, threshold crossing, and post-threshold branch selection.

All three post-threshold branches begin from

$$\Theta \geq \Theta_{\text{crit}}.\tag{4.41}$$

The environmental parameter Σ then selects the electric or magnetic regime. If the magnetic regime is selected, A determines whether continuity can be retained through reassignment.

For the narrow book column, the complete branch logic is most clearly written as three separate conditions:

$$\Sigma < \Sigma_{\text{crit}} \Rightarrow \text{electric breakdown},\tag{4.42}$$

$$\Sigma \geq \Sigma_{\text{crit}}, A = 1 \Rightarrow \text{reassignment},\tag{4.43}$$

and

$$\Sigma \geq \Sigma_{\text{crit}}, A = 0 \Rightarrow \text{magnetic breakdown}.\tag{4.44}$$

Equations 4.42 through 4.44 apply after Equation 4.41 has already been satisfied.

This is the central branch structure of the unified framework.

The electric branch represents localized failure of an electric chain-supported configuration and the development of a new or substantially enhanced conducting pathway.

The magnetic-reassignment branch preserves an admissible continuation by changing the organization or connectivity of the existing magnetic structure.

The magnetic-breakdown branch occurs when such continuation is no longer available and collective magnetic support is lost.

The three outcomes are therefore not physically identical. What they share is a common logical sequence: loading increases, the existing structure crosses a stability boundary, the environmental conditions select an accessible branch, and electromagnetic energy is released or redistributed.

4.8.1 Common Energy-Release Relation

Although the post-threshold branches differ structurally, each involves a change in the electromagnetic energy associated with the pre-event configuration.

Let E_{pre} denote the relevant organized electromagnetic energy immediately before the transition and E_{post} the corresponding energy remaining afterward. Then,

$$\Delta E = E_{\text{pre}} - E_{\text{post}}. \quad (4.45)$$

Equation 4.45 is the common energetic statement linking the three branches.

It does not determine the form into which the energy is converted. Depending on the regime, the released or redistributed energy may appear as current, ionization, heating, electromagnetic radiation, particle acceleration, mechanical motion, plasma motion, magnetic restructuring, or another organized electromagnetic state.

The spatial character also differs. Electric breakdown is commonly concentrated around a discharge path or localized failure region. Magnetic reassignment may redistribute energy through an extended current-supported structure while preserving magnetic continuity. Magnetic breakdown may involve a broader loss of magnetic organization where such continuity can no longer be maintained.

Poynting energy transport provides the conventional macroscopic connection between these transitions and the surrounding electromagnetic system. Before threshold crossing, Poynting flux can supply the loaded region. During and after the transition, $\mathbf{S}$ can change in magnitude

and direction as energy is transported away or transferred locally through $\mathbf{J} \cdot \mathbf{E}$.

The complete physical progression may therefore be summarized as

$$\begin{aligned} \text{energy input} &\rightarrow \text{loading} \\ &\rightarrow \text{threshold} \\ &\rightarrow \text{branch} \\ &\rightarrow \Delta E. \end{aligned} \quad (4.46)$$

This compact sequence separates quantities that are often combined when an event is described only after it has occurred. Stored energy defines the reservoir. Θ identifies loss of stability. Σ characterizes environmental support. A distinguishes magnetic continuation from magnetic failure. The selected branch determines how the energy is released or redistributed.

4.9 Observable Signatures and Testability

The unified framework is useful only if its proposed structural variables can be related to observable quantities.

The quantities Θ , Σ , and A are not presently measured directly as fundamental physical observables. Their proposed effects must therefore be compared with conventional measurements of electric and magnetic field strength, voltage, current density, conductivity, carrier density, plasma properties, geometry, radiation, and energy-release timing.

The central prediction is that the post-threshold branch should be identifiable from the pre-event state, rather than only from the event observed afterward.

4.9.1 Testing the Electric Branch

Electric breakdown provides a comparatively direct laboratory test.

Relevant pre-event measurements include electric-field strength, applied voltage, gap spacing, permittivity, conductivity, current, surface or space charge, electrode geometry, and the temporal approach to discharge.

The framework predicts that the electric branch should remain strongly associated with localized geometry and the development of a conductive pathway. Increasing electric loading should occur while distributed current support remains limited. Threshold crossing should then be followed by a rapid local increase in conductivity.

Observable signatures can include abrupt current increase, field redistribution, ionization, heating, optical emission, channel development, surface flashover, spark formation, or arc development.

Systematic variation of gap spacing, material permittivity, conductivity, surface condition, electrode geometry, and applied field can therefore test whether the pre-event state follows the predicted localized electric branch.

4.9.2 Testing the Magnetic Branch

The magnetic regime requires a different set of measurements.

Relevant quantities include magnetic-field strength, current density, electron or plasma density, magnetic shear, fibril or flux-tube geometry, curvature, twist, compression, confinement, interaction, and the rate of electromagnetic energy input.

Within the proposed framework, increasing magnetic loading should be indicated not only by increased stored magnetic energy but also by changes in the geometry and support of the current-carrying structure.

For solar and laboratory plasma applications, possible measurable proxies include increasing magnetic shear or twist, changes in loop curvature, current concentration, interaction between magnetic structures, precursor heating, changes in spectral emission, particle acceleration, and changes in the timing or direction of Poynting transport.

A magnetic event should develop within an already conducting and magnetically organized region rather than through the creation of the first conductive pathway.

4.9.3 Testing Reassignment Against Breakdown

The distinction between magnetic reassignment and magnetic breakdown provides a more demanding test.

If the unstable magnetic organization can move into an admissible continuation, the event should exhibit reorganization and redistribution of energy while sufficient coherent magnetic support remains.

If no admissible continuation exists, the breakdown branch predicts a more fundamental loss of structural continuity and associated release or redistribution of stored magnetic energy.

The critical observational question is therefore not simply whether magnetic connectivity changes. The relevant issue is whether the pre-event state and earliest rapid release indicate that the structure retained an admissible reorganization pathway or lost the ability to maintain continuity.

Temporal resolution is especially important. The general test sequence is

$$\begin{aligned} \text{loading} &\rightarrow \text{precursor change} \\ &\rightarrow \text{threshold} \\ &\rightarrow \text{branch-specific release.} \end{aligned} \tag{4.47}$$

A strong evaluation therefore requires time-resolved measurements capable of separating pre-threshold loading from the impulsive release.

Poynting flux provides a common measurable bridge between the electric and magnetic regimes. Measurements or reconstructions of E , B , and associated energy transport can help determine whether electromagnetic energy is entering, leaving, or being redistributed within the region approaching instability.

The strongest validation of the proposed framework would occur if pre-event measurements consistently separate into the branch structure defined above. Conversely, repeated failure of measured support conditions to predict the resulting branch would require revision of the framework.

4.10 Limitations and Discussion

The present framework is a structural synthesis rather than a completed constitutive theory.

Maxwell's equations, conventional electromagnetic energy relations, conductivity, Poynting energy flow, dielectric-breakdown theory, magnetohydrodynamics, plasma physics, and established reconnection models remain the quantitative and experimental foundation.

The Photony contribution is the proposed interpretation of electric and magnetic organization in terms of chains, electron transduction, structural loading, threshold crossing, environmental selection, reassignment, and breakdown.

The principal quantitative limitation is that neither $\Theta(\mathbf{r}, n)$ nor $\Sigma(\mathbf{r}, n)$ has yet been reduced to a unique experimentally calibrated functional form. Their present definitions establish the logical structure of the model but do not provide universal numerical thresholds applicable across different materials, plasmas, geometries, and scales.

The same limitation applies to A . The binary notation provides a compact logical distinction between magnetic reassignment and magnetic breakdown, but real physical admissibility must eventually be determined from measurable geometry, electron support, confinement, magnetic structure, current, energy flow, and suitable proxies for the local transduction capacity.

Chain densities and transduction variables are also not directly measured by present electromagnetic instrumentation. Evaluation must therefore proceed initially through conventional quantities such as field strength, current density, conductivity, electron density, temperature, geometry, radiation, plasma motion, and Poynting energy transport.

The immediate theoretical task is to develop calibrated relationships between these observables and the normalized quantities represented by Θ and Σ . The experimental task is to obtain sufficiently time-resolved pre-event measurements to determine whether electric and magnetic events occupy distinguishable regions of the proposed loading and environmental-

support space.

The framework is therefore falsifiable. If measured pre-event conditions systematically fail to predict the branch distinctions proposed here, the corresponding assumptions must be modified or rejected.

4.11 Conclusions

This chapter develops a unified threshold framework for electric and magnetic energy release while preserving the physical differences between the two regimes.

Both begin with electromagnetic energy storage and progressive loading. The structures carrying that loading, however, are different. Low-conductivity environments favor persistent charge separation and localized electric organization. Highly conducting environments permit distributed current and extended magnetic organization.

The effective functional Θ separates stable evolution from threshold crossing. The environmental parameter Σ then separates electric from magnetic post-threshold support. Within the magnetic regime, the availability of an admissible continuation, represented by A , distinguishes magnetic reassignment from magnetic breakdown.

The resulting logical structure is

$$\begin{aligned} \text{loading} &\rightarrow \Theta_{\text{crit}} \\ &\rightarrow \Sigma \\ &\rightarrow \text{branch} \\ &\rightarrow \Delta E. \end{aligned} \tag{4.48}$$

All branches remain subject to conventional electromagnetic energy conservation.

Electric breakdown is characterized by localized charge-supported loading followed by the formation or substantial enlargement of a conducting pathway. Magnetic events begin within already conducting, current-supported structures. They separate further according to whether magnetic continuity can be preserved through reassignment or whether the existing magnetic organization loses structural continuity.

The central testable prediction is that these post-threshold branches should be distinguishable from measurable pre-event conditions rather than only from the observable event afterward.

Quantitative calibration of Θ , Σ , and the physical admissibility condition remains neces-

sary. Nevertheless, the framework provides a common structure for comparing electric breakdown, magnetic reassignment, and magnetic breakdown while retaining their distinct topologies, support conditions, spatial scales, and release mechanisms.

Chapter 5

Alternating Electromagnetic Energy Transport in AC Systems

Purpose: To examine how electromagnetic energy is repeatedly organized, stored, transported, returned, dissipated, and rebuilt during alternating-current operation, and to relate conventional AC quantities to the proposed Photony interpretation of cyclic electric and magnetic organization.

5.1 Introduction

Alternating-current systems are among the most mature and extensively modeled technologies in electrical engineering. Their behavior is routinely described using voltage, current, impedance, reactance, phase, electric and magnetic fields, transmission-line theory, and the Poynting vector. These tools provide accurate quantitative predictions across a wide range of scales, from lumped circuits and power systems to high-frequency conductors and distributed electromagnetic structures. Yet the physical interpretation of how electromagnetic energy is organized and transported during an AC cycle can remain less intuitive than the circuit equations themselves. In particular, conventional descriptions often emphasize oscillating field quantities without explicitly following the repeated transfer of energy among the source, surrounding electromagnetic structure, conductors, dielectric regions, and loads.

The distinction becomes especially important because alternating systems are not simply direct-current systems with periodically reversing current. During each cycle, electric and magnetic energy storage can rise and fall at different times, energy can be temporarily returned toward the source, and the instantaneous direction of power flow can differ from the direc-

tion suggested by average power alone. Capacitors store energy predominantly in electric form, inductors store energy predominantly in magnetic form, and practical conductors support frequency-dependent distributions of current and electromagnetic energy. Reactive power, phase displacement, skin effect, and transmission-line behavior therefore reflect a continuously changing spatial organization of electromagnetic energy rather than only a periodic variation in circuit variables.

The Poynting-vector formulation provides the conventional field-based description of this transport. In differential and integral forms, it relates electromagnetic energy density, energy flux, and the work done on matter. For AC systems, however, the instantaneous Poynting vector can vary strongly throughout the cycle, including intervals in which energy is transported toward a load, redistributed locally, stored temporarily, or returned toward the source. The familiar time-averaged power quantities used in engineering compress this richer temporal behavior into net active and reactive components. This averaging is essential for practical analysis, but it can obscure the sequence of physical energy exchange occurring within each cycle.

The present chapter develops a complementary interpretation of that sequence within the Beyond Field-Line framework established in the

preceding chapters. Chapters 2 through 4 examined electric breakdown, magnetic energy release, and the unified threshold structure governing electric and magnetic post-threshold regimes. The present chapter shifts attention deliberately away from breakdown and rapid release and instead examines normal, stable alternating operation below those instability thresholds.

Within the proposed Photony interpretation, electric and magnetic field lines are treated as macroscopic representations of organized electric and magnetic chain structures. Free electrons are proposed to function as transducers and regulators of those structures. Surface-associated electrons participate predominantly in electric chain organization, while internal electrons associated with current participate predominantly in magnetic chain organization. Alternating voltage and current therefore correspond not only to changing field amplitudes, but also to repeated formation, growth, reduction, reversal, and rebuilding of the corresponding electromagnetic organization.

This interpretation does not replace conventional AC circuit or electromagnetic theory. Voltage, current, impedance, phase, capacitance, inductance, active power, reactive power, Maxwell's equations, Poynting energy transport, transmission-line theory, and established frequency dependent conductor behavior remain the quantitative foundation [Poynting \(1884\)](#); [Jackson \(1998a\)](#); [Griffiths \(2017\)](#); [Pozar \(2012a\)](#); [Sadiku \(2018\)](#). The proposed Photony framework instead provides a structural narrative for the physical organization that may underlie those measurable quantities.

The same perspective applies to reactive components. In a capacitor, energy is accumulated and released primarily through electric chain organization associated with separated charge and the dielectric region between conductors. In an inductor, energy is accumulated and released primarily through magnetic chain organization associated with current and its surrounding magnetic structure. In an AC system containing both, energy can be transferred repeatedly between these forms while only a fraction of the source energy is irreversibly dissipated during each cycle. The resulting phase relationships

between voltage and current are therefore not merely mathematical offsets. They are measurable signatures of the timing of energy storage, transport, return, and dissipation.

This cyclic interpretation also provides the connection between normal AC operation and the energy-release regimes developed in Chapters 2 through 4. Stable alternating operation requires electromagnetic organization to remain within the structural and material limits of the system as energy is repeatedly exchanged. In the notation developed in Chapter 4, normal periodic operation generally satisfies:

$$\Theta(t) < \Theta_{\text{crit}} \quad (5.1)$$

throughout the operating cycle.

As voltage, current density, frequency, local field concentration, thermal loading, or geometric confinement increases, a system may approach conditions in which orderly cyclic transport can no longer be maintained. Electric breakdown, magnetic instability, arcing, dielectric failure, localized heating, and other rapid energy-release processes can therefore be regarded as departures from the normally stable alternating cycle rather than as intrinsic features of AC operation.

A central objective of this chapter is consequently to follow the physical sequence occurring within one AC cycle. The cycle will be examined in terms of four recurring processes:

1. establishment and growth of electric chain organization,
2. establishment and growth of magnetic chain organization,
3. reduction or return of previously established organization, and
4. reversal and rebuilding during the succeeding portion of the cycle.

Conventional AC theory already describes these processes through voltage, current, phase, stored energy, instantaneous power, and electromagnetic energy flow. The Photony interpretation proposes a more explicit structural mechanism in which the same stages correspond to electron transduction, chain growth, reduction, reversal, and re-establishment [Martin \(2022\)](#).

The discussion that follows begins with the conventional description of AC energy storage and transport, including instantaneous power, electric and magnetic energy density, capacitive and inductive storage, phase relationships, reactive power, and Poynting flow. It then develops the proposed electron-mediated interpretation of one complete AC cycle and applies that interpretation to resistive, capacitive, inductive, and RLC systems. Later sections examine transmission conductors, frequency-dependent current distribution, skin effect, and the spatial transport of energy around conductors. The final portions of the chapter return to the boundary between stable periodic operation and the threshold behavior developed in Chapter 4.

Terminology: alternating current, AC cycle, electromagnetic energy transport, electric energy, magnetic energy, electric chain organization, magnetic chain organization, electric chains, magnetic chains, static elemental charge photons, dynamic elemental charge photons, electron transduction, surface-electron transduction, internal-electron transduction, transduction, reduction, pinch-off, reversal, rebuilding, voltage, current, impedance, reactance, phase, phase displacement, instantaneous power, active power, reactive power, apparent power, capacitive energy storage, inductive energy storage, electric-field energy, magnetic-field energy, Poynting vector, Poynting flux, electromagnetic energy density, energy return, energy exchange, displacement current, skin effect, proximity effect, current distribution, transmission-line behavior, frequency dependence, stable periodic operation, structural loading, instability threshold, Θ , Θ_{crit}

5.2 Conventional Energy Description of AC Systems

Alternating-current systems are conventionally described using voltage, current, impedance, phase, electric and magnetic fields, and electromagnetic power flow. These quantities provide highly successful engineering descriptions of AC behavior, but they represent several aspects of the same physical process. Circuit theory de-

scribes relationships among terminals and interconnected components, while electromagnetic field theory describes how energy is stored and transported throughout the spatial regions associated with those components [Griffiths \(2017\)](#); [Jackson \(1998a\)](#). A central objective of this chapter is to connect these descriptions more explicitly and then examine their physical interpretation beyond conventional field-line representations.

An AC source periodically reverses voltage polarity and current direction. For steady-state sinusoidal excitation, voltage and current may be represented as

$$v(t) = V_m \cos(\omega t), \quad (5.2)$$

and

$$i(t) = I_m \cos(\omega t - \phi), \quad (5.3)$$

where V_m and I_m are peak voltage and current, ω is angular frequency, and ϕ is the phase difference between voltage and current.

The instantaneous electrical power is

$$p(t) = v(t)i(t), \quad (5.4)$$

which gives

$$p(t) = \frac{V_m I_m}{2} [\cos \phi + \cos(2\omega t - \phi)]. \quad (5.5)$$

This expression is particularly important because electromagnetic energy transfer in an AC system is not constant. Even under sinusoidal steady-state conditions, instantaneous power varies throughout the cycle. Depending on the load and phase relationship, energy may be transported toward a load during one portion of the cycle and returned toward the source during another.

Using root-mean-square quantities, conventional active power is

$$P = V_{\text{rms}} I_{\text{rms}} \cos \phi, \quad (5.6)$$

while reactive power is

$$Q = V_{\text{rms}} I_{\text{rms}} \sin \phi. \quad (5.7)$$

The apparent power magnitude is:

$$S = V_{\text{rms}} I_{\text{rms}}. \quad (5.8)$$

These quantities form the conventional basis for describing sinusoidal AC power flow [IEE](#)

(2010). Active power represents the net transfer of energy that can ultimately be converted into heat, mechanical work, radiation, or other irreversible forms. Reactive power represents periodic energy exchange associated primarily with capacitive and inductive storage. Consequently, zero average real power does not imply that no energy moves. An ideal reactive element may repeatedly receive and return substantial energy during every cycle.

The distinction between net energy transport and alternating energy exchange is also emphasized in the Photony treatment of real and reactive power [Martin \(2022\)](#). The existence of reactive energy exchange itself, however, is an established result of conventional AC theory and does not depend on the Photony interpretation.

5.2.1 Electric and Magnetic Energy Storage

Conventional electromagnetic theory associates stored energy with both electric and magnetic fields. For linear media, the electric energy density is:

$$u_E = \frac{1}{2}\epsilon E^2, \quad (5.9)$$

and the magnetic energy density is:

$$u_B = \frac{B^2}{2\mu} \quad (5.10)$$

The total electromagnetic energy density may, therefore, be written as:

$$u_{EM} = \frac{1}{2} \left(\epsilon E^2 + \frac{B^2}{\mu} \right) \quad (5.11)$$

for the corresponding linear, nondispersive case [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

These contributions can vary continuously in an AC system. Their relative magnitudes and timing depend on circuit geometry, material properties, excitation frequency, and the phase relationship between voltage and current.

At the lumped-circuit level, the same behavior is represented through capacitive and inductive energy storage. The energy stored in an ideal capacitor is

$$W_C = \frac{1}{2}CV^2, \quad (5.12)$$

while the energy stored in an ideal inductor is:

$$W_L = \frac{1}{2}LI^2 \quad (5.13)$$

A capacitor is therefore associated primarily with electric energy storage, while an inductor is associated primarily with magnetic energy storage [Sadiku \(2018\)](#).

For an ideal capacitor, current leads voltage by 90°. Energy enters the capacitor as the magnitude of its voltage increases and returns to the surrounding circuit as the stored electric energy decreases. For an ideal inductor, voltage leads current by 90°. Magnetic energy increases as current magnitude increases and is released as current magnitude decreases.

Within the proposed Photony framework, these established forms of energy storage receive an additional physical interpretation. Capacitive energy storage is associated with organized electric chains, while inductive energy storage is associated with magnetic chain organization. These chains are proposed to consist of interlinked elemental charge photons and to be transduced and regulated through electron interactions [Martin \(2022\)](#). The conventional energy relations are retained, while the proposed chain model seeks to describe the physical organization represented by the fields.

5.2.2 Power Reversal During an AC Cycle

For a purely resistive load, voltage and current are in phase, such that $\phi = 0$. Instantaneous power becomes:

$$p(t) = V_m I_m \cos^2(\omega t), \quad (5.14)$$

which remains nonnegative throughout the cycle. Energy is continually transferred toward the resistive load and ultimately converted into thermal or other forms.

For an ideal capacitor or inductor, however, instantaneous power alternates between positive

and negative values. During intervals of positive instantaneous power, electromagnetic energy is supplied to the reactive element. During intervals of negative instantaneous power, stored energy is returned toward the source or redistributed elsewhere within the network IEE (2010); Sadiku (2018).

This reversal is a defining characteristic of reactive AC behavior. The system repeatedly establishes stored electromagnetic energy, reduces that stored energy, reverses electrical conditions, and begins another exchange during the succeeding portion of the cycle.

The conventional equations accurately describe this timing and magnitude. They do not require a specification of the physical substructure of the electric and magnetic fields. Photony adds a proposed interpretation by treating the cycle as repeated electron-mediated transduction, chain growth, reduction, reversal, and reconstruction Martin (2022).

5.2.3 The Poynting Description of Energy Transport

Circuit quantities describe power at terminals, while the Poynting vector provides the conventional spatial description of electromagnetic energy transport:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (5.15)$$

The vector $\mathbf{S}$ has units of power per unit area and represents electromagnetic energy flux. Together with electromagnetic energy density, it leads to Poynting's theorem, which expresses local conservation of electromagnetic energy Poynting (1884); Jackson (1998a).

In differential form:

$$\nabla \cdot \mathbf{S} + \frac{\partial u_{EM}}{\partial t} + \mathbf{J} \cdot \mathbf{E} = 0 \quad (5.16)$$

The first term describes the divergence of electromagnetic energy flux, the second describes the temporal change of locally stored electromagnetic energy, and the third represents energy transferred from the electromagnetic field to matter Griffiths (2017).

This formulation demonstrates that electrical energy cannot generally be represented as energy transported solely by charge carriers moving through conductor interiors. Electromagnetic energy occupies the surrounding spatial region and interacts with conductors and components through the electric and magnetic fields established by their geometry and excitation Jackson (1998a); Pozar (2012a).

The circuit treatment in *Photony Foundations* emphasizes the same spatial-energy-flow result using a source-resistor system. Poynting vectors are represented as extending outward from the source, through the electromagnetic region surrounding the conductors, and inward toward the resistor where energy is dissipated Martin (2022).

The distinction between relatively slow electron drift and substantially faster electromagnetic propagation is therefore compatible with the conventional field-energy description, although Photony assigns a different proposed physical substructure to the propagating electromagnetic organization.

The observation that electromagnetic energy can be transported predominantly through regions surrounding conductors is not uniquely a Photony proposition. It follows from Maxwellian electrodynamics and Poynting's theorem. The Beyond Field-Line question concerns what physical organization may correspond to the fields through which that energy is transported.

5.2.4 Conductors as Guides Rather Than Simple Energy Channels

Circuit diagrams can create the impression that conductors themselves form the complete channels through which energy moves from a source to a load. Field theory gives a more spatially distributed picture. Conductors carry current and establish boundary conditions, while electromagnetic energy may propagate primarily through the dielectric or free-space regions between and around those conductors Pozar (2012a).

This behavior is especially clear in distributed systems such as coaxial cables, parallel-wire transmission lines, microstrip structures,

and waveguiding systems. In a coaxial transmission line, for example, the electric and magnetic fields occupy primarily the dielectric region between the conductors, and their cross-product gives the direction of power propagation along the line [Pozar \(2012a\)](#).

The same principle remains applicable at lower frequencies even when lumped-element approximations obscure the spatial structure. Making a circuit electrically small compared with wavelength simplifies its mathematical treatment. It does not eliminate the electromagnetic field structure associated with power transfer.

Conductor loss provides another useful example. In a resistive conductor, the electric field has a component that supports power transfer into the material. In Poynting-vector terms, this appears as electromagnetic energy flux directed into the conductor, where energy is ultimately converted into thermal motion through the $\mathbf{J} \cdot \mathbf{E}$ term of Poynting's theorem [Jackson \(1998a\)](#); [Griffiths \(2017\)](#).

Within the proposed Photony interpretation, much of the circuit-associated electric and magnetic chain organization resides near or outside conductor surfaces, while electromagnetic energy can enter resistive material and be dissipated [Martin \(2022\)](#).

Photony further assigns conductors an additional proposed role. Their electrons are treated not merely as mobile charge carriers but as transducers and regulators interacting with dynamic elemental charge photons to establish electric and magnetic chains [Martin \(2022\)](#). This proposed electron-mediated function becomes particularly important under alternating excitation.

5.2.5 Capacitors and Electric Energy Exchange

A capacitor provides one of the clearest examples of reversible electromagnetic energy exchange. As charge separation increases, the electric field between the plates increases and stored electric energy increases according to:

$$W_C = \frac{1}{2} CV^2 \quad (5.17)$$

When the capacitor discharges, the stored energy decreases and can be returned to the surrounding circuit. The familiar current-voltage relation

$$i = C \frac{dv}{dt} \quad (5.18)$$

shows that current is associated with the rate of change of capacitor voltage rather than simply with the instantaneous magnitude of voltage [Sadiku \(2018\)](#).

The associated electromagnetic description follows directly from Maxwell's displacement-current term, which permits a changing electric field between capacitor plates to participate in the magnetic field relationship even though ordinary conduction current does not cross an ideal dielectric gap [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

Within the Photony interpretation, electric field energy is associated with electric chains transduced primarily through surface electrons. As capacitor electric chain organization decreases, those chains are proposed to retract and unlink through reverse transduction or reduction, returning elemental charge photons to the surrounding electromagnetic system [Martin \(2022\)](#).

Under alternating conditions, reversal of electron orientation associated with polarity reversal is proposed to pinch off the existing chain organization and permit new transduction in the opposite orientation [Martin \(2022\)](#). This mechanism is not part of conventional capacitor theory. It is a proposed structural interpretation of the physical reorganization underlying the established electrical behavior.

The phase relationship is therefore useful in two ways. Conventionally, it describes the timing among voltage, current, and stored electric energy. Within the proposed chain interpretation, it may also describe the timing of chain establishment, reduction, and reversal.

5.2.6 Inductors and Magnetic Energy Exchange

Inductive behavior provides the complementary magnetic case. When current through an inductor increases, magnetic energy increases. When

current decreases, that magnetic energy is released and produces an electromotive response opposing the change in current, consistent with Faraday's law and Lenz's law [Griffiths \(2017\)](#); [Sadiku \(2018\)](#).

The stored magnetic energy is:

$$W_L = \frac{1}{2}LI^2 \quad (5.19)$$

Thus, capacitive behavior represents predominantly electric energy storage and release, while inductive behavior represents predominantly magnetic energy storage and release.

In circuits containing both capacitance and inductance, energy can be exchanged repeatedly between these forms. This oscillatory exchange becomes especially clear in an ideal lossless LC system. At one stage of the oscillation, electric energy is maximal while magnetic energy is minimal. Approximately one quarter-cycle later, the relationship reverses. The total stored energy remains constant in the ideal lossless case, while its organization alternates between predominantly electric and predominantly magnetic forms [Sadiku \(2018\)](#).

Within the proposed Photony framework, magnetic fields are represented as looped chains of interlinked elemental charge photons transduced through internal free electrons [Martin \(2022\)](#). Accordingly, conventional growth and release of inductive energy are interpreted as changes in the population, extent, and organization of magnetic chains.

5.2.7 Voltage Reversal, Current Reversal, and the Half-Cycle

Voltage and current reversals do not necessarily occur simultaneously when phase displacement exists. Consequently, electric and magnetic energy quantities need not reach maxima, minima, or zero crossings at the same instants.

Conventional sinusoidal and phasor descriptions capture these phase relationships accurately. In distributed electromagnetic systems, time-dependent source behavior also produces propagating electromagnetic structures whose temporal and spatial relationships depend on

wavelength, geometry, material properties, and boundary conditions [Pozar \(2012a\)](#).

Within Photony, voltage reversal is proposed to produce electric chain pinch-off, while current reversal separately produces magnetic chain pinch-off. New structures then form with reversed orientation during the succeeding half-cycle [Martin \(2022\)](#).

The temporal motivation is consistent with established electrodynamics: electric and magnetic quantities vary with time, phase relationships depend on the system, and time-varying sources produce propagating electromagnetic disturbances [Griffiths \(2017\)](#); [Jackson \(1998a\)](#). The proposed chain structure underlying those fields is specific to Photony.

The AC cycle therefore contains substantially more physical information than can be represented by a single static field-line diagram. Such a diagram does not show the complete sequence of growth, maximum amplitude, reduction, reversal, rebuilding, propagation, and energy return that occurs continuously during alternating operation.

5.2.8 Frequency and Redistribution of Current and Energy

Frequency introduces an additional dimension to electromagnetic transport. At sufficiently low frequencies and for sufficiently small conductor dimensions, current density can often be approximated as relatively uniform across the conductor cross-section. As frequency increases, current becomes progressively concentrated near conductor surfaces through the skin effect [Sadiku \(2018\)](#).

For a good conductor under the usual approximation, skin depth is:

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} \quad (5.20)$$

where δ is skin depth, ω is angular frequency, μ is permeability, and σ is electrical conductivity [Sadiku \(2018\)](#); [Pozar \(2012a\)](#).

Because skin depth decreases approximately as $1/\sqrt{f}$, progressively less of the conductor

cross-section carries substantial current as frequency increases. Current density decreases exponentially with depth into a good conductor, reaching approximately $1/e$ of its surface value at one skin depth [Jackson \(1998a\)](#); [Sadiku \(2018\)](#).

This redistribution has practical consequences for effective resistance, conductor geometry, transmission-line loss, and high-frequency design [Pozar \(2012a\)](#).

Within Photony, the same frequency-dependent redistribution is interpreted in terms of dynamic elemental charge-photon flow interacting with electron-mediated magnetic chain organization within the conductor [Martin \(2022\)](#). The proposed interpretation does not replace the conventional skin-depth relation, which remains the measurable basis of the phenomenon.

At still higher frequencies, Photony proposes an additional transition. Increasing electron rotational frequency is proposed to shorten the electric and magnetic chains that can be transduced until the framework approaches its far-infrared transition. Beyond that transition, propagating energy is described primarily as classical-photon chainlets rather than longer electromagnetic chains [Martin \(2022\)](#). This proposed transition is distinct from conventional skin-effect theory.

5.2.9 Time Averaging and the Hidden AC Cycle

Much of practical electrical engineering appropriately relies on root-mean-square quantities, phasors, complex impedance, average power, and frequency-domain representations. These methods efficiently compress repetitive AC behavior into quantities convenient for analysis and design [IEE \(2010\)](#); [Sadiku \(2018\)](#).

That efficiency, however, removes much of the instantaneous energy sequence from view.

A capacitor can have zero average real power while repeatedly receiving and returning electromagnetic energy. An inductor can behave similarly through magnetic storage. A resonant circuit can exchange electric and magnetic energy repeatedly while requiring only enough average source power to compensate for losses. A

transmission structure can simultaneously contain forward- and backward-propagating waves following reflection from an impedance discontinuity [Pozar \(2012a\)](#).

Near reactive structures, instantaneous electromagnetic energy flux can therefore differ substantially from the direction and magnitude suggested by average power alone.

For purposes of this chapter, these details are central to understanding alternating electromagnetic energy transport. The AC cycle is therefore treated as a sequence of electromagnetic energy organization rather than merely as repeating voltage and current waveforms.

5.3 Electron Transduction and the Alternating Electric-Magnetic Cycle

The preceding section established that alternating electromagnetic systems repeatedly store, transport, return, and dissipate energy. Voltage and current may be displaced in phase, electric and magnetic energy densities vary throughout the cycle, and instantaneous power may reverse direction in reactive systems. Maxwellian electrodynamics describes these changes through time-dependent electric and magnetic fields and their associated Poynting flux [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

The Beyond Field-Line framework asks a more specific physical question: What process may underlie the repeated formation and reorganization of those fields?

Within Photony, the proposed answer centers on the electron. Rather than treating the electron only as a mobile carrier of conventionally defined charge, the electron is proposed to act as a transducer and regulator of elemental charge photons. Elemental charge photons interacting with the electron are proposed to become interlinked into organized electric or magnetic chains, depending on the electron's location, orientation, and surrounding electromagnetic conditions [Martin \(2022\)](#).

Alternating excitation therefore requires repeated reorganization of the structures that the

electrons are proposed to transduce.

5.3.1 The Electron as an Electromagnetic Transducer

In conventional electrodynamics, conduction electrons respond to applied electric fields and contribute to current density. For the simplest linear, isotropic conducting case,

$$\mathbf{J} = \sigma \mathbf{E} \quad (5.21)$$

Their collective motion is associated with magnetic fields through Ampere-Maxwell relationships, while charge accumulation and redistribution contribute to electric potential and electric-field boundary conditions [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

Photony introduces an additional proposed level of description. The electron is modeled as a toroidal structure with a transduction cavity through which elemental charge photons can pass. Under appropriate conditions, the electron is proposed to interlink these elemental photons into chains. The geometry of the resulting chain depends on electron orientation and whether the electron is primarily internal or surface-associated. With increased skin effect, internal free electrons participating in current become increasingly concentrated near the conductor surface [Martin \(2022\)](#).

The essential proposed sequence is

$$\begin{aligned} \text{elemental photons} &\rightarrow \text{electron transduction} \\ &\rightarrow \text{organized chain.} \end{aligned} \quad (5.22)$$

An electron is not assumed to supply the energy contained in the chain. Rather, it functions as a structural transducer. The elemental charge-photon population supplies the underlying energy, while the electron organizes part of that population into a larger electromagnetic structure.

This distinction becomes particularly important in alternating operation because the electron's transduction condition changes continuously. As the source waveform changes sign and magnitude, the population, orientation, extent, and direction of the transduced structures must also change.

5.3.2 Internal and Surface Electron Roles

Photony distinguishes between the dominant roles of free electrons within a conductor and free electrons located at or near its surface.

Internal free electrons are proposed to become preferentially aligned by dynamic elemental charge-photon flow. Their transduction activity produces closed magnetic chain loops that collectively form the magnetic organization associated with current [Martin \(2022\)](#).

Surface-associated free electrons have a different dominant role. Their orientation relative to local potential and elemental charge-photon conditions allows them to transduce predominantly electric chains extending away from the conductor surface [Martin \(2022\)](#).

The distinction can be summarized as

$$\begin{aligned} \text{internal electrons} &\Rightarrow \text{magnetic organization,} \\ \text{surface electrons} &\Rightarrow \text{electric organization.} \end{aligned} \quad (5.23)$$

This should not be interpreted as a strict spatial separation in which electric phenomena exist only outside conductors and magnetic phenomena only within them. Conventional electromagnetic fields are spatially continuous and satisfy boundary conditions across interfaces [Jackson \(1998a\)](#). The distinction concerns the dominant proposed Photony transduction mechanism.

5.3.3 Alternating Excitation Requires Continuous Reorganization

Under direct-current steady-state conditions, the proposed electron orientations and associated chain structures can remain approximately stationary after the initial transient. Alternating excitation is fundamentally different because both voltage and current periodically change magnitude and direction.

Consequently, the electric and magnetic organizations associated with voltage and current cannot remain fixed. They must grow, diminish, reverse, and rebuild repeatedly.

The fields are also not necessarily in phase. A resistive system may have voltage and current nearly coincident, while capacitive and in-

ductive systems introduce substantial phase displacement. The proposed electric and magnetic chain populations therefore need not reach maxima, minima, or reversal conditions simultaneously.

Photony does not require that an electric chain simply turn into a magnetic chain or that a magnetic chain simply turn into an electric chain. The two organizations are proposed to arise through related but distinct electron-mediated transduction processes driven by the same changing electrical system.

This interpretation is consistent with Jefimenko's caution that time-varying electric and magnetic fields should not necessarily be treated as one field locally creating the other in a simple sequential causal process. Both arise from changing charge and current distributions governed by the electromagnetic system as a whole [Jefimenko \(1989\)](#).

5.3.4 A Complete AC Cycle

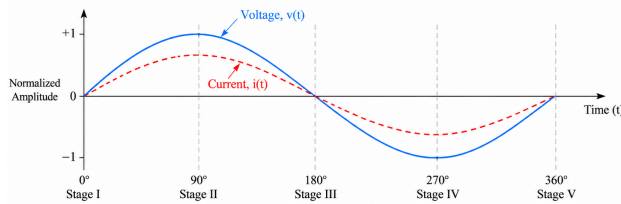


Figure 5.1: Proposed electron-mediated electromagnetic organization during one AC cycle.

Starting near a zero crossing (see Figure 5.1), increasing voltage supports the growth of surface-associated electric chains, while increasing current supports magnetic chain growth through internal electrons. After the positive maximum, both organizations reduce as their supporting quantities decline. Voltage and current reversals produce separate electric and magnetic pinch-off events, depending on phase, followed by rebuilding with reversed orientation during the succeeding half-cycle.

The proposed mechanism can be understood most clearly by following one complete cycle. Consider first a nearly resistive system in which voltage and current are approximately in phase.

Reactive phase differences are considered subsequently.

A complete cycle can be divided into four representative stages.

Stage I: Positive Build-Up. Beginning near a zero crossing, voltage and current increase in the positive direction. Rising voltage establishes an increasing surface-associated electric condition. Within Photony, surface electrons respond by transducing increasing electric chain organization. At the same time, increasing current produces increasing magnetic chain organization through internal electrons.

The electric and magnetic structures therefore grow together, although their geometries differ.

Stage II: Positive Maximum and Reduction. As the sinusoidal waveform approaches its positive maximum, the rate of change of voltage decreases. Current behavior depends on system impedance, but for a nearly resistive example it also approaches a maximum.

The corresponding electric and magnetic organizations reach their largest extent or density for that half-cycle. After the maximum, voltage and current magnitudes decline. Previously established chain organization must therefore also decrease.

Photony describes this decreasing stage using the complementary process of *reduction*. Instead of continuing to interlink elemental charge photons into increasingly extensive chains, electrons are proposed to begin reducing previously established organization as the supporting condition diminishes [Martin \(2022\)](#).

Stage III: Zero Crossing and Reversal. The zero crossing is structurally important. As voltage approaches zero and changes polarity, the existing electric chain organization can no longer remain in its previous orientation. Similarly, when current crosses zero and reverses, the existing magnetic chain organization can no longer remain with its previous circulation.

Photony Foundations proposes that these reversals produce *pinch-off* of the existing chains [Martin \(2022\)](#). For electric organization, voltage polarity reversal controls the relevant pinch-

off condition. For magnetic organization, current reversal controls the corresponding magnetic pinch-off condition. These events need not occur simultaneously when voltage and current are phase shifted.

Stage IV: Negative Build-Up. After reversal, voltage and current increase in magnitude with opposite polarity and direction. Surface-electron orientation reverses, and electric chain transduction begins in the opposite orientation. Internal-electron organization likewise reverses relative to current direction, producing magnetic chain circulation of opposite orientation.

The cycle then progresses through a negative maximum, subsequent reduction, and another zero crossing before the original orientation is restored.

The recurring proposed sequence is

$$\begin{aligned}
 &\text{transduction} \rightarrow \text{growth} \\
 &\quad \rightarrow \text{reduction} \\
 &\quad \rightarrow \text{pinch-off} \\
 &\quad \rightarrow \text{reversal} \\
 &\quad \rightarrow \text{new transduction.}
 \end{aligned} \tag{5.24}$$

Figure 5.1 should therefore be read as a structural interpretation of the waveform, not as an alternative waveform. The conventional voltage and current functions remain unchanged.

5.3.5 Transduction and Reduction

The distinction between transduction and reduction is necessary for an energy-conserving cyclic model.

If electrons only transduced elemental charge photons into increasingly large chains, field organization would continuously accumulate. Alternating steady-state operation requires that previously established structure also be reduced, released, redistributed, or transported away.

Photony therefore treats reduction as the complementary process to transduction [Martin \(2022\)](#).

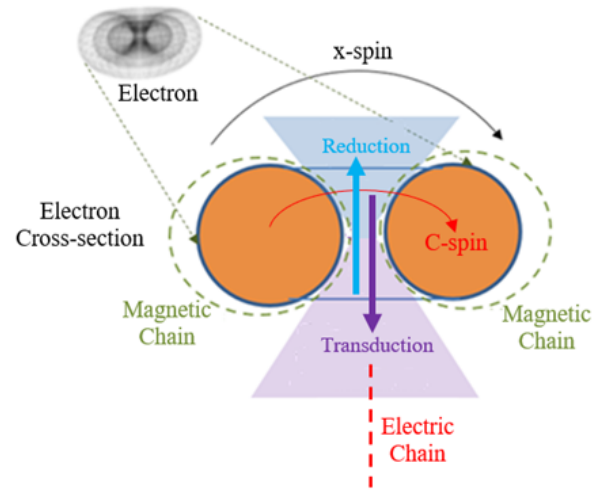


Figure 5.2: Proposed electron-mediated transduction and reduction of electromagnetic chains. During increasing excitation, electrons transduce elemental charge photons into organized electric and magnetic chains.

Surface-associated electrons predominantly support electric chain organization, while internal electrons predominantly support magnetic chain organization.

During decreasing excitation, previously organized chains are reduced or unlinked, returning elemental charge photons to the surrounding electromagnetic system. The illustration is schematic and represents the proposed Photony interpretation.

The cross-sectioned electron in Figure 5.2 also identifies two distinct rotational behaviors used in the Photony model. The internal *c*-spin represents the intrinsic rotational motion associated with the electron's toroidal structure, while the *x*-spin represents an externally driven axial rotation of the electron during alternating excitation.

The *c*-spin is therefore part of the persistent electron structure, whereas the *x*-spin changes in response to the applied AC condition. Within the proposed transduction mechanism, changing *x*-spin orientation helps govern whether the electron predominantly transduces new electric or magnetic chains, reduces previously formed chains, or reverses the direction of chain formation as the waveform passes through a polarity or current reversal.

Conceptually,
 unlinked photons $\xrightarrow{\text{transduction}}$ organized chains,
 organized chains $\xrightarrow{\text{reduction}}$ unlinked photons.
 (5.25)

This reversible structural interpretation is particularly applicable to reactive energy exchange. A capacitor or inductor can repeatedly receive and return energy without consuming all energy that enters it.

Loss mechanisms modify this idealized reversibility. Resistance, dielectric loss, magnetic loss, radiation, and other dissipative processes remove energy from reversible exchange and convert it into heat, emitted radiation, mechanical work, or other forms [Jackson \(1998a\)](#); [Pozar \(2012a\)](#). The important requirement is conservation of energy at the system level.

5.3.6 Extrinsic Electron Rotation During AC Operation

A second feature of the proposed Photony AC mechanism is extrinsic rotation of the electron.

Photony Foundations describes an AC-responsive electron as capable of rotating about its polar axis, referred to as axial or x-spin rotation. The extent and rate of this extrinsic rotational response are proposed to participate in determining the chain structure transduced by the electron [Martin \(2022\)](#).

The x-spin should be distinguished from intrinsic gamma ray structural motions used in the Photony electron model. It is an externally driven response of electron orientation to the alternating electromagnetic environment.

At low AC frequencies, the electron has comparatively more time during each half-cycle to build extended structures. As excitation frequency increases, the available interval for chain construction during each portion of the cycle decreases. Photony consequently proposes that the length of transduced chains decreases with increasing frequency [Martin \(2022\)](#).

The conceptual sequence is

$$\begin{aligned} f \uparrow &\Rightarrow T \downarrow \\ &\Rightarrow \text{transduction interval} \downarrow \\ &\Rightarrow \text{chain length} \downarrow. \end{aligned} \quad (5.26)$$

5.3.7 Electric and Magnetic Pinch-Off Are Distinct

One of the most important consequences of the proposed model is that electric and magnetic pinch-off should not automatically be treated as one simultaneous event.

Electric chain organization tracks predominantly with voltage and surface condition. Magnetic chain organization tracks predominantly with current.

If voltage and current are separated by phase angle ϕ , their zero crossings are likewise displaced in time.

For an ideal capacitor, current leads voltage by 90° . Current, therefore, crosses zero when the capacitor voltage is near an extremum, while the voltage crosses zero when the current magnitude is near a maximum. In the proposed interpretation, magnetic chain reversal and electric chain reversal, therefore, occur at distinctly different stages of the cycle.

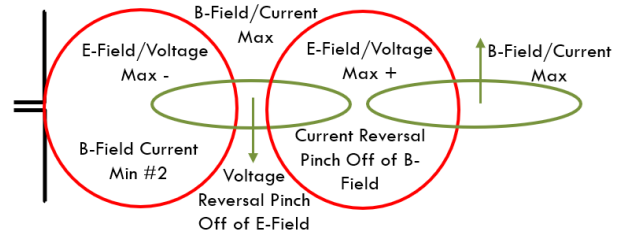


Figure 5.3: Conceptual relationship between voltage-associated electric-chain organization and current-associated magnetic-chain organization during phase-shifted AC operation. The illustration is schematic and represents the proposed Photony interpretation.

Electric-chain organization reaches its extremes with voltage, while magnetic-chain organization reaches its extremes with current. Voltage reversal produces electric-chain pinch-off, whereas current reversal produces magnetic-chain pinch-off. Because voltage and current may be phase displaced, the two pinch-off events need not occur simultaneously.

Figure 5.3 illustrates the temporal separation between electric and magnetic reversal events. The same reasoning applies, with the opposite ordering in an ideal inductor. For a sinusoidal

system, the time displacement corresponding to phase angle ϕ is

For a sinusoidal system, the time displacement corresponding to phase angle ϕ is

$$\Delta t = \frac{\phi}{\omega}, \quad (5.27)$$

when ϕ is expressed in radians.

Thus, increasing voltage-current phase displacement increases the temporal separation between the corresponding electric and magnetic reversal events.

5.3.8 Relation to Electromagnetic Energy Transport

Electron transduction is local, but electromagnetic power transport is spatial.

The Poynting vector,

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad (5.28)$$

describes the conventional direction and density of electromagnetic energy flux [Poynting \(1884\)](#); [Jackson \(1998a\)](#). Any proposed chain mechanism must therefore remain consistent with the spatial power flow represented by $\mathbf{S}$.

Photony interprets the electric and magnetic components entering the Poynting relation as proposed physical chain organizations rather than only field-line representations [Martin \(2022\)](#). Where electric and magnetic organizations coexist orthogonally, their combined geometry establishes a preferred direction for electromagnetic energy transport that corresponds macroscopically to the Poynting vector.

The Poynting vector itself is not replaced by a chain. It remains the established electromagnetic energy-flux quantity,

$$\mathbf{S} = \mathbf{E} \times \mathbf{H},$$

with units of power per unit area. Its surface integral gives the net electromagnetic power crossing a specified boundary. In conventional electromagnetics, this description is sufficient for determining the direction and magnitude of measurable energy transport.

An important consequence is that electromagnetic energy need not be carried primarily

through the interior of a conductor by the drifting charge carriers. In transmission lines, cables, circuit structures, and even simple resistive conductors, a substantial fraction, and in ideal transmission-line treatments essentially all, of the transmitted electromagnetic power is associated with the electric and magnetic fields in the space surrounding and between the conductors.

The conductor establishes the boundary conditions, supports charge and current, and guides the field configuration, while electromagnetic power transport is described by the Poynting vector in the surrounding field region ([Feynman et al., 1964](#); [Jackson, 1998b](#); [Zahn, 2003](#)).

The same field-based interpretation applies locally to resistive conductors. Because a finite electric field exists along the conductor and a magnetic field surrounds the current, the corresponding Poynting vector has a component directed into the conductor. This inward electromagnetic energy flux accounts for the energy subsequently converted into Joule heating within the resistive material ([Feynman et al., 1964](#)). Thus, even where dissipation occurs inside the conductor, the conventional energy-flow description identifies the surrounding electromagnetic field as the immediate source of the delivered energy.

The Photony interpretation must remain consistent with this established result. The proposed electric and magnetic chains are intended to provide a physical structural interpretation of the electromagnetic organization associated with the measured fields. They are not substitutes for the Poynting vector and should not be identified with energy flux itself. The distinction is therefore between proposed electromagnetic structure and measurable electromagnetic energy transport. In this interpretation, the chain organization provides the proposed underlying structure, while the energy-flux, $\mathbf{S}$, continues to describe the macroscopic rate and direction at which electromagnetic energy moves through space.

5.3.9 Near-Field Organization and Released Electromagnetic Structure

Close to a conductor or reactive component, some electric and magnetic organization remains

closely coupled to the source. Energy in this region may be stored temporarily and returned, as occurs in reactive near fields and lumped reactive elements [Jackson \(1998a\)](#); [Pozar \(2012a\)](#).

Farther from an efficient radiating structure, electromagnetic energy becomes predominantly outward propagating. Conventional antenna theory describes this transition through near-field and far-field behavior, with radiation-zone electric and magnetic fields becoming mutually orthogonal and transverse to the propagation direction [Balanis \(2016\)](#).

Photony proposes a structural interpretation of this transition. During each half-cycle, changing electron orientation generates finite electric and magnetic chain structures. As reversals occur, portions of these structures are proposed to pinch off and become increasingly independent of the electron population that originally transduced them [Martin \(2022\)](#).

This interpretation provides the conceptual bridge to the antenna behavior developed later in this Compendium.

5.4 Electric Chain Organization During the AC Cycle

The preceding section introduced electron transduction and reduction as complementary processes underlying the proposed alternating electromagnetic cycle. The present section considers the electric part of that cycle more closely.

The central conventional variables are voltage, electric potential, surface-charge distribution, electric-field strength, capacitance, and electric energy density. Within the Beyond Field-Line framework, these quantities are retained, but the electric field is interpreted as an organized population of electric chains established predominantly through surface-associated electrons.

Voltage and electric-field organization are strongly linked to conductor surfaces and dielectric regions. In an ideal electrostatic conductor, excess charge resides at the surface and the electrostatic field within the conducting material vanishes. In practical conductors carrying current,

a finite internal electric field can exist because finite conductivity requires a field to sustain current. The boundary conditions on conductor surfaces nevertheless remain central to determination of the external electric field [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

For alternating systems, those surface conditions continuously increase, decrease, reverse, and rebuild as applied voltage changes. Electric organization should therefore be considered a dynamic surface-associated process rather than a static field pattern.

5.4.1 Voltage as the Driver of Electric Chain Organization

The electric field is related conventionally to scalar and vector potentials by

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad (5.29)$$

where ϕ is scalar electric potential and $\mathbf{A}$ is magnetic vector potential [Jackson \(1998a\)](#); [Griffiths \(2017\)](#).

Under electrostatic or sufficiently slowly varying conditions,

$$\mathbf{E} \approx -\nabla\phi. \quad (5.30)$$

Within Photony, increasing voltage is interpreted as increasing surface-associated organization and increasing electric chain organization [Martin \(2022\)](#).

This does not replace the conventional relationship between voltage and electric field strength. It proposes a physical structure corresponding to the field represented mathematically by $\mathbf{E}$.

5.4.2 Electric Energy and Chain Organization

The conventional electric energy density remains

$$u_E = \frac{1}{2}\epsilon E^2, \quad (5.31)$$

and capacitor energy remains

$$W_C = \frac{1}{2}CV^2. \quad (5.32)$$

Within the proposed interpretation, increasing electric energy corresponds to greater electric chain organization. This may involve increased chain number, extent, local density, or a combination of these quantities.

No one-to-one proportionality between a drawn conventional field line and a physical chain is implied. Conventional field lines are visualization tools, while the proposed chain population represents an underlying physical distribution.

5.4.3 Capacitive Storage

A parallel-plate capacitor provides a particularly useful system for examining electric chain organization because most of the stored energy lies in the dielectric region between the plates.

For an ideal parallel-plate capacitor,

$$C = \frac{\epsilon A}{d}, \quad (5.33)$$

where A is the plate area and d is the plate separation. Away from the edge regions,

$$E \approx \frac{V}{d}. \quad (5.34)$$

As the voltage magnitude across the plates increases, the electric field magnitude and stored electric energy increase. As voltage decreases, stored energy decreases and may be returned to the circuit [Griffiths \(2017\)](#); [Sadiku \(2018\)](#).

Photony interprets this familiar charging process as the progressive transduction of electric chains between and around the capacitor plates [Martin \(2022\)](#).

Maxwell's displacement-current term,

$$J_D = \frac{\partial D}{\partial t}, \quad (5.35)$$

accounts conventionally for the changing electric field in the Ampere-Maxwell relation, even though conduction current does not traverse an ideal dielectric gap [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

5.4.4 Electric Chain Organization Through One Capacitive Cycle

Consider an ideal capacitor driven sinusoidally:

$$v(t) = V_m \sin(\omega t) \quad (5.36)$$

The capacitor current is:

$$i(t) = \omega C V_m \cos(\omega t). \quad (5.37)$$

Current therefore leads voltage by 90° .

At a voltage zero crossing, stored electric energy is minimal, but dV/dt is maximal, so capacitor current is at maximum magnitude. The system is therefore changing its electric chain organization most rapidly precisely when stored electric energy passes through its minimum.

During the next quarter-cycle, $|V|$ increases. Electric field magnitude and stored energy increase. Within Photony, this interval corresponds predominantly to electric chain transduction and growth.

At voltage maximum, stored electric energy is maximal while ideal capacitor current has fallen to zero. In the proposed interpretation, electric chain organization has reached its greatest extent or density for that polarity.

During the following quarter-cycle, voltage magnitude decreases. Stored electric energy is returned to the surrounding circuit. Photony interprets this stage predominantly as reduction of existing electric chain organization [Martin \(2022\)](#).

At the next zero crossing, the previous electric chain organization reaches its minimum, and the voltage reverses sign. The electric chain organization then begins to grow with opposite polarity.

The magnitude of voltage is therefore associated with the amount of electric energy organization, while dV/dt is associated with the rate at which that organization changes.

5.4.5 Electric Transduction and Reduction

During an interval in which $|V|$ increases, energy is added to the electric configuration. Conventionally,

$$\frac{dW_C}{dt} = CV \frac{dV}{dt} = vi. \quad (5.38)$$

Within Photony, positive growth of electric chain organization corresponds to net transduction. Surface-associated electrons are proposed to interlink elemental charge photons into electric chains [Martin \(2022\)](#).

Any proposed microscopic mechanism must reproduce the conventional energy relationship in Equation 5.38.

When $|V|$ decreases, stored capacitor energy decreases. For an ideal reactive capacitor, that energy is returned to the surrounding circuit.

Photony proposes that decreasing electric chain organization occurs through reduction. Previously formed electric chains are progressively unlinked through the surface electrons that participated in their transduction [Martin \(2022\)](#).

5.4.6 Polarity Reversal and Electric chain Pinch-Off

A voltage zero crossing does more than reduce electric-field magnitude. It changes the sign of the potential difference.

The conventional electric field, therefore, reverses direction:

$$E_+ \rightarrow 0 \rightarrow E_-. \quad (5.39)$$

The Photony interpretation requires a corresponding structural reversal. An electric chain established in one orientation cannot remain unchanged while the macroscopic field reverses.

Photony Foundations proposes that the existing electric chain structure is pinched off as surface-electron orientation reverses, followed by formation of new chains having the opposite orientation [Martin \(2022\)](#).

This proposed pinch-off is distinct from simply changing the arrow direction on a field-line drawing. It implies structural termination or release of organization associated with the preceding half-cycle before the next organization is established.

5.4.7 Electric Chains Between Conductors

In conventional field theory, electric field lines extend from regions of higher potential toward regions of lower potential, with geometry determined by boundary conditions. Between large parallel plates, most field lines are approximately direct across the dielectric gap, while fringe fields curve outward near the plate edges [Griffiths \(2017\)](#).

Photony interprets these paths as possible electric chain structures. Chains transduced by surface electrons may terminate on a suitable opposing surface, while other chains may extend farther into the surrounding region when direct termination is unavailable [Martin \(2022\)](#).

The capacitor therefore provides a useful distinction between surface-to-surface chain organization and fringe-chain organization.

Any proposed chain model must reproduce the conventional geometric dependencies on conductor shape, spacing, permittivity, and applied potential.

5.4.8 Displacement Current and Changing Electric Structure

Maxwell's introduction of displacement current resolved the apparent discontinuity between conduction current in capacitor leads and the changing electric field within the dielectric gap.

The Ampere-Maxwell law can be written

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (5.40)$$

Within an ideal dielectric gap, $\mathbf{J} \approx 0$ while $\partial \mathbf{D} / \partial t \neq 0$ during charging or discharging [Jackson \(1998a\)](#); [Griffiths \(2017\)](#).

Photony proposes that this changing electric chain organization corresponds physically to construction and reduction of electric chains across the dielectric region [Martin \(2022\)](#).

The chain interpretation is not mathematically required by Maxwell's displacement current. Maxwellian electrodynamics is complete at the measured-field level. The chain model is a proposed physical substructure that must remain

consistent with the established displacement-current relationship.

5.4.9 Electron Motion Along a Previously Transduced Electric Chain

Photony Foundations introduces an additional proposed possibility: under suitable conditions, an electron may travel along a previously established electric chain rather than creating that chain solely for field organization [Martin \(2022\)](#).

This distinguishes two proposed electron roles:

electron as chain transducer

and:

electron traveling on an existing chain.

This behavior is proposed to occur for both electric and magnetic chain organizations, although the resulting physical manifestations differ. For an electric chain, electron motion along a previously established chain provides the proposed Photony interpretation associated with electron transport in the photoelectric effect, in which energy supplied through the photon-chain organization can permit an electron to leave its initially bound state and move along the available chain structure.

For a magnetic chain, an electron can likewise remain associated with and ride a previously transduced closed-loop chain. Within the Photony description of atomic structure, this provides the proposed physical basis for an electron occupying and moving around a nucleus along an established atomic-shell chain.

Chain transduction and electron riding should therefore be treated as distinct processes. Transduction establishes or reorganizes the photon-chain structure; riding describes the electron's subsequent interaction and motion along an existing chain. This distinction allows the same underlying chain framework to participate both in field organization and in electron transport or confinement without identifying the electron itself with the chain.

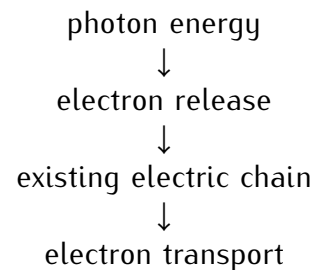
A particularly important proposed example of the second role is the photoelectric effect. In

Einstein's conventional description, an incident photon transfers an energy quantum $E = h\nu$ to an electron, and electron emission becomes possible when the photon energy is sufficient to overcome the relevant binding or work-function requirement. Photony retains this measured energy relationship and does not replace the conventional photoelectric threshold condition.

The proposed Photony interpretation adds a structural description of what happens after the electron has acquired sufficient energy to leave its initially bound state. The incident classical photon supplies the required energy, while a previously established electric-chain organization provides a proposed pathway along which the released electron can subsequently move.

The electron is therefore not interpreted as creating the electric chain as it departs. Instead, the chain has already been established through surface-electron transduction, and the energized electron becomes capable of interacting with and riding that pre-existing structure.

The distinction may be summarized as:



In this interpretation, Einstein's photoelectric relation determines the energy condition required for electron release, while the electric chain provides the proposed structural pathway for subsequent electron motion. Thus, the Photony description does not replace the conventional photoelectric effect; it proposes a physical organization underlying the transport stage that follows photon-induced electron liberation.

This distinction is important because photon absorption, electron release, chain transduction, and electron riding are separate processes. The incident classical photon supplies energy to the electron; surface-associated electrons establish or maintain the electric-chain organization; and the released electron may then travel along that already existing chain under suitable conditions. The electron riding the chain is not itself a link

in the chain and need not be the electron that originally transduced it.

This mechanism should not be equated with conventional displacement current. In conventional electromagnetics, displacement current does not require electrons to cross the dielectric. The proposed chain-riding mechanism represents an additional physical possibility under conditions permitting actual charged-particle transport.

5.4.10 From Reversible Electric Storage to Dissipation

An ideal capacitor returns all its stored electrical energy. Real systems do not.

Dielectric loss, conductor resistance, leakage current, polarization loss, radiation, and other dissipative mechanisms remove some energy during each cycle. These effects can be represented using equivalent series resistance, complex permittivity, dielectric loss tangent, and related conventional quantities [Sadiku \(2018\)](#); [Jackson \(1998a\)](#).

Within the proposed Photony interpretation, reversible operation corresponds to orderly transduction and reduction of electric chain organization. Dissipation indicates that part of the participating energy is diverted into other processes rather than being returned through the same macroscopic cycle.

At sufficiently high electric loading, stable alternating organization may no longer be maintained. Dielectric ionization, partial discharge, corona, arcing, or full voltage breakdown may then occur. Those regimes were developed in Chapter 2 and unified with the broader threshold framework in Chapter 4.

Normal AC electric behavior and electric breakdown therefore occupy different regions of loading. Stable operation repeatedly builds and reduces electric organization. Breakdown occurs when that organization can no longer be maintained through the normal reversible or weakly dissipative cycle.

5.5 Magnetic Chain Organization During the AC Cycle

Electric chain organization is associated primarily with voltage, conductor surfaces, and capacitive energy storage. Magnetic chain organization is associated primarily with current, internal-electron transduction, magnetic loops, and inductive storage.

In conventional electromagnetics, current is associated with magnetic fields, changing magnetic flux produces induced electric fields, and inductive structures store energy magnetically [Griffiths \(2017\)](#); [Jackson \(1998a\)](#); [Sadiku \(2018\)](#). Within the proposed Photony framework, these established relations are retained, while the magnetic field is interpreted as an organized population of closed magnetic chains transduced predominantly through internal free electrons [Martin \(2022\)](#).

As current magnitude rises, falls, passes through zero, and reverses, the associated magnetic chain organization must likewise grow, reduce, reach a minimum, and rebuild with opposite circulation.

5.5.1 Current as the Driver of Magnetic Organization

For a long straight conductor carrying current I , the conventional magnetic field intensity outside the conductor is

$$H_\phi = \frac{I}{2\pi r}, \quad (5.41)$$

with

$$B_\phi = \mu H_\phi, \quad (5.42)$$

for a linear medium [Griffiths \(2017\)](#); [Sadiku \(2018\)](#).

The field magnitude and direction track current. When current increases in one direction, magnetic field magnitude increases with one circulation. When current decreases, field magnitude decreases. When current reverses, field circulation reverses.

Within Photony, this familiar field pattern is interpreted as an organized population of magnetic chain loops [Martin \(2022\)](#).

5.5.2 Internal Electrons and Magnetic chain Transduction

Surface-associated electrons are primarily associated with electric chain organization because of their relationship to conductor potential and surface charge.

Internal free electrons are proposed to respond to dynamic elemental charge-photon flow through the conductor and to transduce closed magnetic chain loops [Martin \(2022\)](#).

The proposed sequence is:

$$\begin{aligned} \text{dynamic photon flow} &\rightarrow \text{electron alignment} \\ &\rightarrow \text{magnetic transduction} \end{aligned} \quad (5.43)$$

The electron acts as a structural transducer rather than as the ultimate source of magnetic energy. The energy belongs to the electromagnetic system, while the electron provides the proposed mechanism by which that energy becomes organized into looped magnetic structures.

5.5.3 Magnetic Energy Storage

The conventional magnetic energy density remains

$$u_B = \frac{B^2}{2\mu}, \quad (5.44)$$

while the total magnetic energy stored in an ideal inductor is

$$W_L = \frac{1}{2}LI^2. \quad (5.45)$$

These relations show that magnetic energy depends on current magnitude rather than current sign [Jackson \(1998a\)](#); [Sadiku \(2018\)](#). Positive and negative currents of equal magnitude correspond to equal stored magnetic energy but opposite magnetic-field orientation.

Within the proposed Photony interpretation, the amount of magnetic chain organization is associated primarily with $|I|$, while the orientation or circulation of that organization depends on current direction.

The magnetic zero crossing is therefore not merely a low-energy point. It is the point at

which the previous circulation reaches its minimum and the succeeding circulation begins in the opposite direction.

5.5.4 Inductive Storage

An ideal inductor provides the clearest conventional example of magnetic energy storage.

Its voltage-current relation is:

$$v(t) = L \frac{di(t)}{dt}. \quad (5.46)$$

For sinusoidal current,

$$i(t) = I_m \sin(\omega t), \quad (5.47)$$

the voltage is

$$v(t) = \omega LI_m \cos(\omega t), \quad (5.48)$$

so, voltage leads current by 90° in an ideal inductor [Sadiku \(2018\)](#).

The instantaneous power satisfies:

$$\frac{dW_L}{dt} = Li \frac{di}{dt} = vi \quad (5.49)$$

Whatever microscopic magnetic chain organization is proposed, its rate of growth or reduction must remain consistent with the measured energy exchange represented by Equation 5.49.

5.5.5 Magnetic Chain Organization Through One Inductive Cycle

At a current zero crossing,

$$I = 0, \quad (5.50)$$

and therefore:

$$W_L = \frac{1}{2}LI^2 = 0 \quad (5.51)$$

Stored magnetic energy is minimal. However, dI/dt is maximal in an ideal sinusoidal inductor, so the inductor voltage is at or near its maximum magnitude. Magnetic chain organization, therefore, changes most rapidly when stored magnetic energy is smallest.

During the next quarter-cycle, current magnitude increases. Magnetic field strength grows, magnetic energy increases, and energy flows into the inductive structure. Within the Photony

interpretation, this interval predominantly corresponds to magnetic chain transduction and growth through internal electrons [Martin \(2022\)](#).

At the current maximum, stored magnetic energy is maximal while $dl/dt = 0$ and ideal-inductor voltage pass through zero.

During the following quarter-cycle, current magnitude decreases. Magnetic energy falls and is returned to the circuit. Photony interprets this interval predominantly as magnetic chain reduction.

At the next current zero crossing, the previous magnetic organization reaches its minimum. Current reverses direction, and a new magnetic chain population is established with opposite circulation.

5.5.6 Magnetic Transduction and Reduction

When $|I|$ increases, conventional magnetic energy increases. Within Photony, increasing current corresponds to increasing dynamic elemental charge-photon flow through the conductor. Internal electrons respond by transducing additional or more extensive magnetic chain loops [Martin \(2022\)](#).

When $|I|$ decreases, stored magnetic energy decreases. In an ideal inductor, this energy is returned to the surrounding circuit rather than dissipated. The inductor develops a voltage polarity opposing the change in current, consistent with Faraday's law and Lenz's law [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

The corresponding Photony interpretation is net magnetic chain reduction. Previously established chains are proposed to be progressively unlinked or shortened as the supporting current condition diminishes [Martin \(2022\)](#).

5.5.7 Current Reversal and Magnetic chain Pinch-Off

Current reversal has a particularly clear magnetic consequence because the conventional magnetic-field orientation is tied directly to current direction.

As current passes through zero,

$$I_+ \rightarrow 0 \rightarrow I_-, \quad (5.52)$$

the corresponding magnetic-field orientation reverses.

$$B_+ \rightarrow 0 \rightarrow B_- \quad (5.53)$$

Within Photony, this reversal is proposed to involve magnetic chain pinch-off [Martin \(2022\)](#). The prior loop organization cannot remain in its previous circulation when the current-producing condition reverses.

The proposed structural sequence is

$$\begin{aligned} &\text{reduction} \rightarrow \text{pinch-off} \\ &\rightarrow \text{reorientation} \\ &\rightarrow \text{new transduction.} \end{aligned} \quad (5.54)$$

Electric and magnetic pinch-off should not be assumed to occur simultaneously. In a resistive circuit, voltage and current reversals may be nearly coincident. In reactive circuits, they can be substantially separated in phase.

5.5.8 Faraday-Lenz Response and Magnetic Reduction

The electric field resulting from induced voltage is conventionally described by Faraday's law,

$$\mathcal{E} = -\frac{d\Phi_B}{dt}, \quad (5.55)$$

where Φ_B is magnetic flux [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

The negative sign expresses Lenz's law: the induced response opposes the change in magnetic flux that produces it.

This conventional result constrains the proposed chain interpretation. If magnetic chain organization decreases, associated energy cannot simply disappear. The reduction must contribute to an electromagnetic response consistent with conventional inductive energy return.

Magnetic reduction, therefore, has both a proposed structural consequence and a measured energetic consequence.

5.5.9 Magnetic Loops and Field Geometry

Magnetic field lines differ geometrically from electric field lines because magnetic flux is divergence-free:

$$\nabla \cdot \mathbf{B} = 0 \quad (5.56)$$

This expresses the absence of observed magnetic monopoles and the continuous nature of magnetic flux [Griffiths \(2017\)](#); [Jackson \(1998a\)](#).

Photony adopts a structurally similar picture by treating magnetic fields as closed chains or loops rather than single-ended structures [Martin \(2022\)](#).

For a straight conductor, magnetic loops encircle the current path. For a solenoid or coil, the individual current-associated magnetic contributions combine into a stronger organized magnetic field concentrated through the coil interior and returning outside the structure.

For more complex conductor geometries, the proposed magnetic chain organization must satisfy the same macroscopic constraints imposed by the Ampere–Maxwell law and magnetic boundary conditions. The chain model cannot assign arbitrary trajectories to magnetic organization.

5.5.10 Magnetic Chain Organization Within and Around a Conductor

The magnetic field of a current-carrying conductor is not confined to its interior. For a cylindrical conductor with approximately uniform current density at sufficiently low frequency, the conventional magnetic field increases from zero at the axis to its surface value and then decreases with radial distance outside the conductor [Griffiths \(2017\)](#); [Sadiku \(2018\)](#).

Within Photony, magnetic chains are proposed to be transduced by electrons within the conductor and to extend as closed loops through the surrounding region [Martin \(2022\)](#).

This feature connects internal current-associated organization with external electromagnetic energy storage. The conductor therefore serves as both a region of electron-mediated magnetic chain organization and a boundary

through which the resulting magnetic loops extend into surrounding space.

5.5.11 Inductors as Concentrated Magnetic Organization

An inductor provides a geometry deliberately designed to enhance magnetic organization. By winding a conductor into multiple turns, magnetic contributions from individual current paths reinforce one another. The result is increased inductance and greater magnetic energy storage for a given current.

For an ideal solenoid,

$$B \approx \mu n I, \quad (5.57)$$

where n is the number of turns per unit length [Griffiths \(2017\)](#).

Within Photony, coil geometry corresponds to a concentrated and mutually reinforcing magnetic chain population. The chains generated through current-associated electron transduction are organized by conductor geometry into a stronger collective magnetic structure.

Inductance can therefore be interpreted structurally as a measure of how effectively a geometry supports magnetic energy organization for a given current. This interpretation remains constrained by conventional inductance L . Photony does not replace inductance with chain count.

5.5.12 Magnetic Chain Organization and Energy Return

The ability of an inductor to return stored energy is essential to AC behavior. When current decreases, the inductor develops the voltage polarity required by the surrounding circuit to oppose that decrease. In a lossless idealization, all magnetic energy can eventually be returned.

Within Photony, chain reduction provides the proposed structural mechanism for that release. As magnetic chains are reduced, organized energy becomes available for re-entry into the broader electromagnetic cycle.

This interpretation is particularly important because magnetic chain reduction must be distinguished from magnetic breakdown. In normal

AC operation, reduction is orderly and reversible or nearly reversible. It occurs every cycle.

Magnetic breakdown, by contrast, involves threshold instability and rapid structural failure under substantially different loading conditions, as developed in Chapter 3 and incorporated into the unified framework of Chapter 4.

5.13 Traveling Waves, Reflections, and Standing-Wave Organization

The preceding treatment of an AC cycle is most directly applicable when the electrical dimensions of the system are sufficiently small that voltage and current can be treated as approximately uniform over the structure at a given instant. As conductor length becomes an appreciable fraction of wavelength, that approximation is no longer sufficient. Voltage and current become functions of both position and time, and electromagnetic energy must be treated as a distributed traveling-wave phenomenon (Pozar, 2012b; Zahn, 2003).

For a uniform transmission structure, forward and reflected waves may be written conventionally as

$$V(z) = V^+ e^{-j\beta z} + V^- e^{j\beta z},$$

and

$$I(z) = \frac{V^+}{Z_0} e^{-j\beta z} - \frac{V^-}{Z_0} e^{j\beta z},$$

where Z_0 is the characteristic impedance and β is the phase constant. These established relations show that local voltage and current need not reach maxima, minima, or zero crossings at the same spatial locations.

Within the proposed Photony interpretation, this means that electric and magnetic chain organization must also vary spatially along the conductor. Regions of large voltage magnitude correspond to stronger local electric-chain organization, while regions of large current magnitude correspond to stronger local magnetic-chain organization. The conductor therefore does not possess a single uniform electric or magnetic chain state during distributed operation. Instead, the local balance between electric and magnetic

organization changes continuously with position and time.

When a traveling wave encounters an impedance discontinuity or termination, part or all of the incident wave may be reflected. The superposition of forward and reflected waves can produce a standing-wave pattern. In a standing wave, fixed spatial regions of relatively high and low voltage and current develop. Voltage antinodes correspond to locations of relatively strong electric-field organization, while current antinodes correspond to locations of relatively strong magnetic-field organization. Because voltage and current standing-wave patterns are spatially displaced, the corresponding electric and magnetic chain organizations are likewise displaced.

This distinction is particularly important within the Photony framework. A standing wave should not be interpreted as electromagnetic energy becoming motionless. Rather, it results from the superposition of oppositely propagating waves.

A standing-wave pattern does not necessarily imply the absence of net electromagnetic energy transfer from the overall structure. In an ideal lossless transmission line with complete reflection, equal forward and reflected waves produce zero time-averaged net power flow along the line, even though electric and magnetic energy continue to exchange locally (Pozar, 2012b; Zahn, 2003).

A radiating antenna is fundamentally different. A resonant antenna may support an approximately standing-wave distribution of voltage and current along its conducting structure while simultaneously transferring electromagnetic energy into an outward-propagating radiation field. The standing-wave pattern describes the spatial organization associated with the conductor and its boundary conditions; it does not imply that the electromagnetic field surrounding the antenna is everywhere standing.

Conventionally, the energy transferred away from the antenna is represented by an outward time-averaged Poynting flux and, at the antenna terminals, by radiation resistance. Part of the electromagnetic energy near the antenna is reactive and is repeatedly stored and returned during the AC cycle, whereas the radiative compo-

ment separates from the antenna and carries net energy outward.

Within the Photony interpretation, the corresponding distinction is between electromagnetic chain organization that remains coupled to the conductor and participates in cyclic near-field storage and reduction, and organization that becomes released from the conductor-associated structure and propagates outward. Thus, spatially standing electric and magnetic organization along the antenna can coexist with outward electromagnetic energy transmission.

The same distributed behavior provides the transition from transmission-line operation to antenna behavior. When conductor geometry, termination, and electrical length permit part of the electromagnetic organization to separate from the guiding structure and propagate outward, the system can radiate. Within the Photony interpretation, this corresponds to electric and magnetic organization that is no longer confined primarily to the conductor-guided near-field structure and instead becomes part of a propagating electromagnetic field. The quantitative conditions for radiation, antenna impedance, radiation resistance, polarization, gain, and radiation pattern remain those of conventional antenna theory and are developed further in Chapter 6.

5.6 Discussion

The electric chain and magnetic chain organizations developed in this chapter are complementary but physically distinct parts of the same alternating electromagnetic system. Electric chain organization tracks primarily with voltage, while magnetic chain organization tracks primarily with current.

Within the proposed Photony interpretation, the electric side is predominantly associated with surface-electron transduction of electric chains, whereas the magnetic side is predominantly associated with internal-electron transduction of magnetic loops [Martin \(2022\)](#).

The two forms of organization coexist and interact but should not be collapsed into one undifferentiated process.

This distinction is especially important at re-

versal. A voltage zero crossing defines an electric reversal condition, while a current zero crossing defines a magnetic reversal condition. These events coincide only when voltage and current are in phase. In reactive systems they are separated in time, meaning that electric chain reversal and magnetic chain reversal need not occur simultaneously.

The magnetic cycle can be summarized as a recurring current-associated sequence complementing the voltage-associated electric cycle. As current magnitude increases, conventional magnetic-field magnitude and magnetic energy increase, while Photony interprets the same interval as net magnetic chain transduction through internal electrons.

At current extrema, stored magnetic energy and proposed magnetic chain organization reach their greatest values for that half-cycle. As current magnitude decreases, magnetic energy is released and the proposed chain population undergoes net reduction. At current zero crossing, the previous magnetic chain organization reaches a minimum and its prior circulation ends. Current reversal then initiates magnetic chain transduction with opposite circulation.

This leads to an important distinction between the amount of magnetic organization and the rate at which that organization changes. Current magnitude corresponds to the degree of magnetic energy organization, while dI/dt corresponds to the rate of change of that organization. Maximum magnetic chain organization therefore occurs near current extrema, while the most rapid magnetic reorganization occurs near current zero crossings.

An analogous distinction applies to electric organization. Voltage magnitude corresponds to the degree of electric energy organization, while dV/dt corresponds to the rate at which that organization changes.

Together, these electric and magnetic descriptions provide the central physical interpretation of alternating electromagnetic behavior developed in this chapter. AC operation is treated as repeated organization, reduction, reversal, and rebuilding of distinct but coupled electric and magnetic structures, mediated in the proposed Photony framework by different elec-

tron populations and orientations.

The conventional voltage, current, field, phase, energy, and power relationships remain unchanged. The proposed contribution is a structural narrative for what may underlie their cyclic behavior.

Normal AC operation also provides an important contrast with the energy-release regimes developed in Chapters 2 through 4. During stable operation,

$$\Theta(t) < \Theta_{\text{crit}}, \quad (5.58)$$

and transduction, reduction, reversal, and rebuilding proceed without threshold failure.

If electromagnetic loading increases sufficiently that

$$\Theta(t) \geq \Theta_{\text{crit}}, \quad (5.59)$$

the stable periodic description is no longer sufficient. The environmental-selection and branch structure developed in Chapter 4 then becomes relevant.

Thus, AC cycling and breakdown should not be treated as separate unrelated subjects. Stable alternating operation describes repeated organization below the instability threshold. Chapters 2 through 4 describe what can occur when that stable organizational capacity is exceeded.

5.7 Conclusions

This chapter has developed a physical interpretation of alternating electromagnetic behavior in which conventional voltage, current, field, energy, phase, and power relations are retained while electric and magnetic fields are interpreted, within the proposed Photony framework, as organized chain structures.

Surface-associated electrons predominantly transduce electric chains, internal electrons predominantly transduce magnetic loops, and AC operation consists of repeated transduction, reduction, reversal, and rebuilding as voltage and current vary through each cycle.

Electric chain organization follows primarily from voltage and surface conditions, whereas magnetic chain organization follows primarily

from current and internal electron response. Their maxima and reversal events need not occur simultaneously because voltage and current may be displaced in phase.

Capacitive systems demonstrate repeated electric energy storage and return. Inductive systems demonstrate repeated magnetic energy storage and return. Systems containing both forms of storage can transfer energy repeatedly between electric and magnetic organizations, while active losses remove only a portion of that energy during each cycle.

Poynting-vector analysis provides the conventional spatial description of this transport. Electromagnetic energy need not be confined to conductor interiors, and conductors should therefore be understood not simply as channels carrying energy, but as structures that establish boundary conditions and guide electromagnetic transport through surrounding space.

Frequency modifies this organization through established effects such as skin depth and current redistribution. Within the proposed Photony interpretation, increasing frequency also changes the time available for electron transduction; therefore, it alters the proposed extent of the resulting chain structures.

The resulting picture treats AC operation as a continuously evolving electromagnetic process rather than merely as a pair of oscillating circuit variables. Electric and magnetic chain organization is repeatedly established, stored, reduced, returned, reversed, and rebuilt.

Under normal conditions, this cyclic process remains below the common instability threshold developed in Chapter 4. When electric, magnetic, thermal, geometric, or frequency-dependent loading exceeds the capacity of the system to maintain stable organization, the behavior can depart from ordinary AC cycling and enter one of the energy-release regimes developed in the preceding chapters.

Conventional electromagnetic and circuit theory remain the quantitative foundation throughout. The proposed Photony interpretation provides a complementary structural account of the repeated electromagnetic organization underlying those established relationships.

Chapter 6

Unintended Antennas, Electromagnetic Compatibility, and Signal Integrity

Purpose: To examine how ordinary conductors, interconnects, cables, traces, return paths, enclosures, and related structures can develop unintended electromagnetic coupling and radiation, and to relate established signal-integrity and electromagnetic-compatibility behavior to the proposed Photony interpretation of distributed electric and magnetic organization.

6.1 Introduction

Unintended electromagnetic coupling and radiation are persistent challenges in electronic systems. Conductors that are designed primarily to carry power or signals can also support spatially distributed current and voltage patterns that permit them to behave partially as antennas. Printed-circuit traces, cables, connector leads, enclosure seams, grounding structures, and return paths can therefore contribute to radiated emissions, susceptibility, crosstalk, waveform distortion, ringing, and other electromagnetic-compatibility and signal-integrity problems. Conventional engineering analysis treats these effects through transmission-line theory, parasitic capacitance and inductance, common-mode and differential-mode currents, impedance discontinuities, resonance, shielding, grounding, and near-field and far-field coupling.

These methods are quantitatively successful and remain the engineering foundation for electromagnetic-compatibility and signal-integrity design. They identify the conditions under which unwanted currents develop, resonances form, energy couples between structures, and radiation escapes from a system [Ott \(2009\)](#); [Paul \(2006\)](#); [Bogatin \(2018\)](#). The question ad-

dressed in this chapter is complementary and physical: what structural process might underlie the transition from ordinary conducted excitation to unintended electromagnetic coupling and radiation?

Chapter 5 developed alternating electromagnetic operation as a recurring process of electric and magnetic organization, transduction, reduction, reversal, and rebuilding. Electric organization was associated primarily with voltage and surface-associated electrons, while magnetic organization was associated primarily with current and internal electrons. That chapter also established the distinction between electromagnetic organization that remains closely coupled to a source and organization that can propagate outward as electromagnetic energy.

The same underlying processes become especially important when the conducting structure itself is electrically distributed. An intended antenna deliberately uses conductor geometry, current distribution, voltage distribution, wavelength, and boundary conditions to promote controlled radiation. The same physical conditions can arise unintentionally in structures that were never designed to function as antennas.

A conductor therefore need not be deliberately shaped as a resonant radiator before it can support distributed electromagnetic behav-

ior. If its length, geometry, return-path configuration, frequency, impedance discontinuities, or surrounding structures allow substantial spatial variation of current and voltage, the conductor can develop localized regions of magnetic and electric organization similar in principle to those found in an intended antenna. The distinction is not that an intended antenna possesses electromagnetic organization while an unintended conductor does not. Rather, the distinction lies in whether the geometry and excitation were deliberately selected to control that organization and its release.

Within the proposed Photony interpretation, unintended antennas are treated as accidental electron-transduction systems. Internal electrons interacting with dynamic elemental charge photons predominantly support magnetic organization associated with current, while surface-associated electrons interacting with static elemental charge photons predominantly support electric organization associated with voltage and surface charge [Martin \(2022\)](#). Changes in conductor geometry, termination, return-path continuity, or surrounding dielectric and conductive structures alter where these transduction processes become concentrated and how the resulting chains grow, reduce, redistribute, or become released.

This perspective is particularly relevant to common-mode behavior. Differential-mode currents are intended to flow through a defined signal and return structure, whereas common-mode currents can involve a much larger unintended electromagnetic geometry that includes cables, chassis structures, reference planes, shields, and surrounding space. Conventional EMC theory recognizes that relatively small common-mode currents can produce substantial radiated emissions when they excite electrically long conductors or poorly controlled return structures [Ott \(2009\)](#); [Paul \(2006\)](#). The Beyond Field-Line interpretation asks how that behavior may be understood structurally in terms of redistributed electron transduction and electromagnetic organization.

Signal integrity provides a second, closely related perspective. Signal integrity asks whether the intended signal reaches its desti-

nation with the required amplitude, timing, edge shape, phase, and noise margin. Electromagnetic compatibility asks whether electromagnetic energy remains sufficiently confined to the intended structures and whether the system avoids disturbing, or being disturbed by, its electromagnetic environment. The two disciplines are therefore distinct but inseparable. A change in electromagnetic organization that degrades the intended signal can also create unwanted coupling or radiation, while externally coupled electromagnetic energy can alter the signal carried by the same structure.

Reflections, overshoot, undershoot, ringing, resonances, return-path discontinuities, and crosstalk are conventionally described through distributed impedance and electromagnetic coupling. In the proposed interpretation, these phenomena correspond to redistribution or reorganization of the evolving electric and magnetic structures associated with the propagating signal. An impedance discontinuity does not merely change a numerical impedance. It changes the local electromagnetic conditions under which energy propagates, current and voltage redistribute, and internal and surface-associated electron populations participate in transduction.

One particularly important pathway connects signal-integrity behavior directly to electromagnetic compatibility:

$$\begin{aligned} \text{differential signal} &\rightarrow \text{discontinuity} \\ &\rightarrow \text{mode conversion} \\ &\rightarrow \text{common-mode current} \\ &\rightarrow \text{unintended radiation.} \end{aligned} \tag{6.1}$$

The reverse pathway is equally important:

$$\begin{aligned} \text{external field} &\rightarrow \text{receiving structure} \\ &\rightarrow \text{common-mode coupling} \\ &\rightarrow \text{mode conversion} \\ &\rightarrow \text{signal degradation.} \end{aligned} \tag{6.2}$$

These two sequences illustrate why signal integrity and electromagnetic compatibility should not be treated as unrelated engineering problems. Both concern the spatial and temporal

organization, guidance, redistribution, coupling, and release of electromagnetic energy.

Terminology: electromagnetic compatibility, signal integrity, unintended antenna, intended antenna, electromagnetic coupling, radiated emissions, conducted emissions, susceptibility, immunity, differential-mode current, common-mode current, mode conversion, transmission-line behavior, characteristic impedance, propagation delay, forward path, return path, return-current distribution, impedance discontinuity, reflection, reflection coefficient, standing wave, resonance, termination, crosstalk, aggressor conductor, victim conductor, near-end crosstalk, NEXT, far-end crosstalk, FEXT, mutual capacitance, mutual inductance, parasitic capacitance, parasitic inductance, shielding, grounding, chassis coupling, enclosure seam, reference plane, cable shield, connector discontinuity, via discontinuity, common-mode radiation, differential-mode radiation, electric coupling, magnetic coupling, Poynting vector, electromagnetic energy transport, skin effect, current localization, edge rate, frequency content, electrically distributed structure, wavelength, electric organization, magnetic organization, electric chains, magnetic chains, static elemental charge photons, dynamic elemental charge photons, electron transduction, surface-electron transduction, internal-electron transduction, electromagnetic redistribution, unintended electromagnetic release

6.2 Conventional Electromagnetic Description of Unintended Antennas and Coupling

Unintended radiation and coupling occur when structures intended primarily to conduct power or signals also support spatially distributed electromagnetic behavior. Conventional electromagnetic-compatibility and signal-integrity theory describes these effects through current and voltage distributions, common-mode and differential-mode excitation, transmission-line behavior, impedance discontinuities, resonance, parasitic capacitance and inductance,

return-path geometry, shielding, grounding, frequency-dependent current distribution, and electromagnetic coupling [Ott \(2009\)](#); [Paul \(2006\)](#); [Bogatin \(2018\)](#). These relationships remain the quantitative foundation for the analysis developed in this chapter.

The purpose of this section is therefore not to replace established EMC or signal-integrity theory, but to identify the conventional behaviors that any proposed structural interpretation must reproduce. Particular attention is given to the conditions under which ordinary interconnects become unintended antennas, the distinction between differential-mode and common-mode behavior, the role of return paths and discontinuities, frequency-dependent conductor behavior, and the conventional description of electromagnetic energy transport and radiation.

6.2.1 When a Conductor Becomes Electromagnetically Distributed

At sufficiently low frequency, a conductor or interconnect can often be approximated as a lumped element whose voltage and current are effectively uniform over its physical dimensions. As frequency increases, however, propagation delay becomes significant and the structure must be treated as distributed.

A useful criterion is based on the ratio of physical length L to wavelength λ ,

$$\frac{L}{\lambda}. \quad (6.3)$$

The wavelength is

$$\lambda = \frac{v_p}{f}, \quad (6.4)$$

where v_p is the propagation velocity in the relevant medium and f is frequency.

As L/λ increases, phase differences along the conductor become increasingly important. The conductor can then support traveling waves, reflections, standing-wave patterns, and spatially nonuniform current and voltage distributions [Pozar \(2012a\)](#); [Bogatin \(2018\)](#).

This transition is fundamental to unintended antenna behavior. A cable, trace, shield, enclosure seam, or ground structure need not be

designed as an antenna before it can support an electrically significant current distribution. Once the physical dimensions become sufficiently large relative to wavelength, the structure can interact strongly with its surrounding electromagnetic environment.

6.2.2 Differential-Mode and Common-Mode Excitation

One of the most important distinctions in EMC analysis is between differential-mode and common-mode current.

In an ideal differential structure, equal and opposite currents flow in the signal and return conductors,

$$I_1 = -I_2. \quad (6.5)$$

The associated electric and magnetic fields are concentrated primarily between or near the conductors. Because the opposing currents and charges are closely spaced, far-field radiation can be comparatively small when the geometry is well balanced [Paul \(2006\)](#); [Ott \(2009\)](#).

Common-mode excitation occurs when currents on nominally paired conductors are no longer equal and opposite. A common-mode current may be represented as

$$I_{CM} = \frac{I_1 + I_2}{2}, \quad (6.6)$$

while the differential-mode current may be represented as

$$I_{DM} = \frac{I_1 - I_2}{2}. \quad (6.7)$$

These relationships show that imbalance between the two conductors produces a current component that is not locally canceled by an equal opposing current.

This distinction is central to unintended radiation because a common-mode current can involve a much larger effective current path or radiating structure than the intended differential geometry. Cables, shields, chassis structures, reference planes, and attached conductors can then participate in the electromagnetic system.

Relatively small common-mode currents can therefore produce substantial emissions when

they excite an electrically long structure [Ott \(2009\)](#); [Paul \(2006\)](#).

6.2.3 Return Paths and Electromagnetic Geometry

The phrase “current path” is incomplete unless the return path is also specified. At high frequency, return current does not necessarily follow the geometrically shortest conductive route. Instead, the electromagnetic system establishes a return distribution determined by impedance and strongly influenced by inductance, capacitance, conductor geometry, and nearby reference structures [Bogatin \(2018\)](#); [Paul \(2006\)](#).

For a well-controlled transmission structure, the signal and return paths remain close together, reducing loop area and confining much of the electromagnetic field.

If the return path is interrupted, displaced, or forced around a discontinuity, the effective loop area increases. The surrounding electromagnetic field then occupies a larger region, increasing the opportunity for coupling and radiation.

For the narrow-column format, this relationship is conveniently written as

$$\begin{aligned} \text{return separation} \uparrow &\rightarrow \text{loop area} \uparrow \\ &\rightarrow \text{coupling tendency} \uparrow. \end{aligned} \quad (6.8)$$

This relationship is especially important in printed-circuit boards, cable assemblies, connectors, and enclosure transitions. A slot in a reference plane, a poorly bonded shield, or a connector transition can force the return current away from its preferred local path and thereby convert an otherwise controlled structure into a stronger unintended radiator.

6.2.4 Transmission-Line Behavior and Reflections

When an interconnect becomes electrically distributed, its voltage and current may be represented as combinations of forward- and backward-traveling waves. For an ideal transmission line,

$$V(z) = V^+ e^{-j\beta z} + V^- e^{j\beta z}, \quad (6.9)$$

and

$$I(z) = \frac{V^+}{Z_0} e^{-j\beta z} - \frac{V^-}{Z_0} e^{j\beta z}, \quad (6.10)$$

where Z_0 is the characteristic impedance and β is the phase constant Pozar (2012a).

At a load impedance Z_L , the voltage reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \quad (6.11)$$

Equation 6.11 shows that impedance discontinuities produce reflected electromagnetic energy whenever the terminating impedance differs from the characteristic impedance.

In signal-integrity analysis, the resulting reflections can produce overshoot, undershoot, ringing, multiple transitions, and timing errors. In EMC analysis, the same reflected and standing-wave behavior can increase local current or voltage maxima and thereby enhance coupling or radiation.

Thus, impedance discontinuities are important not only because they distort signals but also because they redistribute electromagnetic energy spatially along the structure.

6.2.5 Standing Waves and Resonance

Forward and reflected waves can combine to produce standing-wave patterns. Under resonant conditions, relatively large local current or voltage amplitudes can develop even when the source excitation is modest.

The resonance condition depends on geometry, boundary conditions, loading, and propagation velocity. A simplified open or shorted structure may exhibit strong resonances when its electrical length approaches fractions of a wavelength, such as

$$L \sim \frac{\lambda}{4} \quad \text{or} \quad L \sim \frac{\lambda}{2}, \quad (6.12)$$

depending on the specific boundary conditions.

These relationships explain why cables, traces, enclosure seams, and other conductors can become efficient unintended antennas at particular frequencies. The physical structure may not have been designed for resonance, yet its dimensions and terminations can satisfy electromagnetic boundary conditions that produce strong current or voltage distributions.

The result can be enhanced radiation, susceptibility, or coupling to nearby structures.

6.2.6 Parasitic Capacitance and Inductance

No practical conductor exists in electromagnetic isolation. Nearby conductors, dielectric materials, chassis surfaces, ground planes, and enclosures produce distributed capacitance and inductance.

A small parasitic capacitance C_p has impedance

$$Z_C = \frac{1}{j\omega C_p}, \quad (6.13)$$

while a parasitic inductance L_p has impedance

$$Z_L = j\omega L_p. \quad (6.14)$$

As frequency increases, capacitive impedance decreases while inductive impedance increases. Structures that appear electrically isolated at low frequency can therefore become strongly coupled at high frequency, while seemingly negligible conductor lengths can develop substantial inductive voltage drops.

The resonant frequency of an idealized parasitic LC combination is

$$f_0 = \frac{1}{2\pi\sqrt{L_p C_p}}. \quad (6.15)$$

Equation 6.15 illustrates why unintended resonances can arise from ordinary physical construction. Connector pins, package leads, vias, cable shields, enclosure seams, and board geometries all contain distributed inductive and capacitive behavior.

6.2.7 Near-Field Electric and Magnetic Coupling

Electromagnetic coupling between structures can occur before either structure behaves as an efficient far-field antenna. In the near field, electric and magnetic coupling are often analyzed separately according to the dominant source geometry.

Electric coupling is associated primarily with voltage and capacitance. A time-varying voltage can drive displacement current through a mutual capacitance,

$$i_C = C_m \frac{dv}{dt}, \quad (6.16)$$

where C_m is the mutual capacitance between the structures.

Magnetic coupling is associated primarily with current and mutual inductance. A changing current in one circuit can induce voltage in another according to

$$v_M = M \frac{di}{dt}, \quad (6.17)$$

where M is the mutual inductance.

Equations 6.16 and 6.17 provide the conventional first-order descriptions of electric and magnetic crosstalk.

Both coupling mechanisms become increasingly important as edge rates become faster because large dv/dt and di/dt can occur even when the fundamental repetition frequency is relatively low.

For two nearby transmission lines, the coupled disturbance is commonly observed as an unwanted voltage waveform on the victim line. Near-end crosstalk, NEXT, is the coupled disturbance measured at the end of the victim line nearest the source end of the aggressor line, whereas far-end crosstalk, FEXT, is measured at the opposite end of the victim line.

Although crosstalk is commonly measured as a voltage, its physical origin is not exclusively voltage based. A changing voltage on the aggressor produces capacitive, or electric-field, coupling through the mutual capacitance between the conductors, while the associated changing current produces inductive, or

magnetic-field, coupling through their mutual inductance. NEXT and FEXT therefore result from the combined distributed electric and magnetic coupling along the coupled length of the transmission lines [Paul \(2006\)](#); [Bogatin \(2018\)](#).

The relative magnitudes and polarities of NEXT and FEXT depend on conductor geometry, propagation velocity, coupling length, termination conditions, and the relationship between mutual capacitance and mutual inductance. Because voltage and current in a propagating transmission-line mode are related through the characteristic impedance, the electric and magnetic coupling contributions are normally coupled aspects of the same electromagnetic propagation process.

6.2.8 Frequency Content and Edge Rate

Digital signals are not single-frequency waveforms. A rapidly changing transition contains a broad spectrum of frequency components, and the highest significant frequencies can be much greater than the nominal clock or repetition frequency.

Consequently, an interconnect that appears electrically short at the fundamental frequency can behave as a distributed electromagnetic structure for the higher-frequency components of a fast edge [Bogatin \(2018\)](#).

This distinction is important for EMC because unwanted radiation is often governed more strongly by transition time than by nominal clock frequency. Faster edges increase high-frequency spectral content, which can excite smaller structures and shorter unintended antennas.

The progression may be summarized as

$$\begin{aligned} \text{faster edge} &\rightarrow \text{higher frequencies} \\ &\rightarrow \text{shorter wavelengths} \\ &\rightarrow \text{greater distributed behavior.} \end{aligned} \quad (6.18)$$

The physical dimensions of the system must therefore be compared with the wavelengths of the significant spectral components rather than with the clock frequency alone.

6.2.9 Frequency-Dependent Current Localization

Frequency also changes the distribution of current within the conductor itself. For a good conductor, the conventional skin depth is

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}, \quad (6.19)$$

where ω is angular frequency, μ is permeability, and σ is conductivity. At 1MHz, the skin depth of copper is approximately 66 microns.

Equation 6.19 shows that skin depth decreases approximately as the inverse square root of frequency. Consequently, increasing frequency progressively concentrates current closer to conductor surfaces.

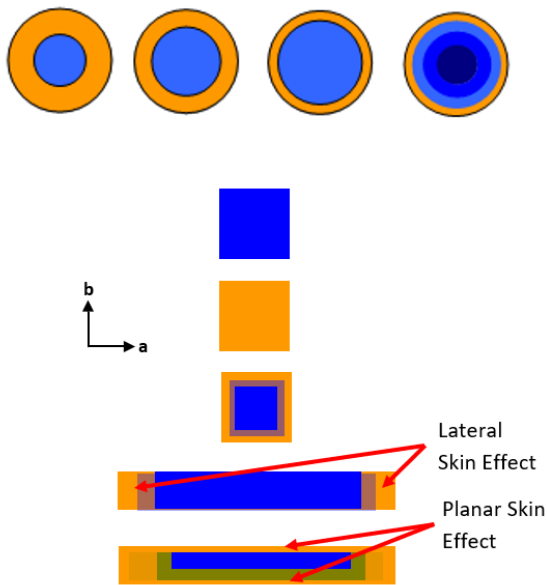


Figure 6.1: Frequency-dependent current localization in conducting structures. Adapted from Figures 91 and 93 in *Photony Foundations* Martin (2022).

Figure 6.1 The distribution of current within a conductor changes with frequency. At sufficiently low frequency, current can occupy much of the available cross-section. As frequency increases, the highest current density moves progressively toward the surface.

Panel (a) illustrates the conventional progression of current density in a round conduc-

tor cross-section from a broadly distributed low-frequency condition (left) toward increasing surface localization (right) as frequency rises.

Panel (b) illustrates how conductor cross-sectional geometry modifies that localization in rectangular cross-sections, including lateral and planar skin-effect behavior. For rectangular Panel B structures, the geometry produces nonuniform localization across different dimensions, so the effective current distribution can differ substantially from that of a round wire.

Lateral vs. Planar Skin EFFECT: In Panel B, the top square conductor shows no current (blue). The next square conductor shows direct current (DC) (orange). The following square conductor shows the skin effect (blue + orange). The top rectangular conductor shows lateral skin effect (orange at the ends). The bottom rectangular conductor shows both lateral and planar skin effect. The lateral skin effect appears first as the frequency increases past the skin effect start frequency, which is dimensionally dependent. The skin depth start frequency corresponds to a skin depth of 1/2 the dimension of interest. A larger conductor thickness (in this case, the width (a) or x-dimension) corresponds to a lower start frequency.

These effects are particularly relevant to cables, traces, shields, bus conductors, and other structures encountered in electromagnetic compatibility and signal-integrity analysis because increasing frequency changes both the effective conducting region and the spatial distribution of the associated electromagnetic fields.

This behavior is particularly important for EMC and signal integrity because many practical high-frequency conductors are not round. Printed-circuit traces, planes, shields, enclosure walls, connector structures, and bus conductors have strongly geometry-dependent current distributions. Changes in the location and concentration of current alter resistance, inductance, loss, field confinement, return-path behavior, and the degree to which neighboring structures can be electromagnetically coupled.

The frequency-dependent surface localization illustrated in Figure 6.1 also becomes important when the conventional behavior is reinterpreted in terms of electron transduction and the

spatial relationship between current-associated magnetic organization and surface-associated electric organization.

6.2.10 Conventional Radiation from an Unintended Structure

Once currents and charges are distributed over an electrically significant structure, the system can radiate electromagnetic energy.

The conventional energy-flux description remains the Poynting vector,

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad (6.20)$$

with total outward radiated power given by

$$P_{\text{rad}} = \oint_A \langle \mathbf{S} \rangle \cdot d\mathbf{A}. \quad (6.21)$$

Equations 6.20 and 6.21 apply whether the structure was intentionally designed to radiate or became an antenna accidentally.

This is an important point. Maxwell's equations do not distinguish between an intended antenna and an unintended one. The distinction is an engineering distinction involving geometry, function, excitation, and design intent.

6.2.11 Conventional Constraints on the Proposed Interpretation

Any structural interpretation of EMC and signal-integrity behavior must reproduce several established results.

First, electrically long conductors support distributed voltage and current rather than lumped behavior. Second, impedance discontinuities produce reflections and standing waves. Third, differential-mode and common-mode currents can have substantially different radiation characteristics. Fourth, return-path geometry strongly influences electromagnetic confinement and coupling. Fifth, parasitic capacitance and inductance become increasingly important as frequency rises. Sixth, increasing frequency alters the current distribution within the conductor through skin effect and related geometry-dependent localization. Seventh, near-field coupling can occur through both electric and mag-

netic mechanisms. Finally, unintended radiation must remain consistent with electromagnetic energy conservation and the outward Poynting flux described by Equation 6.21.

The Beyond Field-Line interpretation developed in the following sections does not alter these relationships. Instead, it asks how the same behavior may be understood structurally in terms of dynamic elemental charge-photon flow, surface accumulation, electron transduction, electric and magnetic organization, redistribution at discontinuities, frequency-dependent surface localization, and unintended electromagnetic release.

6.3 Electron Transduction in Unintended Electromagnetic Structures

The conventional analysis of Section 6.2 establishes the measurable behavior that any proposed structural interpretation must reproduce. Electrically significant conductors support distributed voltage and current, discontinuities produce reflection and redistribution, changing voltages and currents produce capacitive and inductive coupling, common-mode currents can radiate efficiently, and increasing frequency alters both current localization and electromagnetic confinement.

Within the proposed Photony framework, these established behaviors are interpreted in terms of electron-mediated electric chain and magnetic chain organization. The purpose of this section is not to replace conventional transmission-line, antenna, EMC, or signal-integrity theory. Rather, it asks what physical reorganization may accompany the measured voltage, current, coupling, and radiation described by those theories.

The central distinction developed earlier in this Compendium remains applicable. Dynamic elemental charge photons participating in current transport are proposed to interact predominantly with internal electrons that transduce magnetic chain organization. Static elemental charge photons associated with surface accumu-

lation are proposed to interact predominantly with surface-associated electrons that transduce electric chain organization [Martin \(2022\)](#).

Thus, an electrically active conductor can support both forms of organization simultaneously:

current transport → magnetic chain organization,
surface potential → electric chain organization.
(6.22)

In a well-controlled interconnect, geometry constrains these organizations so that electromagnetic energy remains predominantly associated with the intended forward and return structure. In an unintended electromagnetic structure, discontinuities, asymmetry, changing return geometry, resonance, or electrical length can permit that organization to redistribute into regions and conductors that were not intended to participate.

6.3.1 Electric chain and Magnetic chain Organization in Distributed Conductors

A conductor does not become electromagnetically distributed merely because its physical construction changes. It becomes distributed when propagation delay, wavelength, geometry, and frequency cause voltage and current to vary significantly with position. Conventional transmission-line theory describes this behavior through distributed inductance, capacitance, resistance, conductance, characteristic impedance, and propagation velocity.

Within the proposed Photony interpretation, the same transition can be viewed as a spatial variation in electric chain and magnetic chain organization along the structure.

Where local voltage increases, surface accumulation and the associated electric chain organization can increase. Where local current increases, internal-electron transduction and the associated magnetic chain organization can increase. A propagating signal therefore does not require a uniform electromagnetic state along the conductor. The local organization changes with position and time as the waveform propagates.

Reflections provide a particularly useful example. At an impedance discontinuity, conventional theory requires incident and reflected voltage and current waves whose amplitudes are determined by the reflection coefficient. Within Photony, the same event corresponds to redistribution of the local electric chain and magnetic chain organization required to satisfy the new boundary condition.

The discontinuity therefore does not create a separate class of electromagnetic physics. It changes the organization permitted by the geometry and impedance of the structure.

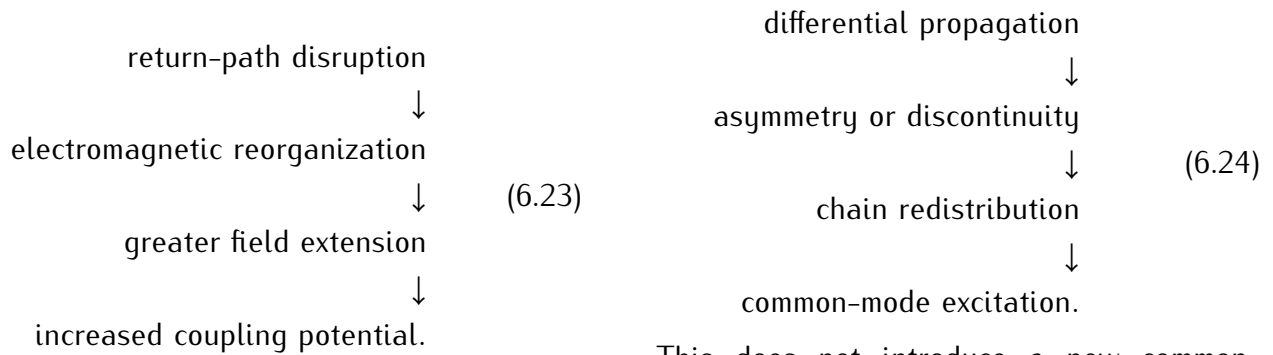
6.3.2 Return-Path Disruption and Electromagnetic Reorganization

Return-path geometry is one of the most important determinants of electromagnetic confinement. For a transmission structure with a continuous, closely coupled return path, the electromagnetic fields associated with forward and return currents remain comparatively confined to the intended geometry [Paul \(2006\)](#); [Ott \(2009\)](#); [Bogatin \(2018\)](#).

A slot in a reference plane, an interrupted shield, a connector transition, a poorly placed return via, or a change between reference structures forces the return current to follow a different distributed-impedance path. The resulting increase in loop area can increase inductance, field extension, coupling, and common-mode excitation.

Within the proposed Photony interpretation, such a return-path discontinuity requires both magnetic chain and electric chain reorganization. Current-associated magnetic chain organization must adapt to the altered current path, while surface charge and electric chain organization must redistribute to establish the voltage conditions required by the new geometry.

The important structural sequence is therefore



This interpretation is consistent with the conventional result that a poor return path can degrade signal integrity and increase electromagnetic emissions even when the forward conductor itself is unchanged.

6.3.3 Differential-to-Common-Mode Conversion

Differential-mode propagation is comparatively well confined when equal and opposite currents occupy closely coupled conductors with appropriate return geometry. Common-mode excitation, by contrast, can involve current flowing with respect to a much larger surrounding structure, including shields, chassis, cables, enclosure surfaces, or parasitic return paths.

Conventional EMC analysis identifies asymmetry as an important mechanism for conversion between differential and common modes. Unequal impedances, unbalanced geometry, connector transitions, reference-plane changes, and parasitic coupling can prevent the two sides of a nominally differential structure from responding identically.

Within the proposed Photony framework, mode conversion corresponds to a redistribution of the electromagnetic organization that had previously been confined to the differential structure. Electric chain and magnetic chain organization that was approximately balanced between closely coupled paths can become asymmetrically distributed. Part of that organization can then extend onto larger conductive structures.

The progression can be represented as

This does not introduce a new common-mode law. The measurable voltage, current, impedance, and radiation remain those of conventional EMC theory. The Photony interpretation proposes a physical reorganization beneath those measured quantities.

6.3.4 Connectors, Vias, Shields, and Chassis Transitions

Connectors, vias, shield terminations, reference-plane transitions, and chassis interfaces are especially important because they are regions in which electromagnetic geometry changes abruptly.

A schematic may show a signal conductor passing continuously through such a transition, but the surrounding electromagnetic structure may change substantially. The return current may be forced onto a different reference surface, the local impedance may change, parasitic capacitance and inductance may become significant, and the spatial separation between forward and return currents may increase.

Within the proposed Photony framework, these locations are interpreted as electromagnetic reorganization regions. Internal-electron magnetic chain transduction must conform to the changing current distribution, while surface-associated electric chain transduction must conform to the changing surface-potential and charge distribution.

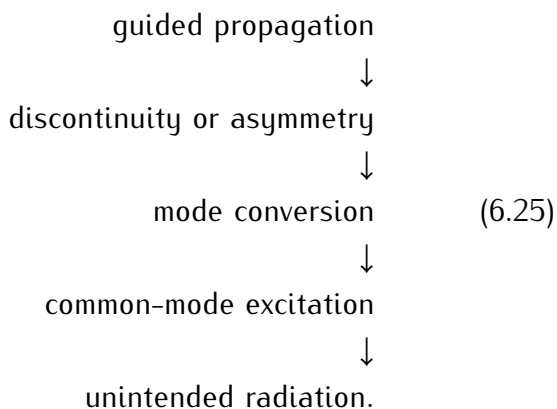
A transition that preserves the intended electromagnetic geometry allows the organization to continue with relatively little redistribution. A transition that introduces asymmetry, excess loop area, or an abrupt impedance change can cause part of the organized energy to occupy additional structures.

This interpretation helps explain why apparently small physical details can have large EMC consequences. A connector shell, via field, shield pigtail, reference-plane opening, or chassis bond may occupy only a small fraction of the physical system while determining how the electromagnetic organization is transferred between much larger structures.

6.3.5 Unintended Launch and Reception

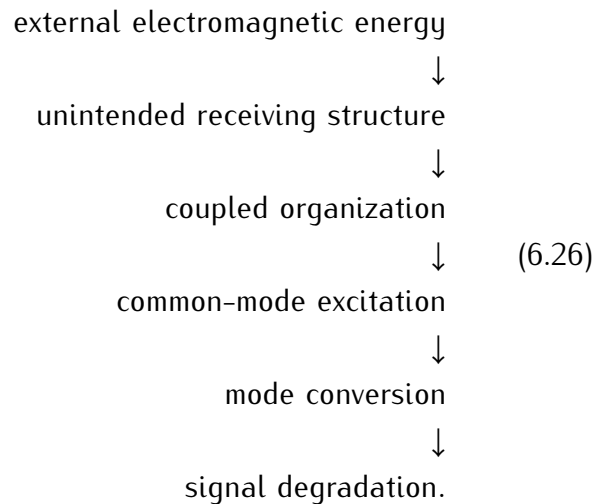
An unintended antenna can operate in either direction. A structure that allows electromagnetic energy to leave the intended system can often also allow external electromagnetic energy to enter it.

For emission, the conventional progression may be written as



Within the proposed Photony interpretation, this sequence represents electromagnetic organization that was initially confined to the intended interconnect becoming redistributed onto a structure capable of coupling energy into the surrounding region.

The reverse process describes susceptibility:



The same cable, trace, shield, enclosure seam, or chassis structure can therefore participate in both egress and ingress. Conventional reciprocity and antenna theory provide the quantitative basis for this behavior. The Photony interpretation adds the structural view that electromagnetic organization can be launched from or transduced into the same physical geometry depending on the direction of energy transfer.

6.3.6 Frequency-Dependent Transduction and Current Localization

Frequency changes not only the electrical length of a structure but also the region of the conductor in which current is concentrated. As shown in Figure 6.1, conventional skin effect progressively moves current toward conductor surfaces as frequency increases.

This has an important consequence for the proposed Photony interpretation. At sufficiently low frequency, current-associated internal-electron transduction can occupy a substantial fraction of the available conductor cross-section. As frequency increases and current becomes increasingly surface localized, the region in which magnetic chain transduction is most strongly supported likewise becomes increasingly concentrated toward the surface.

At the same time, surface-associated electric chain organization is already linked to voltage and surface charge. Increasing frequency therefore brings the dominant current-associated magnetic organization and the

surface-associated electric organization into increasingly close spatial proximity.

This does not alter the conventional skin-depth relation,

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}, \quad (6.27)$$

which remains the quantitative description of current localization. The Photony interpretation asks what structural change accompanies that measured redistribution.

This frequency dependence is particularly important in traces, shields, connectors, planes, enclosure walls, and other EMC structures because the same increase in frequency that localizes current also reduces wavelength, increases distributed behavior, strengthens the influence of parasitic geometry, and allows smaller structures to participate in coupling and radiation.

Thus, frequency changes both the dimensions that are electromagnetically significant and the physical region within those structures where electron-mediated transduction is proposed to occur.

6.4 Discussion

The analysis of this chapter shows that signal integrity and electromagnetic compatibility are closely related consequences of distributed electromagnetic behavior. A discontinuity that produces reflection, ringing, overshoot, or timing error can also redistribute voltage and current in ways that promote common-mode excitation and radiation. Conversely, an external field can couple to the same structures and degrade the intended signal.

Conventional transmission-line theory, Maxwellian electrodynamics, impedance analysis, parasitic modeling, antenna theory, and Poynting energy transport remain the quantitative foundation for these effects. The proposed Photony framework adds a structural interpretation in which changes in geometry, return-path continuity, frequency, and mode are accompanied by redistribution of electric chain and magnetic chain organization through surface and internal electron transduction.

The practical implication is that electromagnetic design cannot be reduced to the forward signal conductor alone. Signal paths, return paths, impedance transitions, connectors, vias, shields, chassis structures, enclosure apertures, and neighboring conductors together determine where electromagnetic energy is guided, stored, coupled, reflected, or radiated.

6.5 Conclusions

Unintended antennas, electromagnetic compatibility, and signal integrity are closely related consequences of distributed electromagnetic behavior. Conductors designed to carry power or information can become electrically significant structures when their dimensions, geometry, return paths, frequency content, discontinuities, or surrounding environment permit nonuniform voltage and current distributions to develop.

Conventional theory provides the established quantitative description of these processes. Transmission-line equations describe propagation and reflection, mutual capacitance and inductance describe near-field coupling, common-mode and differential-mode analysis describe current organization, skin effect describes frequency-dependent current localization, and the Poynting vector describes electromagnetic energy transport and radiation. These relationships remain fully applicable whether a structure is intended to radiate or becomes an antenna unintentionally.

Several practical conclusions follow. Controlled forward and return paths reduce loop area and help confine electromagnetic energy. Impedance discontinuities redistribute energy and can produce reflections, standing waves, and resonant enhancement. Common-mode current is particularly important because even relatively small current amplitudes can excite large electrically significant structures. Fast edge rates increase high-frequency spectral content and therefore increase the number of physical structures that can behave as distributed electromagnetic elements.

The principal engineering objective is consequently not merely to suppress fields or currents

in isolation, but to control the complete electromagnetic geometry of the system. Signal paths, return paths, impedance transitions, connectors, vias, shields, chassis structures, enclosure apertures, and neighboring conductors all participate in determining where electromagnetic energy is guided, stored, coupled, reflected, or radiated.

Chapter 6 therefore extends the earlier treat-

ment of electromagnetic energy transport into the practical domains of signal integrity and electromagnetic compatibility. It also establishes the foundation for the final synthesis in Chapter 7, where electric chain organization, magnetic chain organization, alternating transport, coupling, radiation, and energy release are brought together within a unified Photony interpretation.

Chapter 7

Synthesis: A Unified Photony View of Electromagnetic Systems

Purpose: To synthesize the preceding treatment of electric organization, magnetic organization, alternating transport, coupling, radiation, and threshold energy release into a unified Photony interpretation of electromagnetic systems. The chapter advances from individual field phenomena toward a system-level view in which geometry, conductivity, electron transduction, energy storage, propagation, concentration, and release form a continuous physical sequence extending from ordinary electrical devices to large-scale solar and terrestrial electromagnetic events.

7.1 Introduction

Electromagnetic systems are usually introduced through separate concepts: electric fields, magnetic fields, voltage, current, capacitance, inductance, radiation, transmission lines, antennas, breakdown, and energy dissipation. Each concept is useful, and the conventional equations describing them are among the most successful tools in physics and electrical engineering. Nevertheless, a physical system does not experience these quantities as separate textbook topics. They occur together, evolve together, and exchange energy continuously.

A capacitor does not know that it belongs to electrostatics. An inductor does not know that it belongs to magnetics. A transmission line does not decide where circuit theory ends and field theory begins. A spark does not wait for an engineer to classify it as dielectric breakdown, and a solar flare does not distinguish between plasma physics, magnetohydrodynamics, radiation physics, and electrodynamics. Nature presents one evolving electromagnetic system.

The central objective of this chapter is to view that system as a whole.

The preceding chapters developed several

pieces of this picture. Electric organization was associated with voltage, charge separation, surface conditions, and electric energy storage. Magnetic organization was associated with current, internal electron transduction, and magnetic energy storage. Alternating-current operation showed that these organizations can grow, reduce, reverse, and rebuild repeatedly without instability. Signal-integrity and electromagnetic-compatibility analysis showed that geometry determines whether electromagnetic energy remains guided, becomes redistributed, couples into neighboring structures, or escapes as radiation. Electric and magnetic breakdown introduced the additional possibility that organization can be driven beyond a stable operating regime and release energy rapidly.

The synthesis proposed here is that these are not isolated phenomena. They are different outcomes of a common electromagnetic architecture.

Within the proposed Photony framework, that architecture begins with elemental energy organization. Dynamic elemental charge photons participate in current transport and are proposed to support magnetic chain formation through predominantly internal-electron transduction. Static elemental charge photons accu-

mulate primarily at surfaces and are proposed to support electric chain formation through predominantly surface-associated electron transduction [Martin \(2022, 2026b\)](#). Electric and magnetic structures can then extend through space, interact with matter, exchange energy, and reorganize as the macroscopic electrical system evolves.

The conventional quantities remain unchanged. Voltage is still measured in volts, current in amperes, electric and magnetic fields retain their established definitions, Maxwell's equations remain the governing macroscopic field relations, and the Poynting vector remains the conventional description of electromagnetic energy flux [Maxwell \(1865\)](#); [Poynting \(1884\)](#); [Griffiths \(2017\)](#); [Jackson \(1998a\)](#). Photony is introduced here as a proposed physical interpretation beneath those quantities, not as a replacement for them.

The important change is one of perspective.

Instead of beginning with the question, “What field exists here?”, the Photony interpretation encourages a sequence of deeper questions: What electromagnetic organization has been established? What maintains it? Where is its energy concentrated? How is that organization connected to surrounding structures? What happens when its loading increases, decreases, reverses, or loses continuity?

These questions connect phenomena that otherwise appear unrelated.

The ignition coil in an automobile stores magnetic energy and transforms a low-voltage electrical input into a high-voltage condition capable of producing electric breakdown across a spark-plug gap. A solar active region stores enormous magnetic free energy in stressed, current-supported magnetic structures. When those structures become unstable, part of that magnetic energy can be rapidly converted into heating, particle acceleration, radiation, plasma motion, and large-scale disturbances. Those disturbances can eventually interact with Earth's magnetosphere and induce electric fields in long terrestrial conductors. In extreme cases, the resulting voltages and currents can contribute to insulation stress, heating, arcing, equipment damage, and ignition [Martin \(2026a\)](#).

The scales differ enormously. The governing

geometries differ. The available energy differs by orders of magnitude. Yet an important pattern survives:

- energy input → electromagnetic organization
- energy concentration
- redistribution or threshold
- energy release.

(7.1)

This sequence is one of the most important ideas developed in this Compendium. The detailed mechanism depends on material, conductivity, geometry, frequency, boundary conditions, and scale, but electromagnetic systems repeatedly organize energy before they transport or release it.

The chapter therefore moves beyond the individual mechanisms developed earlier and asks what can be learned by viewing electromagnetic systems as connected structures of energy organization. Particular attention is given to the relationship between electric and magnetic organization, the role of electrons as proposed transducers, the importance of geometry, the distinction between stable cycling and threshold release, and the striking continuity between familiar engineered devices and large-scale natural electromagnetic systems.

Terminology: electromagnetic system, electromagnetic organization, energy architecture, energy loading, energy concentration, energy storage, energy transport, energy redistribution, energy release, electric organization, magnetic organization, electric chains, magnetic chains, magnetic fibrils, static elemental charge photons, dynamic elemental charge photons, classical photon, electron transduction, surface-electron transduction, internal-electron transduction, transduction, reduction, reversal, rebuilding, chain continuity, chain density, electric-field energy, magnetic-field energy, Poynting flux, current, voltage, capacitance, inductance, conductivity, impedance, geometry, boundary condition, return path, coupling, radiation, electric breakdown, voltage breakdown, magnetic breakdown, magnetic reassignment, instability threshold, stable operation, source-

coupled organization, free-space propagation, electromagnetic compatibility, geomagnetic induction, geomagnetically induced current, solar magnetic energy, solar storm, terrestrial induction, ignition, arcing, fire

7.2 From Components to Electromagnetic Pathways

The preceding chapters examined electric and magnetic behavior through individual physical mechanisms. The synthesis developed here begins with a different question: instead of asking what each component does, ask what path the electromagnetic energy follows.

This change in viewpoint is important. A circuit diagram divides a system into resistors, capacitors, inductors, conductors, switches, loads, and sources because that representation is extremely useful for calculation. The electromagnetic system itself is not divided in that way. Energy moves through a continuously connected physical environment whose behavior is determined by geometry, materials, boundaries, frequency, conductivity, surface conditions, and available return paths.

A useful system-level sequence is

$$\begin{aligned} \text{source} &\rightarrow \text{organization} \\ &\rightarrow \text{transport} \\ &\rightarrow \text{interaction} \\ &\rightarrow \text{destination.} \end{aligned} \quad (7.2)$$

The destination may be a conventional load, but it may also be a dielectric gap, another conductor, a capacitor, an antenna, a plasma, a shield, a resonant structure, or surrounding free space.

Within conventional electromagnetics, this pathway is constrained by Maxwell's equations, constitutive relations, boundary conditions, circuit relations, and energy conservation. The Poynting vector,

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad (7.3)$$

provides the established description of electromagnetic energy flux [Poynting \(1884\)](#); ?.

Within the proposed Photony interpretation, the same pathway is viewed as an evolving physical organization of elemental charge photons and the electron-mediated structures they support. The conductor is therefore not treated simply as a pipe through which electromagnetic energy passes. Rather, it helps establish the conditions under which electromagnetic organization can form, propagate, couple, reduce, reverse, or become released.

This leads to an important distinction:

the conductor helps define the path;
it is not the complete path

(7.4)

The electromagnetic pathway usually includes the space surrounding the conductors, the dielectric regions between them, the return structure, and any neighboring matter with which the fields interact.

The Photony interpretation therefore shifts attention from isolated components to connected electromagnetic architecture.

7.3 The Life Cycle of Electromagnetic Organization

A second unifying idea is that electromagnetic organization has a life cycle.

It is established, strengthened, maintained, transported, reduced, redistributed, transformed, or released.

This life cycle appears in systems that otherwise look unrelated.

In a capacitor, electric chain organization increases as voltage rises, reaches a maximum near the voltage maximum, decreases as energy is returned, and reverses as polarity changes.

In an inductor, magnetic chain organization increases with current, reaches a maximum near the current maximum, reduces as current falls, and rebuilds with opposite orientation after reversal.

In a transmission line, electric chain and magnetic chain organization propagate together while maintaining a relationship established by the line geometry and characteristic impedance.

In an antenna, part of the source-coupled organization becomes separated from the source structure and propagates outward.

In electric breakdown, charge-supported electric chain organization increases until the medium can no longer sustain the existing non-conducting state.

In magnetic breakdown, the proposed failure occurs within an already current-supported magnetic structure whose organization has exceeded its stable capacity.

These processes can be summarized as

formation → growth → maintenance
 → reduction or redistribution
 → return, propagation, or release.
 (7.5)

This sequence is more general than any single circuit element.

Within Photony, the physical object of interest is therefore not merely the field value at one position and time, but the evolving organization that produces that field behavior.

7.4 Electric and Magnetic Chain Organization Are Complementary

One of the most important conclusions of this Compendium is that electric and magnetic chain organization should not be treated as competing explanations.

They are complementary forms of electric chain and magnetic chain organization.

The conventional electric energy density is

$$u_E = \frac{1}{2} \epsilon E^2, \quad (7.6)$$

while the conventional magnetic energy density is

$$u_B = \frac{B^2}{2\mu}. \quad (7.7)$$

These relationships remain the measurable energy constraints on any proposed microscopic interpretation.

Within Photony, the distinction lies principally in how the corresponding electric chain and magnetic chain organization is established.

Static elemental charge photons accumulated predominantly at surfaces are proposed to participate in electric chain organization through surface-associated electron transduction.

A conductor carrying current can simultaneously possess a surface-charge distribution that establishes an electric field and an internal current distribution that supports magnetic chain organization. [Martin \(2022, 2026b\)](#).

The macroscopic system normally contains both.

A conductor carrying current can simultaneously possess a surface-charge distribution that supports electric chain organization and an internal current distribution that supports magnetic chain organization.

The important question is, therefore, not

Is the system electric or magnetic?

but rather

What form of electric chain or magnetic chain organization dominates the local energy, transport, or instability process at this location and time?

That question becomes especially useful when following energy through a system that contains several different physical regimes.

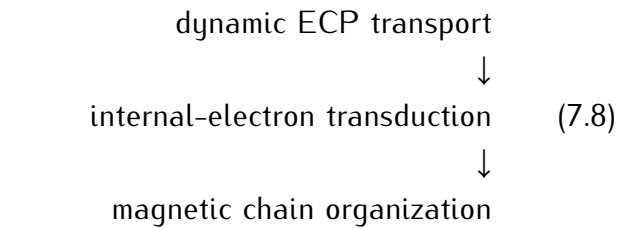
7.5 Electron Transduction as a System-Level Interface

Photony assigns the electron a broader role than that of a simple mobile charge carrier.

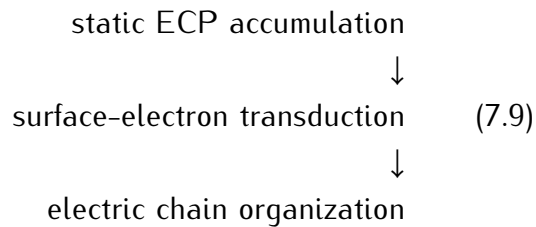
The electron is proposed to function as an electromagnetic transducer and regulator.

This proposed role provides a physical interface between the microscopic transport of elemental charge photons and the macroscopic electric and magnetic organization measured in circuits and fields.

For current-supported magnetic organization,



For surface-supported electric organization,



The importance of this interpretation becomes more apparent at the system level.

A current entering a conductor is not merely associated with a number of electrons drifting through metal. It establishes a distributed electromagnetic condition involving current, magnetic organization, surface charge, electric organization, and surrounding energy transport.

Likewise, a voltage applied between two conductors is not merely a numerical potential difference. It corresponds to a physical redistribution of surface charge and an accompanying reorganization of the surrounding electric structure.

The electron therefore sits at an important boundary between matter and field.

Within Photony, it is one of the principal agents through which matter constructs and responds to electromagnetic organization.

7.6 Geometry Determines What Electromagnetic Organization Is Possible

Geometry is one of the most powerful variables in electromagnetics.

- It determines capacitance.
- It determines inductance.
- It determines characteristic impedance.
- It determines field concentration.
- It determines coupling.
- It determines resonance.

It determines radiation efficiency.

It often determines where breakdown begins.

These effects are conventionally calculated through field solutions, distributed circuit parameters, transmission-line relations, numerical methods, and boundary conditions.

The Photony interpretation adds a structural statement: geometry constrains the configurations that electric and magnetic chains can assume.

A parallel-plate capacitor encourages electric organization between opposing surfaces.

A coil concentrates magnetic organization through and around its turns.

A closely coupled differential pair constrains much of the electromagnetic organization to the region between signal and return conductors.

A large loop permits a larger region of magnetic interaction.

A quarter-wave conductor can establish a standing-wave distribution that supports efficient radiation.

A sharp electrode concentrates electric loading into a small region.

These are very different applications of one principle:

$$\boxed{\begin{array}{c} \text{geometry selects the admissible} \\ \text{chain organization} \end{array}} \quad (7.10)$$

This principle also explains why geometry becomes increasingly important during transients and at high frequency. The system cannot redistribute energy arbitrarily. It must reorganize through pathways allowed by its physical boundaries.

A poorly placed slot in a reference plane, an abrupt connector transition, or an exposed cable shield may therefore alter far more than the apparent shape of the conductor. It changes the electromagnetic architecture available to the system.

7.7 Boundaries Are Conversion Regions

Many of the most important electromagnetic events occur at boundaries.

A boundary may separate conductors from dielectrics, one dielectric from another, a transmission line from a connector, a cable from a chassis, a solid conductor from free space, or one plasma region from another.

At such boundaries, one or more support conditions change.

The conductivity may change.

The permittivity may change.

The permeability may change.

The geometry may change.

The available electron population may change.

The allowable current path may change.

The electromagnetic organization must respond.

Within conventional electromagnetics, this response appears through boundary conditions and discontinuities in field or material properties. Within Photony, the same location becomes a natural site for chain redirection, concentration, transduction, reduction, reassignment, or release.

This suggests a useful broader principle:

boundaries force chain organization to reorganize

(7.11)

The word “renegotiated” is used here descriptively rather than as a formal physical term. It emphasizes that the electromagnetic structure supported on one side of a boundary may not be able to continue unchanged on the other side.

This provides a common way to think about seemingly unrelated phenomena.

A transmission-line reflection occurs because the electromagnetic organization encounters a different impedance.

A dielectric interface redirects electric organization.

A spark gap interrupts conduction until the electric loading becomes large enough to create a conducting pathway.

An antenna end forces the source-supported current and voltage distribution to satisfy a new boundary condition.

A plasma current sheet separates regions of differing magnetic organization.

The details differ, but the structural question is the same: what forms of organization can continue across the boundary?

7.8 Frequency Changes the Physical Region of Interaction

Frequency is normally introduced through waveform repetition, impedance, reactance, phase, or wavelength. At the system level, its influence is even broader.

Frequency determines how quickly electromagnetic organization must be constructed, reduced, and rebuilt.

It determines how far a disturbance propagates during one cycle.

It changes skin depth,

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}, \quad (7.12)$$

and therefore changes the region of a conductor carrying significant current.

It changes capacitive impedance,

$$Z_C = \frac{1}{j\omega C}, \quad (7.13)$$

and inductive impedance,

$$Z_L = j\omega L. \quad (7.14)$$

It changes wavelength and therefore determines whether a physical structure behaves as a lumped component, transmission line, resonator, or antenna.

Within Photony, these conventional results suggest a physical interpretation that is especially important: increasing frequency changes not merely the rate of the electromagnetic process, but also the region in which that process is concentrated.

At low frequency, current can occupy a larger conductor cross-section.

At higher frequency, current becomes more strongly localized near the surface.

As this occurs, current-associated magnetic transduction moves closer to the same boundary

region in which surface charge, electric organization, dielectrics, return paths, nearby conductors, and free-space coupling are important.

This offers a structural reason why high-frequency engineering cannot be understood simply by extending low-frequency circuit intuition.

The physical electromagnetic interaction increasingly becomes a surface, boundary, and geometry problem.

7.9 Electromagnetic Systems Possess Geometric Memory

An electromagnetic system carries the influence of its geometry into its future behavior.

This may be described as *geometric memory*.

The term does not imply a new fundamental memory mechanism. It means that the electromagnetic state that develops at any moment depends on the physical pathways and boundary conditions established by the system.

A coil retains the consequence of its winding geometry through its inductance and magnetic configuration.

A capacitor retains the consequence of its plate area, separation, and dielectric through its capacitance and electric-field distribution.

A transmission line retains the consequence of its conductor geometry through characteristic impedance and propagation velocity.

An antenna retains the consequence of its length and shape through its current and voltage distributions.

A PCB retains the consequence of a reference-plane slot because the return current must reorganize around it.

A stressed solar magnetic structure retains the consequence of prior plasma motion and current organization through its accumulated magnetic free energy and topology.

The importance of this idea appears most strongly during rapid changes.

A transient does not encounter an abstract circuit. It encounters the physical geometry already present.

That geometry determines where energy can travel next.

7.10 Stable Operation Is Continuous Reorganization

A stable electromagnetic system is not necessarily static.

In fact, many of the most important electromagnetic systems operate by continuous reorganization.

An AC circuit repeatedly constructs and reduces electric and magnetic organization.

A resonator repeatedly exchanges energy between electric and magnetic forms.

A transmission line continually establishes advancing electric and magnetic structure.

An antenna continually builds, reduces, reverses, and releases source-supported electromagnetic organization.

Stability therefore should not be confused with lack of motion or lack of change.

A stable system is one in which the required reorganization remains within the available physical pathways.

Using the conceptual threshold notation introduced earlier,

$$\Theta < \Theta_{\text{crit}} \quad (7.15)$$

represents an operating condition in which the organization can continue through admissible states.

Once

$$\Theta \geq \Theta_{\text{crit}}, \quad (7.16)$$

a qualitatively different response may become necessary.

The variable Θ remains a conceptual organizational measure rather than an established directly measurable physical quantity. Its value lies in separating two ideas that are often confused:

large electromagnetic energy does not necessarily imply instability; instability occurs when the available structure can no longer support the required organization.

A superconducting magnet may store enormous energy while remaining stable.

A tiny spark gap may fail with far less stored energy.

The difference lies not merely in how much energy exists, but in whether the local structure can support it.

7.11 Five Possible Destinations for Organized Electromagnetic Energy

Once electromagnetic energy has been organized, it must eventually do something.

At the system level, five broad outcomes are especially useful.

The first is *return*. Energy stored temporarily in electric or magnetic organization can be returned to the source or exchanged with another reactive element.

The second is *useful transfer*. Energy can be delivered to a load and converted into mechanical motion, heat, light, chemical change, information, or another desired form.

The third is *radiation*. Source-supported organization can become outward-propagating electromagnetic energy.

The fourth is *dissipation*. Organized electromagnetic energy can be converted into heat through resistive and other loss mechanisms.

The fifth is *threshold release*. The organization can become unsustainable and undergo electric breakdown, magnetic reassignment, or magnetic breakdown, depending on the physical regime.

This system-level classification may be summarized as

$$\text{organized EM energy} \rightarrow \begin{cases} \text{return,} \\ \text{useful transfer,} \\ \text{radiation,} \\ \text{dissipation,} \\ \text{threshold release.} \end{cases} \quad (7.17)$$

The importance of this classification is that all five outcomes begin with organization.

The observable event is therefore not the entire electromagnetic history.

It is the final stage of an energy pathway.

7.12 The Upstream Energy Source and the Downstream Failure Need Not Match

One of the most important ideas developed in this synthesis is that the form in which electromagnetic energy is initially stored need not be the form in which failure eventually appears.

A magnetic process can produce an electric failure.

The ignition system provides a familiar example.

Current in the primary winding establishes magnetic energy,

$$W_L = \frac{1}{2} L I^2. \quad (7.18)$$

When the current changes rapidly, the changing magnetic state produces an inductive voltage,

$$v = L \frac{di}{dt}. \quad (7.19)$$

Through transformer action, a much larger secondary voltage is established. The spark-plug gap then experiences increasing electric-field loading until the gas becomes conducting.

The final visible event is an electric discharge.

The upstream reservoir was magnetic.

The process can be represented as

$$\begin{aligned} \text{magnetic storage} &\rightarrow \text{rapid magnetic change} \\ &\rightarrow \text{high voltage} \\ &\rightarrow \text{electric breakdown.} \end{aligned} \quad (7.20)$$

This example is valuable because it provides a small, familiar demonstration of a much broader electromagnetic principle.

7.13 From the Sun to a Spark

The same general causal logic can operate across vastly larger scales.

The comparison must be made carefully. A solar active region is not an ignition coil, and the terrestrial power grid is not a spark plug. The geometries, energies, materials, and timescales are entirely different.

What can be compared is the energy path-way.

In the solar atmosphere, conventional solar physics identifies magnetic free energy stored in stressed, current-supported magnetic structures as the principal energy reservoir for major eruptive events. Within the proposed Photony interpretation, that magnetic organization is represented structurally by magnetic chains and larger magnetic fibrils.

The article *From Magnetic and Electric Fields to Fires* develops the causal sequence from solar magnetic energy release, through the propagation of a solar disturbance and interaction with Earth's magnetic environment, to ground-level electric-field induction, voltage stress, arcing, and possible ignition [Martin \(2026a\)](#).

The broad sequence is

solar magnetic energy $\rightarrow$ solar disturbance
 $\rightarrow$ geomagnetic change
 $\rightarrow$ induced electric field
 $\rightarrow$ current and voltage
 $\rightarrow$ electrical stress.

(7.21)

The remarkable feature of this sequence is that a magnetic-energy process far from Earth can eventually produce a terrestrial problem whose immediate failure mechanism is electric.

A transformer insulation system does not fail because it is physically located inside a solar flare.

It fails, if conditions become sufficiently severe, because the solar-driven disturbance has been transformed through several stages into a local electrical condition that the terrestrial system can no longer support.

That distinction is critical.

The initiating mechanism and the final failure mechanism can be different.

7.14 The Ignition Coil and the Solar Storm

The ignition-coil and solar-storm examples can now be compared at the level of electromagnetic logic.

For the ignition system,

magnetic storage $\rightarrow$ induced voltage
 $\rightarrow$ electric breakdown.
 (7.22)

For the solar-terrestrial system,

solar magnetic release
↓
geomagnetic induction (7.23)
↓
voltage stress

The scales differ enormously, but the conceptual lesson is the same:

$$\left. \begin{array}{l} \text{upstream magnetic energy} \\ \rightarrow \text{downstream electric stress} \end{array} \right\} \quad (7.24)$$

This is one of the most useful unifying ideas in the entire Compendium.

It explains why an event should not be classified solely by its final visible manifestation.

A spark is electric breakdown, but its energy may have been supplied through an upstream magnetic process.

A terrestrial arc during an extreme geomagnetic disturbance is an electric event at the location of failure, even though the distant energy pathway began with solar magnetic activity.

The distinction between energy source and failure location therefore becomes essential in both engineering and natural systems.

7.15 Conductivity Helps Select the Dominant Regime

Conductivity strongly influences what form of electromagnetic organization a system can sustain.

The conventional dielectric relaxation time is

$$\tau = \frac{\epsilon}{\sigma}, \quad (7.25)$$

where ϵ is permittivity and σ is conductivity.

Low conductivity permits separated charge to persist for relatively long periods. This favors electric-field buildup and substantial voltage differences across insulating or weakly conducting regions.

High conductivity permits charge to redistribute rapidly and supports larger current flow. Under such conditions, large current-supported magnetic structures can become the dominant energy reservoir.

The distinction is therefore not simply

electric material versus magnetic material.

A more useful question is

Which type of electromagnetic organization can this environment sustain and accumulate most effectively?

Within Photony, this becomes a regime-selection question.

Low-conductivity environments tend to favor persistent electric chain loading.

Highly conducting environments can favor current-supported magnetic chain loading.

This distinction helps explain why breakdown through air or insulation and magnetic instability in a highly conducting plasma can belong to a unified energy-release framework without being the same physical process.

Conductivity is not the only selector. Geometry, loading history, current density, curvature, boundary conditions, material state, and available energy also matter.

Nevertheless, conductivity provides one of the clearest bridges between the electric and magnetic branches.

7.16 Failure Often Occurs Where Energy Is Concentrated, Not Where It Originated

Energy source and damage location are frequently separated.

A power supply may be physically distant from the insulation defect that arcs.

A switching transistor may create the transient that produces destructive overvoltage at a connector.

A common-mode current may originate from an asymmetry on a circuit board but radiate efficiently from an attached cable.

A solar disturbance begins millions of kilometers from Earth, while its effects can become concentrated in a transformer winding or grounding network.

This suggests another useful system principle:

damage often occurs where energy is concentrated, not where it originated	(7.26)
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Within Photony, concentration corresponds to a local increase in the density or structural loading of electric or magnetic chain organization.

Within conventional engineering, the same location may appear as a voltage maximum, current maximum, resonant point, field enhancement, thermal hot spot, return-path discontinuity, or dielectric stress concentration.

The language differs.

The physical lesson is the same.

The important location in a failure analysis is often the point where the electromagnetic pathway becomes concentrated.

7.17 A New Diagnostic Question: Where Did the Energy Come From?

Electrical failure analysis often begins at the damaged component.

That is necessary, but it may not be sufficient.

The unified framework suggests beginning with five questions:

Where was the energy stored?
 How was it transported?
 Where was it concentrated? (7.27)
 What changed the available pathway?
 Where was the energy finally released?

These questions apply surprisingly broadly.

For a spark-plug system, the energy was stored magnetically, transformed through the ignition coil, concentrated electrically across the gap, and released through controlled gas breakdown.

For an EMC problem, the energy may originate in a digital switching event, propagate along an interconnect, encounter an asymmetry, convert into common-mode excitation, and radiate from a cable.

For an insulation failure, electric energy may accumulate across a defect until a conductive path forms.

For a solar flare, magnetic free energy accumulates in stressed current-supported structures and is subsequently released through rapid magnetic reorganization.

For a geomagnetic disturbance, energy originating in the solar environment can ultimately appear as induced electrical stress in terrestrial infrastructure.

The five-question framework therefore provides a practical bridge between Photony interpretation and conventional diagnostics.

7.18 Emission and Susceptibility Are the Same Door Viewed From Opposite Sides

The EMC discussion of Chapter 6 provides another important system-level insight.

A structure that couples electromagnetic energy outward can often also couple electromagnetic energy inward.

An unintended antenna is therefore not only a radiation problem.

It can also be a receiving structure.

A long cable, chassis seam, exposed trace, loop, shield discontinuity, or poorly controlled return path can participate in both emission and susceptibility.

This reciprocity is conventionally understood through electromagnetic coupling and antenna theory. Photony adds a structural interpretation: the same geometry that permits electromagnetic organization to leave the intended pathway can also provide a pathway through which external organization couples into the system.

This leads to a concise engineering principle:

a path that leaks outward may also admit energy inward	(7.28)
---	--------

This principle connects signal integrity, EMC, antenna behavior, and even space-weather susceptibility.

A long terrestrial conductor can become an unintended receiving structure for large-scale, slowly varying electromagnetic disturbances just as a much smaller conductor can receive high-frequency radiation.

The frequencies and coupling mechanisms differ, but the pathway concept remains useful.

7.19 Scale Changes the Numbers, Not Necessarily the Logic

Care is required when comparing electromagnetic systems across very different physical scales.

It would be incorrect to claim that a spark gap, transmission line, solar flux structure, and planetary magnetosphere are dynamically equivalent.

They are not.

Their characteristic dimensions, frequencies, conductivities, plasma conditions, timescales, field strengths, and governing approximations can differ enormously.

Yet energy conservation, geometry, material response, boundary conditions, coupling, and threshold behavior remain relevant at all of those scales.

This permits a more modest and supportable conclusion:

scale changes the implementation while preserving the energy pathway	(7.29)
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That is the appropriate level at which the ignition-coil and solar-storm comparisons should be understood.

The value lies in recognizing recurring organization, not in asserting physical identity.

7.20 What Photony Adds at the Macroscopic Level

Photony begins with microscopic propositions, but its greatest practical value may ultimately lie in the way those propositions reorganize macroscopic thinking.

At the macroscopic level, Photony suggests that:

electric fields can be viewed as a physical electric chain organization rather than only as vectors assigned to space;

magnetic fields can be viewed as physical magnetic chain organization rather than only as abstract field geometry;

electrons can be viewed as transducers between elemental transport and larger electromagnetic structures;

voltage can be interpreted as a measure associated with surface-supported electric organization;

current can be interpreted as the directed transport condition that supports magnetic organization;

capacitance and inductance can be viewed as geometries designed to support specific forms of electromagnetic organization;

transmission lines can be viewed as geometries that guide coupled electric and magnetic organization;

antennas can be viewed as structures that permit source-coupled organization to become propagating electromagnetic structures;

EMC problems can be viewed as failures to control where electromagnetic organization is allowed to propagate;

breakdown can be viewed as a threshold at which the existing organization can no longer be supported by the available structure.

These interpretations do not change the conventional equations.

They change the questions being asked of those equations.

7.21 Be the Photon: Following the Energy Instead of the Diagram

Signal-integrity practice has long benefited from the instruction to "**be the signal.**"

The idea is powerful because it asks the engineer to stop looking only at the schematic and to imagine the actual electromagnetic journey through the physical structure.

Where does the signal travel?

Where is the return current?

What impedance does it encounter?

Where will it reflect?

What neighboring structure can it couple with?

Where can it radiate?

The Photony framework extends that thought experiment one level deeper:

Be the photon.

The phrase is not meant literally as a replacement for electromagnetic analysis. It is a physical reasoning tool.

Follow the energy.

Ask where the elemental transport is proposed to occur.

Ask which electrons can transduce it and at what frequency.

Ask what electric or magnetic organization is created.

Ask what geometry supports that organization.

Ask where the organization becomes concentrated.

Ask whether it can continue smoothly.

Ask what happens when it reaches a boundary.

Ask whether it will be reduced, returned, redirected, radiated, reassigned, or broken down.

This viewpoint is particularly useful because it discourages the assumption that the schematic boundary is also the electromagnetic boundary.

It rarely is.

In practical electromagnetic systems, the intended signal path is usually easy to identify, while the return path is often more difficult to follow because it is determined by distributed impedance and surrounding geometry rather than by schematic connectivity alone. At low frequency, the return current may approximately follow the obvious conductive route back to the source. As frequency increases, however, inductance, capacitance, reference planes, conductor spacing, apertures, and discontinuities increasingly determine the actual return-current distribution.

For high-frequency signals, the electromagnetic return path is often the least obvious part of the circuit and one of the most important. The current tends to remain close to the forward path when a nearby continuous reference plane is available because that geometry reduces loop inductance and confines the associated electromagnetic fields. The return path can become much more complicated when the reference structure is interrupted or when the signal moves from one electromagnetic environment to another.

A common example occurs when a printed-circuit trace crosses a split or slot in its reference plane. The forward signal can continue directly across the trace, but the return current cannot cross the gap in the same manner. It must detour around the opening, use a nearby stitching connection, or couple capacitively to another reference structure. The result is a larger effective loop area, greater field spreading, increased inductance, and a greater tendency toward common-mode coupling and radiation.

Another difficult return path occurs at a connector. A signal conductor may pass cleanly from a printed-circuit board into a cable, while the

associated return current must transfer from a board plane to a connector shell, shield, ground pin, or other conductive structure. If that transition is incomplete or asymmetric, part of the energy that was previously confined to the intended signal-return pair can redistribute onto the cable shield, chassis, or other conductors. The connector can therefore become the location where a well-controlled differential signal begins to produce common-mode current.

A via transition can create a similar problem. When a signal changes layers, its return current may also need to change reference planes. If no nearby PCB stitching via or capacitive return path is available, the return current may spread over a larger region before finding a suitable route. The signal via itself may appear electrically simple, while the return transition produces a more significant electromagnetic discontinuity.

Cable shields provide another instructive example. A shield may be intended to provide a low-impedance return or containment path, yet poor termination at one end can force high-frequency return current onto unintended surfaces. A long pigtail connection, for example, can add substantial inductance and prevent the shield from behaving as an effective high-frequency continuation of the chassis or enclosure. The current then redistributes over the cable, connector, chassis, and surrounding capacitances.

Chassis and enclosure currents can be even harder to visualize because the return path may not resemble a conventional wire at all. Current can spread over broad metal surfaces, cross seams through multiple contact points, couple capacitively across small gaps, and divide among several parallel paths. A seam, aperture, poorly bonded panel, or paint-covered contact region can therefore alter the return geometry enough to change both emissions and susceptibility.

A further example occurs in mixed power and signal systems. A digital signal may reference a local ground plane while its attached cable, power supply, or chassis provides an additional return path through parasitic capacitance. At low frequency, that parasitic path may appear negligible. At high frequency, its impedance can become sufficiently small that a significant fraction

of the return current uses it. The actual electromagnetic loop can then extend far beyond the intended circuit.

These examples show why a return path should not be identified simply by following the symbol labeled "ground." Ground is not a single physical location at high frequency. It is a distributed conductive and capacitive network whose geometry determines where return current can flow.

The ambiguity of the word "ground" has long been recognized in electromagnetic-compatibility practice. Archambeault memorably summarized the problem with the statement, "Ground is a place where potatoes and carrots thrive" [Archambeault \(2002\)](#). The humor makes an important engineering point: a schematic symbol labeled "ground" does not, by itself, identify the physical path taken by high-frequency return current.

A further reason to avoid treating "ground" as an absolute electrical reference is propagation time. Electromagnetic information does not travel instantaneously along a grounding conductor. In free space, propagation requires approximately 1 ns per foot, and propagation along a practical conductor or cable is generally slower because of the surrounding dielectric environment. A conductor extending several feet to an Earth ground rod therefore cannot be assumed to place a high-frequency circuit instantaneously at the electrical potential of the Earth connection.

This distinction becomes increasingly important as signal rise time decreases. A grounding conductor that behaves acceptably at power frequency may possess enough inductance and propagation delay to behave very differently during a fast transient. At sufficiently high frequency, the remote end of the wire is not an ideal zero-impedance reference. The phrase "Earth ground" therefore does not guarantee a good high-frequency electromagnetic return path [Ott \(2009\)](#); [Paul \(2006\)](#).

Earth itself is also not necessarily an equipotential reference. Ground rods and grounding grids have finite resistance and impedance, and substantial currents flowing through soil can establish local ground-potential differences. An industrial site containing large electrical loads,

furnaces, welding systems, motors, or other high-current equipment can, therefore, have significant currents flowing through its grounding network and surrounding earth. A ground rod located in such an environment may be connected to Earth, yet the local Earth potential can differ from that at another point in the facility. Connecting two supposedly "grounded" locations can then produce current rather than eliminate it.

The distinction between an Earth connection and an electromagnetic return path is particularly clear in transmission-line operation. A properly constructed transmission line does not require an Earth-ground connection to propagate a signal. What it requires is a controlled forward-and-return electromagnetic geometry. In a coaxial cable, for example, the center conductor and outer conductor establish the transmission structure. In a balanced pair, the two conductors establish the corresponding differential structure. Neither requires the Earth itself to complete the high-frequency signal path.

This can be stated simply:

$$\text{signal return path} \neq \text{Earth ground.} \quad (7.30)$$

Spacecraft provide an especially clear demonstration. An orbiting spacecraft cannot depend on a conducting connection to terrestrial Earth; yet, its electronics, transmission lines, power distribution systems, radios, sensors, and digital circuits operate normally. Their electrical references and return paths are established by conductors, chassis structures, reference planes, cable shields, and distributed capacitances within the spacecraft. The required electromagnetic circuit is completed locally rather than through planet Earth.

The more useful engineering question is, therefore, not, "Is this conductor connected to Earth?" but, "Where does the return current actually flow at the frequency and rise time of interest?" A low-impedance Earth connection may be essential for personnel safety, lightning protection, fault-current management, or other purposes while simultaneously being a poor high-frequency signal return.

This distinction is central to the "Be the Photon" viewpoint. The photon does not recognize

a schematic ground symbol. Within the proposed Photony interpretation, the electric chain and magnetic chain structures follow the physical electromagnetic pathways permitted by conductor geometry, impedance, capacitance, inductance, and surrounding matter. The actual return structure, rather than its schematic name, determines how that organization is completed.

In practical systems, the term “ground” may refer to earth, chassis, safety ground, a signal reference plane, a power reference, or a current return structure. These functions are not interchangeable. At high frequency, current must still return to its source through the distributed impedance of the available conductors and surrounding electromagnetic structure. The physically important question is therefore not simply “Where is ground?” but “What structure actually provides the return path at the frequencies of interest?”

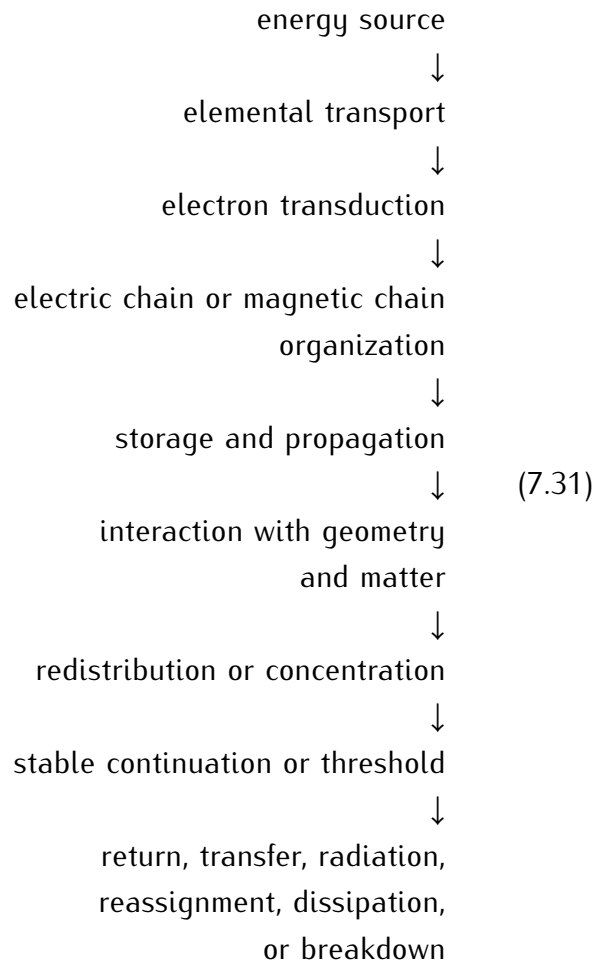
This distinction fits naturally with the Photony interpretation. A ground symbol does not define the electric chain or magnetic chain pathway. The actual conductor geometry, reference surfaces, capacitances, inductances, and discontinuities determine how the electromagnetic organization is established and how the return portion of the pathway is completed.

Within the proposed Photony interpretation, this behavior can be viewed as a reorganization of the electric chain and magnetic chain structure that supports the propagating signal. When the preferred return geometry is continuous and close to the forward path, the chain organization remains comparatively confined. When the return path is interrupted, displaced, or forced through a different structure, the electromagnetic organization must spread, redirect, or couple into additional conductors and surrounding space.

This is why the return path is often more revealing than the forward path. The forward conductor shows where the engineer intended the signal to go. The return current often reveals where the electromagnetic system actually went.

7.22 The Unified Photony Energy Path

The ideas developed throughout this Compendium can now be condensed into one system-level Photony sequence:



Not every system passes through every stage. The microscopic stages remain proposed Photony mechanisms.

The macroscopic stages are constrained by conventional electromagnetic theory and measurement.

The significance of the sequence is that it connects the smallest proposed structural elements to the largest observable behavior without requiring a different conceptual language at every scale.

Electricity, magnetism, propagation, radiation, EMC, breakdown, and space-weather coupling become different branches of one energy pathway.

7.23 The Central Synthesis

The most important idea of this Compendium can now be stated simply.

Electromagnetic systems organize energy.

They do so through matter, geometry, charge separation, current, boundaries, and time-dependent excitation.

That organization determines where energy can be stored.

It determines where energy can move.

It determines what other structures can receive it.

It determines where energy becomes concentrated.

It helps determine whether the system remains stable.

And when stability is lost, the prior organization helps determine how the energy will be released.

The proposed Photony framework gives that organization a physical structure.

Conventional electromagnetics gives it quantitative law.

The two levels should therefore not be treated as competitors.

One answers primarily,

What electromagnetic behavior is measured and calculated?

The other asks,

What physical organization might be producing that behavior?

The value of the Photony interpretation will ultimately depend on whether that second question leads to experimentally distinguishable predictions.

But even before that test is complete, the system-level viewpoint produces a useful discipline of thought:

follow the energy from source to destination, and back	(7.32)
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That principle connects the ignition coil to the spark plug, the transmitter to the antenna,

the PCB trace to the radiating cable, the Sun to the magnetosphere, and the geomagnetic disturbance to the terrestrial conductor.

The systems differ.

The energy pathways tell the story.

7.24 Discussion

The synthesis developed in this chapter suggests that electromagnetic systems are best understood as connected energy pathways rather than as collections of isolated electric and magnetic effects. Energy is introduced, organized, transported, concentrated, transformed, and ultimately returned, dissipated, radiated, or released.

This viewpoint helps separate the location of an energy source from the location of a final failure. An ignition coil, for example, stores magnetic energy and produces the high-voltage condition that leads to electric breakdown at the spark plug. Likewise, solar magnetic energy can drive a disturbance that eventually produces induced electric fields, currents, and voltage stress in terrestrial systems [Martin \(2026a\)](#). The physical systems are very different, but both demonstrate that the form in which energy is initially stored need not be the form in which damage eventually appears.

The same reasoning applies to transmission lines, EMC problems, connectors, return paths, and insulation failures. The visible problem may occur at one location even though the energy was generated, transformed, or concentrated elsewhere. For this reason, electromagnetic analysis should follow the complete energy pathway rather than stop at the damaged or radiating component.

Return paths provide a particularly good example. The forward conductor is usually obvious, while the return current may divide among planes, shields, chassis structures, capacitances, and other conductive paths. A schematic ground symbol does not identify that physical route. Earth ground may be important for safety and fault protection while still being a poor high-frequency return path. Conversely, a properly designed transmission line or spacecraft electrical system can have an excellent return structure

without requiring a direct connection to Earth.

Within the proposed Photony framework, these behaviors are interpreted as changes in electric chain and magnetic chain organization constrained by geometry, material properties, boundaries, frequency, and electron transduction. Conventional electrodynamics remains the quantitative foundation. Photony adds the structural question of what physical organization is changing as the measured electromagnetic state changes.

The practical value of this viewpoint is that it encourages the reader to follow the energy from its source through every transformation, boundary, and concentration point to its final destination.

7.25 Conclusions

This chapter has brought the principal ideas of the Photony Electromagnetic Compendium into a single system-level framework. The central theme is that electromagnetic systems organize energy through electric chain and magnetic chain structures whose behavior is shaped by current, voltage, electron transduction, geometry, conductivity, boundaries, and frequency.

Electric and magnetic phenomena should therefore not be treated as isolated worlds. A magnetic-energy process can produce an electric failure, and an electric disturbance can create

current-supported magnetic organization. The observable outcome depends on the pathway through which energy is stored, transported, transformed, and concentrated.

Geometry and boundaries are especially important because they determine where electromagnetic organization can continue smoothly and where it must reorganize. Return paths are equally important because the actual electromagnetic pathway is determined by distributed impedance and physical structure rather than by schematic labels such as “ground.”

The proposed Photony framework does not replace Maxwell’s equations, transmission-line theory, Poynting energy transport, antenna theory, breakdown physics, or EMC analysis. Those remain the quantitative basis for electromagnetic engineering and physics. Photony instead proposes a physical interpretation beneath them, centered on elemental charge photons, electron-mediated transduction, and organized electric and magnetic chains.

The broad lesson of the Compendium is therefore simple: follow the energy. Determine where it is organized, how it is transported, where geometry and boundaries redirect it, where it becomes concentrated, and how it is ultimately returned, radiated, dissipated, or released. That pathway-based view provides the unifying Photony interpretation developed throughout this book.

Bibliography

- (2010). IEEE standard definitions for the measurement of electric power quantities under sinusoidal, nonsinusoidal, balanced, or unbalanced conditions.
- Archambeault, B. R. (2002). *PCB Design for Real-World EMI Control*. Kluwer Academic Publishers, Boston.
- Balanis, C. A. (2016). *Antenna Theory: Analysis and Design*. John Wiley & Sons, 4 edition.
- Bogatin, E. (2018). *Signal and Power Integrity: Simplified*. Pearson, 3 edition.
- Feynman, R. P., Leighton, R. B., and Sands, M. (1964). *The Feynman Lectures on Physics, Volume II: Mainly Electromagnetism and Matter*. Addison-Wesley, Reading, MA.
- Griffiths, D. J. (2017). *Introduction to Electrodynamics*. Cambridge University Press, Cambridge, UK, 4 edition.
- Hagyard, M. J. and Smith, J. B., J. (1988). Flare onset at sites of maximum magnetic shear. In *Max 1991: Flare Research at the Next Solar Maximum, Workshop 1: Scientific Objectives*. NASA Goddard Space Flight Center.
- Hill, J. D., Uman, M. A., and Jordan, D. M. (2011). High-speed video observations of a lightning stepped leader. *Journal of Geophysical Research: Atmospheres*, 116(D16):D16117.
- Hourdakis, E. S., Simonds, B. J., and Zimmerman, N. M. (2006). Submicron gap capacitor for measurement of breakdown voltage in air. *Review of Scientific Instruments*, 77(3):034702.
- Jackson, J. D. (1998a). *Classical Electrodynamics*. John Wiley & Sons, New York, 3 edition.
- Jackson, J. D. (1998b). *Classical Electrodynamics*. John Wiley & Sons, New York, NY, 3rd edition.
- Jefimenko, O. D. (1989). *Electricity and Magnetism: An Introduction to the Theory of Electric and Magnetic Fields*. Electret Scientific Company, 2 edition.
- Lin, S., Xu, L., Wang, A. C., et al. (2020). Quantifying electron-transfer in liquid-solid contact electrification and the formation of electric double-layer. *Nature Communications*, 11:399.
- Martin, F. L. (2022). *Photony Foundations*. PC Consulting.
- Martin, F. L. (2026a). From magnetic and electric fields to fires. *OAJAST*.
- Martin, F. L. (2026b). *The Photony Mathematical Compendium – Volume VI*. PC Consulting.
- Maxwell, J. C. (1865). A dynamical theory of the electromagnetic field. *Philosophical Transactions of the Royal Society of London*, 155:459–512.

- Meek, J. M. (1940). A theory of spark discharge. *Physical Review*, 57:722–728.
- Moore, R. L., Sterling, A. C., Gary, G. A., Cirtain, J. W., and Falconer, D. A. (2011). Observed aspects of reconnection in solar eruptions. *Space Science Reviews*, 160:73–94.
- Occupational Safety and Health Administration (2026). 29 cfr 1910.106: Flammable liquids. U.S. Department of Labor. See especially 1910.106(e)(6)(ii) and 1910.106(f)(3)(iv) regarding electrical interconnection, bonding, grounding, and protection against static sparks.
- Ott, H. W. (2009). *Electromagnetic Compatibility Engineering*. John Wiley & Sons.
- Paul, C. R. (2006). *Introduction to Electromagnetic Compatibility*. John Wiley & Sons, 2 edition.
- Poynting, J. H. (1884). On the transfer of energy in the electromagnetic field. *Philosophical Transactions of the Royal Society of London*, 175:343–361.
- Pozar, D. M. (2012a). *Microwave Engineering*. John Wiley & Sons, 4 edition.
- Pozar, D. M. (2012b). *Microwave Engineering*. John Wiley & Sons, Hoboken, NJ, 4th edition.
- Saberi, M., Jafarzadeh, S., Wedemeyer, S., Gafeira, R., Szydlarski, M., Jess, D. B., and Stangalini, M. (2025). The sun at millimeter wavelengths. v. magnetohydrodynamic waves in a fibrillar structure. *Astronomy & Astrophysics*, 693:A19.
- Sadiku, M. N. O. (2018). *Elements of Electromagnetics*. Oxford University Press, 7 edition.
- Sobarzo, J. C., Pertl, F., Balazs, D. M., Costanzo, T., Sauer, M., Foelske, A., Ostermann, M., Pichler, C. M., Wang, Y., Nagata, Y., Bonn, M., and Waitukaitis, S. (2025). Spontaneous ordering of identical materials into a triboelectric series. *Nature*, 638(8051):664–669.
- Soler, R., Andries, J., and Goossens, M. (2012). Resonant alfvén waves in partially ionized plasmas of the solar atmosphere. *Astronomy & Astrophysics*, 537:A84.
- Sterling, A. C., Moore, R. L., Berger, T. E., Bobra, M. G., Kubo, M., Lites, B. W., Shine, R., Tarbell, T. D., Title, A. M., Ichimoto, K., Suematsu, Y., Tsuneta, S., Katsukawa, Y., Shibasaki, K., et al. (2007). Hinode observations of the onset stage of a solar filament eruption. *Publications of the Astronomical Society of Japan*, 59(sp3):S823–S829.
- Van Brunt, R. J. (1994). Physics and chemistry of partial discharge and corona: Recent advances and future challenges. *IEEE Transactions on Dielectrics and Electrical Insulation*, 1(5):761–784.
- Wang, X., Patel, A., Bane, S., and Shashurin, A. (2021). Experimental study of atmospheric pressure single-pulse nanosecond discharge in pin-to-pin configuration. *Journal of Applied Physics*, 130:103303.
- Zahn, M. (2003). *Electromagnetic Field Theory: A Problem Solving Approach*. Krieger Publishing Company, Malabar, FL. Originally published by John Wiley & Sons in 1979.